

NAVAL AIR STATION SIGONELLA

INTEGRATED PEST MANAGEMENT PLAN



JUNE 2024

PREPARED BY



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Naval Air Station Sigonella Integrated Pest Management Plan

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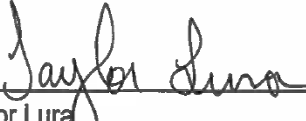
Naval Air Station Sigonella Integrated Pest Management Plan

Technical Review

The Integrated Pest Management Plan has been reviewed in accordance with DoDM 4150.07, Volume 1

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Integrated Pest Management Plan Annual Review

Year	Completion Date	Integrated Pest Management Coordinator	Copies of Changes to the Pest Management Consultant (Annual) Date Completed
2023			
2024			
2025			
2026			
2027	Rewrite		

Scheduled On-Site Pest Management Technical Review

Scheduled Date	Integrated Pest Management Coordinator	Pest Management Consultant Date Review Completed
21 August 2023	Alessandra Contarino	24 August 2023

Note: Technical Reviews should be scheduled approximately 3 years apart and in conjunction with an Environmental Management System (EMS) review if possible.

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Executive Summary

This Integrated Pest Management Plan is a comprehensive, long-range document that captures all of the pest management operations and pesticide-related activities conducted on the installation. It incorporates pest management practices and the local, federal, and Department of Defense regulations, and conforms to the requirements of the Environmental Final Governing Standards Italy (Chapter 11, Pesticides), Department of Defense Instruction 4150.07, DoD Pest Management Program, Chief of Naval Operations Instruction 6250.4C, Navy Pest Management Programs, and Chief of Naval Operations Instruction 5090.1E, Environmental Readiness Program. It supplies comprehensive information about the pest management program to installation staff and internal and external stakeholders and compliance regulators.

The Naval Facilities Engineering Systems Command, Atlantic's Applied Biology Center prepared this plan using information obtained through pest management data collection, on-site observations, installation personnel interviews, and document reviews.

The main goal of the various pest control functions is to support the mission of Naval Air Station Sigonella (NASSIG). Commercial contract pest control operators currently provide pest control services for NASSIG I and NASSIG II. In-house pest control operators provide vegetation control at Naval Radio Transmitting Facility (NRTF) located in Niscemi, Italy. Pest control services are needed on the installation in order to:

1. Provide services that will prolong the life of the structures through structural and nuisance pest control
2. Maintain the safety and security of industrial and storage areas through weed control
3. Provide nuisance pest control to all buildings (except public-private venture housing) and outdoor areas to ensure a good working and living environment
4. Control weed and insect pests in all recreational and lawn areas to maintain aesthetics and provide recreational facilities to personnel
5. Provide control of mosquitoes, flies, and other potential disease vectors to ensure the comfort and well-being of all personnel
6. Provide vertebrate pest control, including rodent control, to all areas of the installation

For the current level of work to be accomplished, a sufficient staff of qualified applicators must be maintained. Contract personnel must meet host nation certification requirements as specified by the contract. Pest management performance assessment representatives must successfully take an initial Pest Management Performance Assessment Representatives course. To maintain their certification, the Pest Management Performance Assessment Representatives and Department of Defense pesticide applicators must successfully pass a Department of Defense pest management training and recertification course every three years. Pest control facilities must comply with current safety standards to provide a safe workplace and to minimize pesticide hazards.

The fundamental pest management requirements are that pesticides be mixed, applied, stored, and disposed of in accordance with the pesticide label and all applicable policies and regulations; installations maintain an up-to-date integrated pest management plan; installations designate an integrated pest management coordinator and a pest management performance

assessment representative (if contracted); all pest management operations be reported; all pesticides be approved prior to use; and all pest management personnel be appropriately certified.

This plan focuses on safe, environmentally-sound, and cost-effective control of pests through integrated pest management. Integrated pest management depends on education, proper surveillance and identification of pests, non-chemical and chemical control methods, and individual responsibility for pest prevention.

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Chapter 1. Introduction

1.1. INTEGRATED PEST MANAGEMENT PLAN

The Integrated Pest Management Plan (IPMP) is a long-range, comprehensive planning and operational document that establishes the strategy and methods for conducting a safe, effective, and environmentally-sound Integrated Pest Management (IPM) program. The IPMP covers all pest management and pesticide-related activities conducted by civilian and military Department of Defense (DoD) personnel and commercial contractors within all functional areas of the installation.

1.1.1. Authority

Department of Defense Instruction (DoDI) 4150.07, DoD Pest Management Program, Chief of Naval Operations Instruction (OPNAVINST) 6250.4C, Navy Pest Management Programs, Chief of Naval Operations (OPNAVINST) 5090.1E, Environmental Readiness Program and the Environmental Final Governing Standards (FGS) Italy (Chapter 11, Pesticides) require that all Navy installations develop and implement an IPMP in accordance to the guidelines. The IPMP will detail all aspects of pesticide management including administration, procurement, contract services, storage, disposal, safety, reporting, vehicles, integrated pest management, and applicable laws and regulations.

1.1.2. Department of Defense Measures of Merit

This plan provides the framework for the installation to meet the DoD's annual goals or measures of merit (MoMs). As established in DoDI 4150.07, and updated by the Armed Forces Pest Management Board, the MoMs are:

1. Measure of Merit 1: All DoD installations will maintain IPM plans that are reviewed and approved by a DoD-certified Pest Management Consultant (PMC) and annually updated by the Integrated Pest Management Coordinator (IPMC).

NASSIG helps meet this goal by implementing (via the commanding officer's signature) this plan.

2. Measure of Merit 2: All DoD installations will adhere to the principals of integrated pest management and the DoD will maintain the goal of minimizing annual pesticide use by both government and contractor pesticide applicators on its installations. The DoD will maintain or reduce total pesticide use on DoD installations to a level no greater than 425,000 pounds (average usage of the fiscal years 2007 and 2009 usage) of active ingredient per year.

NASSIG provides data for this MoM through the reporting requirement ([section 2.3](#)).

3. Measure of Merit 3: All DoD pesticide applicators will be certified. Direct-hire Certification employees, certified in accordance with Department of Defense Manual (DoDM) 4150.07, Volume 2, have up to 2 years to become certified after initial employment. Contracted employees must have appropriate certification in the appropriate categories at the time the contract is awarded.

NASSIG ensures proper certification of all applicators through regular verifications and maintains a list of certifications in [appendix E](#) of this plan. See [section 2.4](#) for training and certification requirements.

4. Measure of Merit 4: By the end of Fiscal Year 2020, all pesticide application on all Archiving DoD installations, or in support of a DoD operation, will be reported. Reports will be entered into a searchable DoD database and permanently archived.

NASSIG ensures proper reporting of pesticide applications by entering applications into the NAVFAC Online Pesticide Reporting System (NOPRS).

1.1.3. Integrated Pest Management Plan Implementation

The IPMP must be reviewed and approved by installation stakeholders and professional pest management consultants (PPMCs) from Naval Facilities Engineering Systems Command (NAVFAC), and the Bureau of Medicine and Surgery (BUMED). The IPMP is implemented upon signature of the installation's commanding officer. The IPMC has the task of implementing, coordinating, and executing the IPMP among each of the functional areas of the installation.

1.1.4. Integrated Pest Management Plan Maintenance

Once the IPMP has been developed and implemented, it must be reviewed annually and updated as necessary. The installation IPMC is responsible for maintaining the IPMP.

1.1.4.1. Internal Review

The IPMC shall conduct an internal review annually in coordination with the pest management service providers (PMSPs) and other functional area points of contact (POCs). The review should include updating contract information, applicator certifications, pesticides, and pest management operations to be used on the installation, as well as updating pesticide use records. The pest management program self-assessment checklist ([appendix C](#)) is available as a tool to review compliance issues during the internal review.

1.1.4.2. Off-Site Review

The IPMC may request that a NAVFAC Applied Biology PPMC perform a review of regulatory requirements, reporting, and pesticide approval procedures.

1.1.4.3. On-Site Review

The NAVFAC Applied Biology PPMC shall perform an on-site review of the entire pest management program every three years to ensure compliance with the IPMP. The review may be performed more frequently if extensive program problems exist. A copy of the last technical review report can be found in [appendix C](#).

1.1.4.4. Integrated Pest Management Plan Rewrite

The IPMP should be rewritten every five years to reflect new contracts, personnel, pest management practices, and regulatory changes.

1.2. INSTALLATION BACKGROUND

Naval Air Station Sigonella is responsible for four sites on the island of Sicily:

1. Naval Air Station Sigonella I (NASSIG I)
2. Naval Air Station Sigonella II (NASSIG II)
3. Naval Radio Transmitting Facility (NRTF) Niscemi
4. Pachino Target Range Complex (PTRC)

1.2.1. Mission

The mission at NASSIG is to provide consolidated operational, command and control, administrative, logistical and advanced logistical support to US and other North Atlantic Treaty Organization (NATO) commands.

1.2.2. Location and Facilities

NASSIG I is located in the province of Catania and occupies 92 acres that are ceded for use by the Italian Ministry of Defense. NASSIG I is the personnel support area that provides the main housing and support facilities for US Navy Personnel. NASSIG I consists of family housing, bachelor officer housing, Navy Exchange (NEX) services, a hospital, a Navy Lodge, a DoD school, administrative/security facilities, community support facilities, and operational facilities.

NASSIG II is located in the province of Syracuse and occupies 2,180 acres that are ceded for use by the Italian Ministry of Defense. NASSIG II contains the North Atlantic Treaty Organization (NATO) Maritime Patrol Airfield, an ordnance area, maintenance and public works functions, bachelor housing and administrative, community, and personnel support facilities.

NRTF Niscemi is located 7 km southeast of the city of Niscemi in the province of Caltanissetta and occupies 670 acres that are ceded for use by the Italian Ministry of Defense. NRTF Niscemi contains antenna fields, and transmitter facilities, including the transmitter building and the helix house. Much of the region is under development restriction for hydrological and forestry reasons.

The PTRC supports limited training for the Sixth Fleet and NATO aircraft operations. There are no personnel at PTRC and it is seldom used. No pesticides are applied at PTRC because it's a protected area. The PTRC compound consists of three sites (Site I, Site II, and the boat landing site) that are located near the shoreline on land leased from the Italian Ministry of Defense. Site I and the boat landing site lie within the province of Ragusa, while Site II is within the province of Syracuse. Site I houses the operations center for the range. Site II is a single tower located east of Site I along the shore towards Pachino in an area known as Grotticelle, and serves as a triangulation point for bomb scoring. The boat landing site located on Punta Castellazzo is used to launch and retrieve boats that aid in ordnance recovery. The boat landing site is within a zone of very limited development and is protected by an archaeological *vincolo* as the site of the ancient city of Apolline.

This plan also applies to the Weapons Compound, all outside housing complexes (including Mineo and Marinai) and Augusta Bay.

1.3. PEST MANAGEMENT PROGRAM OVERVIEW

The pest management program is summarized below.

1. General pest control and grounds maintenance pest services are provided by the Base Operations Support (BOS) contract with Valiant/Alca JV, LLC.
2. Department of Defense Dependents School (DoDDS) pest control services are provided by SKE Pest Control.
3. Navy Gateway Inns and Suites (NGIS) pest control services are provided by TriStar.
4. Pest control services for Marinai Housing are provided by Pizzarotti.
5. Grounds maintenance pest control service is provided by in-house applicators at NRTF Niscemi.

1.3.1. Pest Management Objectives

The objectives of the installation's pest management program are:

1. The prevention of pest-related health and safety problems that affect the mission.

Examples of health-related pests that may affect the mission include, but are not limited to mosquitoes, ants, and cockroaches. Any pest may impact the mission when its numbers become excessive. Prevention of pest-borne disease and injury is a component of force health protection. Force health protection seeks to maintain a healthy and fit military and civilian force in order to maintain the highest levels of readiness. Pest management is a force multiplier for construction battalions, maintenance commands, and other deployable units. Additionally, the military and civilian infrastructure on the installation must be protected in order to provide the necessary support to these units as well.

2. The prevention of pest damage to equipment and subsistence used to support the operational mission of the activities and tenant commands.

Equipment and materials are susceptible to physical damage by pests and the financial costs of such damage can be high. Rodents, for example, can cause considerable damage to electronic equipment through gnawing on electrical components. Examples of pests that may damage equipment and subsistence include, but are not limited to rodents, wood-destroying pests, and stored-product pests.

3. Vegetation management to protect the local environment.

The introduction of non-native/invasive species of plants can increase the risk of fire and degrade the surrounding native environment that is home to a number of endangered and threatened animal and plant species.

4. The protection of government real property, materiel, and aesthetics.

Buildings and roads that form the infrastructure of the installation are susceptible to pests. If not adequately prevented and controlled, termites can cause extensive damage to wood structures. Weeds can cause damage to roadways and increase the risk of fire.

5. The reduction of the use of and dependence on pesticides.

1.3.2. Program Requirements

1.3.2.1. Administration

Proper administration of the pest management program ensures accountability and documentation through planning, record keeping, reporting, training, pesticide and contract approval, and regulatory compliance. Table 1-1 outlines the pest management administrative program requirements.

Table 1-1. Pest management administrative program requirements.

Requirement	Description	Reference	Responsibility	Locator
PLANNING	Review and revise the IPMP annually.	OPNAV M-5090.1 OPNAVINST 6250.4C DoDM 4150.07 Vol 1	IPMC	Section 1.1.4
PESTICIDE APPROVAL	Compile and submit list of new pesticides to NAVFAC Applied Biology for approval for use on the installation.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy DoDM 4150.07 Vol 1	IPMC in coordination with PMPARs and PMSPs	Section 2.2
RECORDING	Record all pest management operations conducted on the installation after each operation.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy DoDM 4150.07 Vol 1 7 CFR §110*	PMSPs	Section 2.3.1
MAINTAINING	Maintain records of all pest management operations conducted on-site indefinitely.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy DoDM 4150.07 Vol 1 7 CFR §110*	IPMC in coordination with PMPARs	Section 2.3.2
REPORTING	Compile and report all pest management operations to NAVFAC Applied Biology monthly.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy DoDM 4150.07 Vol 1 7 CFR §110*	IPMC in coordination with PMPARs	Section 2.3.3

Requirement	Description	Reference	Responsibility	Locator
PESTICIDE APPLICATOR CERTIFICATION	Ensure that all personnel applying pesticides on installations have either current DoD pesticide applicator certification or state commercial applicator certification.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy DoDM 4150.07 Vol 1 40 CFR §171*	IPMC in coordination with PMPARs and PMSPs	Section 2.4
COMPLIANCE	Ensure that all program elements are in compliance with all federal regulations. DoD policy is to voluntarily comply with local/state regulations if compliance does not degrade DoD mission.	OPNAV M-5090.1 OPNAVINST 6250.4C DoDM 4150.07 Vol 1	IPMC in coordination with PMPARs	Section 3.3
CONTRACT REVIEW	Review pest management contract specifications for compliance with the IPMP and submit to NAVFAC Applied Biology for final review and approval prior to advertising.	OPNAV M-5090.1 OPNAVINST 6250.4C DoDM 4150.07 Vol 1	Facilities Support Contracting personnel PMPARs IPMCs	Section 2.5
* (applies to restricted-use pesticides only)				

1.3.2.2. Operations

Operations are the day-to-day management of pests through pesticides and non-chemical means. Pest management on the installation includes the following categories of operations:

1. Ornamental and Turf—Control and management of pests of landscape plants and turf including arthropods, fungi, and weeds.
2. Right-of-way—Control and management of vegetation along roadways as well as vegetation control near fuel farms to reduce fire risk and along fence lines to enhance security.
3. Aquatic Weed Control—Control of vegetation in ponds and ditches.
4. Industrial, Institutional, Structural, and Health-Related—Control and management of pests in and around buildings. Pests may include cockroaches, termites, bees, venomous animals, stored product insects, rodents, and feral animals.
5. Public Health—Control and management of human and animal disease vectors such as rodents, mosquitoes, flies, ticks, and fleas.

6. Nuisance Pest Control—Control of insect pests, such as ants, that are a nuisance or annoyance to base personnel, but do not present a health risk.
7. Invasive Plants—Removal of non-native species of plants that are detrimental to native plant and animal habitats.
8. Vertebrate Control—Control of animal predators that prey upon endangered or threatened animals and their habitats or infest food and material storage.

Each of these operations must meet various requirements that are listed and described in table 1-2.

Table 1-2. Pest management operations program requirements.

Requirement	Description	Reference	Responsibility	Locator
INTEGRATED PEST MANAGEMENT	Federal agencies shall use IPM techniques in carrying out pest management activities and shall promote IPM through procurement and regulatory policies, and other activities.	OPNAV M-5090.1 OPNAVINST 6250.4C DoDI 4150.07 7 U.S.C. § 136r-1	IPMC PMSPs	Section 3.1.1
STORAGE	Pesticides kept on installations must be procured and stored in accordance with installation and federal regulations. DoD Policy is to comply with local/state regulations when meeting those standards does not degrade DoD missions.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy DoDM 4150.07 Vol 1 AFPMB TG 17	Pest control shop supervisor	Section 3.4.3

Requirement	Description	Reference	Responsibility	Locator
APPLICATION	Only registered pesticides will be used. Applicators must apply pesticides in a manner that ensures safety and protects the environment. A copy of the pesticide label shall be available at the application site.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy DoDI 4150.07	Pesticide applicators	Section 3.4.5
CONTAINERS	All containers used to store or transport a pesticide must have the original or copy of the original label attached. Service containers must have attached label identifying: the person responsible for the container, the name of chemical, and the signal word.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy	Pesticide applicators	Section 3.4.5.1
APPLICATOR SAFETY	The installation must provide procedures, medical support, equipment, and supplies to ensure the safety of DoD pesticide applicators during pest control operations. Note: Contractors are responsible for supplying their own PPE and having a medical support plan in place in the event of an accident.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy DoDM 4150.07 Vol 1	Navy Medicine Readiness and Training Unit Safety Department	Section 4.1

Requirement	Description	Reference	Responsibility	Locator
CLEANING AND DISPOSAL	Equipment shall be cleaned to prevent health and environmental hazards due to chemical residues. Prevent water from container and equipment rinsing from entering storm drains and water bodies. Dispose of empty containers properly. Manage and dispose hazardous waste and non-hazardous waste properly.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy	Pesticide applicators	Section 3.4.6
SPILL PREVENTION	Spill kits shall be maintained in pest control shops and on pest control vehicles. Pest management personnel shall be familiar with the installation spill contingency plan.	OPNAV M-5090.1 OPNAVINST 6250.4C FGS Italy	Pesticide applicators Environmental Spill Response Manager	Section 5.3.4

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Chapter 2. Program Administration

2.1. ROLES AND RESPONSIBILITIES

The success of the pest management program depends largely on a clear understanding of the roles and responsibilities for the organizations and personnel involved. The following is a listing of the key organizations and personnel and their duties as presented in DoD policy and guidance documents for the implementation of the IPMP.

2.1.1. Commander, Navy Installations Command

Commander, Navy Installations Command (CNIC) is responsible for the funding and prioritizing of the pest management program.

2.1.2. Installation Commanding Officer

The installation commanding officer (CO) is responsible for the compliance and enforcement of the pest management program. The installation commanding officer delegates compliance and enforcement of the pest management program to the IPMC via the IPMC designation letter. Responsibilities of the installation CO include:

1. Budgeting for IPMPs, training, operations, and facilities in compliance with legal and DoD requirements
2. Designating an integrated pest management coordinator in writing
3. Implementing and supporting the IPMP
4. Ensuring the conduct of all pest management operations are safe and has minimal impact on the environment
5. Ensuring implementation of an IPM program that reduces dependency on the use of pesticides
6. Ensuring compliance of the installation's IPMP and program with all applicable host nation, federal and DoD regulations and, where it does not degrade DoD mission, and local regulations

2.1.3. Integrated Pest Management Coordinator

The installation CO designates, in writing, the IPMC as their advisor and coordinator of all installation pest management activities, ultimately being the first line of defense for promoting strategies and methods for conducting a safe, effective, and environmentally-sound IPM program on DoD installations. The IPMC designation letter is included in [appendix E](#). The individual best suited to fill the IPMC position varies from one installation to another. It is common for IPMCs to wear many hats. For example, an installation IPMC may be a public works or environmental department representative such as a DoD pest controller, a natural resources manager, or a contract performance assessment representative (PAR). What makes sense at one installation does not necessarily fit the mold of another. Whatever position the IPMC holds, they have the pest management program enforcement authority by the installation CO's written designation.

The IPMC actively seeks to familiarize him/herself with all pest management activities on the installation, and be a good communicator.

At a minimum, IPMCs must pass an initial IPMC training course within 1 year of appointment, and maintain this qualification by passing refresher training every 3 years or repeating the initial training course. The primary responsibilities of an IPMC include:

1. Coordinating the installation's IPM program with stakeholders including implementation, maintenance, and annual update of the IPMP
2. Coordinating the revision of the IPMP every 5 years
3. Promoting IPM in the pest management program to cost-effectively and safely manage pests and to prevent adverse environmental impact
4. Coordinating reporting of all pest management operations, to include contracted and credit card purchased pesticide services, on the installation to NAVFAC Atlantic Applied Biology
5. Ensuring current certification and continuing pest management training of pesticide applicators and pest management performance assessment representatives (PMPARs)
6. Receiving and compiling lists of new pesticides and uses from all PMSPs on the installation and submitting them to NAVFAC Atlantic Applied Biology for review and approval
7. Maintaining current list of approved pesticides
8. Acting as liaison between installation, Applied Biology, and local and federal agencies for pest management and pesticide regulatory issues
9. Ensuring the installation contracting officers submit pest management contract specifications to the NAVFAC Applied Biology PPMC for review prior to advertising
10. Participate in internal/external program audits

2.1.4. DoD In-House Pest Control Shop

The in-house pest control shop performs the grounds pest management at the NRTF Niscemi.

1. Practice IPM
2. Maintain and ensure safety and regulatory compliance of pesticide storage and mixing areas
3. Voluntarily comply with state pesticide and pest management laws and regulations when meeting those standards does not degrade DoD missions
4. Ensure compliance with DoD requirements for applicator certification, pesticide approval, and pest management operation reporting

2.1.5. Environmental Department

The installation environmental department, including natural resources, provides oversight on environmental protection and compliance regarding pest management operations.

Responsibilities of the department include:

1. Reviewing and approving new pesticides and pest management operations that may adversely impact the environment
2. Conducting internal compliance assessments of the pesticide and pest management program
3. Providing technical review of the IPMP

As part of the environmental division, the natural resources manager is responsible for managing natural resources at the installation. In this capacity, the manager may be responsible for conducting or contracting some pest management operations (e.g., invasive species management). Natural resources are further described in the installation's 2019 Natural Resources Management Plan (NRMP). A copy of the NRMP is included electronically in the appendices of this plan. Responsibilities of the natural resources manager include:

4. Providing information on protected species, endangered or threatened species, critical habitat, and noxious or invasive species
5. Identifying environmentally sensitive sites in the NRMP and IPM plan where pesticide use and/or pest management activities should be restricted.
6. Providing guidance on the management of nuisance wildlife
7. Reviewing the IPM plan to ensure that the plan is compatible with the NRMP
8. Coordinating, as needed, natural resources programs involving pest management with the PMSPs and the IPMC
9. Ensuring INRMP is consistent with IPM plan and pest management operations comply with all applicable environmental statutes

The cultural resources manager is responsible for managing cultural resources at the installation, and lists properties that are or are eligible for inclusion in the National Register of Historic Places. In this capacity, the manager may be responsible for approving pest control in and around these areas. Likewise, for pest management activities in and around cultural or archeological sites. The cultural resources are further described in the installation's 2023 Cultural Resources Management Plan (CRMP). A copy of the CRMP is included electronically in the appendices of this plan.

2.1.6. Facilities Engineering and Acquisition Division

The Facilities Engineering and Acquisition Division (FEAD) prepares, manages, and assesses pest control and grounds maintenance contracts. Performance assessment representatives (PARs) monitor and evaluate the performance of contracted PMSPs to ensure that pest control measures are properly applied. The PAR serves as liaison between the contractor and the IPMC. PARs assigned to manage a pest management contract must attend formal PAR DoD pest management training and complete refresher training every three years. The PAR must:

1. Assist contract writers with preparing contracts that ensure that all requirements of the IPM plan and DoD policy are included in the contract specifications
2. Coordinate with the NAVFAC Applied Biology PPMC for a review of pest management contract specifications prior to advertisement for bid
3. Maintain copies of each contract on file
4. Monitor pest management contractors; ensuring effective and safe application of pest management practices, identifying and documenting discrepancies, and seeking corrective action with contractors in accordance with the contract
5. Ensure contractors record all pest management activities and submit reports including actual pesticide use through the NAVFAC Online Pesticide Reporting System (NOPRS) or to the PPMC monthly

All PMPARs shall be delegated the authority (in the contract and in the PMPAR appointment letter) to halt any contract pesticide applications that:

1. Endanger or present a hazard to humans, animals, or the environment
2. Violate contract specifications, or applicable federal, DoD, or Navy laws/regulations
3. Violate the pesticide label

2.1.7. Installation Facilities Engineering and Planning

Proper planning and design of installation facilities and landscaping can provide long-term prevention of pests reducing or eliminating the need for pesticide use, pest management services, and future structural or landscape renovations. Architects and engineers should use the following:

1. Consider pest prevention in building and landscape design
2. Incorporate Unified Facilities Guide Specifications (UFGS) for termiticide applications and termite barrier installations in construction projects
3. Incorporate pest prevention in the Unified Facilities Criteria (UFC) for Landscape Architecture into landscaping projects
4. Consult with installation IPMC, PAR, PMSPs, and the PPMC regarding pest prevention in facilities and landscape planning and design and providing input into building exterior and architectural plans

2.1.8. United States Naval Hospital, Sigonella

The U.S. Naval Hospital, Sigonella includes preventive medicine, industrial hygiene, and occupational health departments. This section discusses responsibilities for each department.

2.1.8.1. Preventive Medicine

Preventive Medicine provides public health support to the installation in accordance with Navy Medical (Command) (NAVMED) P-5010, Manual of Naval Preventive Medicine, and OPNAVINST 6250.4C. Preventive medicine personnel must:

1. Act as advisor and liaison to the installation CO for public health pest prevention and management
2. Conduct surveys for pests of medical importance, such as cockroaches, mosquitoes, bed bugs, etc., through habitability and food service sanitation inspections
3. Conduct surveys and surveillance of medically-important pests and vectors in coordination with the installation PMSPs. Based on vector surveillance data, conduct risk assessments and establish triggers or action thresholds for vector management to mitigate human health threats when vectors or medically important pests pose a public health threat, interfere with duty performance, or affect unit morale
4. Establish and maintain liaison with local health agencies as they pertain to vector management and vector-borne and zoonotic disease prevention
5. Maintain current certification as DoD category 8 (public health) pesticide applicators
6. Develop and maintain an emergency vector-borne disease response plan in case of a vector-borne disease outbreak or disaster
7. Provide technical review of the IPMP
8. Provide recommendations for sanitation and pest prevention based on monthly inspections

2.1.8.2. Industrial Hygiene

Industrial hygiene (IH) personnel perform surveys (i.e., for pest management employees) to characterize occupational exposures (i.e., to inherent chemical, physical, ergonomic, and biological stressors) and control measures (e.g., engineering—local exhaust and mechanical dilution ventilation systems; administrative—warning signs, standard operating procedures, training requirements, etc.; and personal protective equipment—respiratory protection and chemical resistant clothing). In addition, IH surveillance information is used to initiate, continue, or end medical surveillance.

2.1.8.3. Occupational Health

Occupational health personnel are responsible for performing all necessary medical surveillance (such as physical examinations and cholinesterase testing) for government pest management personnel, as deemed necessary.

2.1.9. Contract Pest Management Service Providers

Contract PMSPs are required to be certified as pesticide applicators by the local Italian Provincial Agricultural Office. These responsibilities apply to all contractors on the installation. Contract PMSPs must:

1. Conduct pest management operations in accordance with the contract specifications or lease agreements and the IPMP and in compliance with federal and local laws and regulations
2. Submit a list of pesticides proposed for use on the installation to their government representative

3. Communicate all pest management issues and requirements via the government representative
4. Submit daily pest management operation records to the government representative or through NOPRS.

2.1.10. Morale, Welfare, and Recreation

Morale, welfare, and recreation (MWR) provides recreational activities for military and civilian personnel on the installation. This includes athletic fields. Additionally, MWR establishes and manages lease agreements with commercial food concessions. With respect to the athletic field maintenance, MWR activities must:

1. Ensure that all personnel who apply pesticides maintain current certifications in the appropriate categories (see section 2.4 for more information)
2. Ensure the NAVFAC Applied Biology PPMC approves all pesticides prior to use
3. Provide copies of the pesticide labels to the IPMC
4. Maintain and report records of all pesticide applications in accordance with the requirements outlined in this IPMP
5. Ensure that all pesticide applicators practice IPM
6. Ensure that landscape cultural management practices are used to maintain the health of plants and turf to prevent disease and pest infestations
7. Ensure that new plants brought onto the installation for landscaping in recreational areas are not invasive, infested with pests, or infected with disease.

With respect to food establishments, MWR must:

1. Ensure maintenance of proper sanitation in all food handling facilities
2. Submit any contract specifications (outside of the installation pest management contract) for pest management to the IPMC for technical review prior to submitting the contract for bid
3. Ensure that only current, licensed pesticide applicators apply pesticides
4. Ensure reports of all pest management activities in accordance with the requirements outlined in this IPMP (for pest management that is conducted separately from the installation contract)

2.1.11. Navy Exchange

The Navy Exchange (NEX) displays and sells household and garden pesticides. Additionally, the NEX establishes and manages lease agreements with commercial food concessions. With respect to pesticide sales, the NEX must:

1. Ensure display of pesticides in accordance with the pesticide label and other federal, host nation, and local regulations

2. Ensure proper training of store employees on emergency procedures in the event of a pesticide spill
3. Ensure that pesticides sold have labels written in English.

With respect to food concessions, the NEX must:

1. Ensure maintenance of proper sanitation in all food handling facilities
2. Submit any contract specifications (outside of the installation pest management contract) for pest management to the IPMC for technical review prior to submitting the contract for bid
3. Ensure only current, licensed pesticide applicators apply pesticides
4. Ensure reporting of all pest management activities in accordance with the requirements outlined in this IPMP (for pest management that is conducted separately from the installation contract)

2.1.12. Commissary

The commissary not only sells food and healthcare items, but also household pesticide items. The commissary must:

1. Maintain proper sanitation in the store
2. Ensure food items for sale are free from stored product pests
3. Ensure surveillance and control for invading pests occurs in commissary facilities
4. Coordinate with the Army Veterinary Services on pest or sanitation problems
5. Ensure the display of pesticides in accordance with the pesticide label and other federal, and local regulations. Guidance on storage and display requirements for retail sale pesticides can be found in the Tri-Service Food Code, para. 7-301.11 and guidance for meeting these requirements is in [AFPMB TG 45](#)
6. Ensure proper training of store employees on emergency procedures in the event of a pesticide spill
7. Ensure that pesticides sold have labels written in English

2.1.13. United States Army Veterinary Services

The veterinary services department provides clinical support for military working dogs and services for privately owned pets and animals. Veterinary Services also provide food safety inspections for the commissary and for other food items delivered to the installation.

Responsibilities of the Army Veterinary Services often include:

1. Conducting surveillance for pests which damage, destroy, and contaminate food stored in the commissary and installation facilities
2. Ensuring stored field rations (e.g., meals, ready to eat (MREs), etc.) are free from pests

3. Providing review of distribution/shipments when needed for IPM concerns (e.g., commissaries)
4. Advising preventive medicine and the IPMC of any zoonotic diseases that may require pest management
5. Providing advice and education to pet owners on preventing pest infestations

2.1.14. Building Occupants and Barracks/Housing Residents

All installation personnel have the responsibility to:

1. Apply good sanitary and pest exclusionary practices to prevent pest infestations
2. If permitted for personal use, use pesticides in accordance with the pesticide label
3. Coordinate and cooperate fully with PMSPs in scheduling pest management and preparing the areas for pesticide treatment if necessary

2.1.15. Airfield Operations

Bird Aircraft Strike Hazard (BASH) management programs are required at all installations with an airfield and aircraft operations. BASH is an aviation safety program usually led by the airfield operations officer at installations and involves the aircraft squadrons and installation personnel. The installation PMSPs and natural resources manager support the program and are often members of the BASH working group or team. Pest management for BASH involves not only birds, but medium to large mammals (i.e., feral dogs, rabbits, etc.) that can also be struck while crossing taxiways and runways. Pest management also involves not only managing the animal hazards but the environment that attracts them to the airfield. This includes control of vegetation that provides animal harborage and coverage; rodents and insects that serve as food; and birds in and on structures such as hangars, fences and trees. BASH program responsibilities are to:

1. Practice IPM
2. Ensure all airfield PMSPs, including USDA Wildlife Services, comply with federal and local regulations and the requirements of DoD policies and directives
3. Report all pest management operations

2.2. PESTICIDE APPROVAL

Only pesticides approved by both the Environmental Protection Agency (EPA) and Italy shall be used. Additionally, DoD and Department of the Navy (DoN) directives require installations to submit a list of all pesticides that will be used during control operations to the cognizant NAVFAC Applied Biology PPMC for review and approval (OPNAVINST 6250.4C, paragraph 4; OPNAV M-5090.1, paragraph 24-3.4; MCO 5090.2, paragraph 030601). The purpose of this approval process is to ensure that only registered pesticides which are safe, effective, and appropriate for the site will be used on the installation. Requests for pesticide approval will be submitted to the NAVFAC Applied Biology PPMC via the installation IPMC using NOPRS (see [section 2.3.3](#)). Once a pesticide is approved, it may be used on-site as per the label directions. New pesticides may also be added to the list and submitted for approval as needed. The list should be reviewed and updated annually by the IPMC as part of the IPMP maintenance. Pesticides currently approved for use on the installation are listed in [appendix D](#).

The IPMC shall maintain a hard copy or electronic version of the manufacturer's label and safety data sheet (SDS) for each pesticide on the pesticide authorized use list (AUL); these should be made easily accessible to installation personnel upon request. The PMPARs or the PMSPs should also maintain copies. Pesticide labels and their registration status can be found on the EPA's [National Pesticide Information Retrieval System](#).

2.3. RECORDS AND REPORTING

All shore installations and units performing pest control operations shall maintain daily records of pesticide applications and submit reports of pest management operations at least monthly to the cognizant PPMC (OPNAVINST 6250.4C, paragraph 23; OPNAV M-5090.1, paragraph 24-3.4.

2.3.1. Pest Management Record Keeping

All PMSPs shall record pest management operations daily. Records shall include all pest management operations including surveys and non-chemical control operations performed on the installation by commercial contractors as well as work performed by DoD pest management personnel. The records will include the following information: date of application, location and site, type of operation, target pest, area treated, name of applicator, pesticide information (trade name, active ingredient, and formulation), amount of pesticide applied, and calculated pounds of active ingredient applied. The following operations are excluded from the record keeping requirement:

1. Personal use of insect repellent
2. Application of repellent by deployable units during mass treatment of clothing and tentage
3. Application of pesticides for flea and tick control to pets by pet owners and veterinary services

Records shall be submitted via the NAVFAC Online Pesticide Reporting System ([section 2.3.3](#)).

Note: Personnel with access to the Navy and Marine Corps Intranet (NMCI) network and a Common Access Card (CAC) may view a multitude of shore installation maps through GeoReadiness Explorer ([GRX](#)). Several functions within this program may be useful to environmental personnel and pesticide applicators such as the capability to retrieve the coordinates of a location, and measurement tools which the user then can use to calculate the size of a job.

2.3.2. Maintaining Pest Management Operations Records

The installation must archive complete daily pest management operation records on-site indefinitely. Pesticide applications for each building, structure, or outdoor site must be accounted for. Past hardcopy records must be archived so as to prevent them from being destroyed. Electronic records shall be stored to prevent destruction or loss; back-up copies are recommended. All records reported to NAVFAC Applied Biology will be stored and may be used as a back-up. Downloading records from NOPRS at least annually and maintaining them on-site is highly recommended.

2.3.3. Pest Management Service Provider Reporting Procedures

Reports will be reviewed by the IPMC and the NAVFAC Applied Biology PPMC to provide program oversight to the installation and to generate data for tracking overall DoN pesticide usage.

All PMSPs that have internet access must use the NOPRS to record, report, and manage pesticide and pest management records. This system is preferred to other methods because it eliminates the need to send hardcopy or electronic records to the IPMC and then to the NAVFAC Applied Biology PPMC. Records are entered directly into a central database that can be accessed by account holders and downloaded into a spreadsheet. The only computer requirement is reliable internet access. Integrated pest management coordinators must contact the NAVFAC Applied Biology PPMC to establish a supervisor account. Pest management service provider's applicators can then contact the IPMC to request an applicator account. The NOPRS website is: <https://noprs.pestlogics.com/>.

2.4. TRAINING, CERTIFICATION, AND LICENSING

Integrated pest management requires personnel who are properly trained to investigate and diagnose pest problems, select the appropriate pest management method, apply the appropriate pesticide, perform these operations so that they are safe to humans and the environment, and educate and advise their customers on pest prevention methods. All DoD personnel who apply or supervise the application of pesticides shall be trained and certified within two years of employment in accordance with the DoD Plan for the Certification of Pesticide Applicators, or EPA-approved state certification plan (DoDM 4150.07, Volume 1, paragraph 4.2). Additionally, professional pest management personnel shall be certified if their duties include:

1. Making recommendations for the use of pesticides, applying pesticides, or directly supervising the application of pesticides
2. Conducting demonstrations on the proper use and techniques of pesticide application or the supervision of pesticides
3. Conducting field research that includes using or supervising the use of pesticides

An exception to the standard training and certification requirements are those individuals approved by the IPMC to apply ready-to-use pesticides as part of the self-help program.

2.4.1. Verification of Qualifications

Copies of contractor or lessee host nation licenses shall be obtained from all PMSP personnel applying pesticides on the installation. Verification of DoD pesticide applicator certifications, as well as IPMC and PMPAR accreditation, can be obtained from the NAVFAC Applied Biology PPMC. A list of applicator certifications as well as a list of pest control business licenses is found in [appendix E](#).

2.4.2. Requirements for Department of Defense Pesticide Applicators

DoD applicators may be certified in the following categories:

1. Category 2—Forestry
2. Category 3—Ornamental and Turf (e.g., landscape arthropod and vertebrate pests)

3. Category 5—Aquatic (e.g., aquatic weeds in lakes, ponds, rivers, streams, irrigation canals)
4. Category 6—Right-of-Way (e.g., weeds on sidewalks, along fence lines, parking lots, road ways, storage tank grounds)
5. Category 7—Industrial, Institutional, Structural, and Health-Related (e.g., termites and other wood-destroying insects, cockroaches, crickets and other invading organisms)
6. Category 8—Public Health (e.g., mosquitoes, ticks, fleas, rodents)
7. Category 11—Aerial Application (e.g., application of pesticides for any pest by fixed or rotary-wing aircraft)

Preventive medicine technicians (PMTs) are required to be certified only in Category 8 and receive certification during PMT school. Golf course applicators are only required to be certified in categories 3, 5, and 6.

Initial certification in categories 2, 3, 5, 6, 7, and 8 for civilian employees is a three week course conducted by a designated DoD training agency. The Navy course is conducted by the Navy Entomology Center of Excellence (NECE) in Jacksonville, Florida. Initial certification and recertification in category 11 is a one week course conducted by the Air Force Reserve. Certification for all categories is valid for three years. With proper justification, certifications can be extended by the applicator's certifying authority for an additional six months for civilian personnel or an additional 12 months for military personnel. Recertification courses for civilians in all categories except category 11 are conducted annually by NAVFAC Applied Biology. Contact the NAVFAC Applied Biology PPMC for DoD initial and recertification course dates.

2.4.3. Requirements for Commercial Contract Applicators

"Licensed firms shall perform all pest management services procured by contract using only trained operators who are certified in the host nation pest management categories for the work planned. Copies of host nation applicator certifications shall be reviewed prior to award" (OPNAVINST 6250.4C, paragraph 15b). Copies of contractor applicator certificates can be found in appendix E. All contract pesticide applicators applying pesticides on the installation must hold a host nation commercial or government pesticide applicator's license.

2.4.4. Requirements for Natural Resource Management Applicators

Commercial contract applicators applying herbicides for invasive plant control or habitat restoration must hold a host nation license. Personnel using pesticides for wildlife control must also hold an appropriate host nation license. Department of Defense employees applying pesticides for invasive plant control or habitat restoration shall be DoD-certified as a pesticide applicator.

2.4.5. Requirements for Performance Assessment Representatives

Pest management performance assessment representatives assess the performance of contractors in the Performance-Based Acquisition (contracting) Program. PMPARs assigned to manage a pest management contract must attend formal PMPAR DoD pest management training and complete refresher training every three years (DoDM 4150.07, Volume 2, paragraph 3.4). NAVFAC Applied Biology provides initial and refresher PMPAR training annually. Contact the NAVFAC Applied Biology PPMC for course dates.

2.4.6. Requirements for Integrated Pest Management Coordinators and Environmental Personnel

IPMC personnel will be trained in accordance with the requirements and guidelines in DoDM 4150.07 Volume 2, and with any additional DoD Component's requirements or guidelines for Pest Management Quality Assurance Evaluator (PMQAE) and PMPAR training (DoDM 4150.07, para 3.4.b.(1)). Newly designated IPMCs are required to receive training in the administrative and operational requirements of installation pest management. Environmental personnel who have compliance oversight of pesticides on the installation should also receive training. The initial PMPAR and IPMC course provides the necessary training. Naval Facilities Engineering Systems Command, Atlantic conducts these courses annually. If applying pesticides or recommending pesticide applications, the IPMC must be certified as a DoD pesticide applicator. Contact the NAVFAC Applied Biology PPMC for course dates.

2.5. PEST MANAGEMENT CONTRACTING

Contracts requiring the use of pesticides must be reviewed and approved by the NAVFAC Applied Biology PPMC. This includes contracts issued by non-appropriated activities and tenant commands on the installation. Pest control contracts are required to be monitored by a trained PMPAR (DoDM 4150.07, Volume 1, paragraph 4.2.g).

2.5.1. Pest Management Contracts

The installation's pest management contracts are listed below.

2.5.1.1. Naval Facilities Engineering Systems Command Pest Management Contract

The installation has a pest management contract in place under the Base Operations Support (BOS) contract. This contract covers general indoor and outdoor pest management for invertebrate and vertebrate pests. Additionally, the contract requires the contractor to pick up dead vertebrates and dispose of them appropriately. To put in a call for pest management services, personnel submit a request through Maximo. The names of the current contract companies are listed in [appendix E](#).

2.5.1.2. Grounds Maintenance Contract

The installation has a grounds maintenance contract in place under the BOS contract. Grounds maintenance work includes the use of pesticides (herbicides) to control weeds in semi-improved and improved grounds including substations, fence lines, thermal lines, parking lots, ditches, and ponds. The contractor also uses non-chemical methods of weed control such as line trimming. To put in a call for grounds maintenance services, personnel submit a request through Maximo. The names of the current contract companies are listed in [appendix E](#).

2.5.1.3. Miscellaneous Pest Management Contracts

The Department of Defense Dependents School (DoDDS), Navy Gateway Inns and Suites (NGIS) and Marinai Housing perform their own pest management independently of the installation pest management service provider. DoDDS pest management services are provided by SKE Pest Control. NGIS pest control services are provided by TriStar. Pest control services for Marinai Housing are provided by Pizzarotti.

2.5.2. Contract Specifications and Review

Pest management contract specifications must be written to ensure effectiveness, safety, and regulatory compliance. The facilities support contract/base operation support (FSC/BOS) performance-based contract template for pest control (sub-annex [1503020](#)) and grounds maintenance (sub-annex [1503050](#)) is available from NAVFAC Applied Biology or on the NAVFAC Public Works Facility Support Contracts Flank Speed website ([requires login](#)). The facilities contracting officer (KO) or contracting officer representative (COR) can provide additional information. PPMCs will review pest management contracts, including augmentation contracts, for technical compliance, to verify that appropriate IPM and pest management standards are specified before granting the award (DoDM 4150.07, Volume 1, paragraph 4.6.b).

Termite pretreatment contract specifications for new construction shall also be reviewed by the NAVFAC Applied Biology PPMC prior to procurement. The Unified Facilities Guide Specifications (UFGS) 31 31 16 Soil Treatment for Subterranean Termite Control should be included in all new construction contracts for termite pretreatment. See [section 3.2.5.3](#) for more information on termite treatment contracts.

More guidance for personnel preparing or assessing DoD pest management contracts can be found in AFPMB TG No. 39, Preparing DoD Pest Control Contracts and Assessing Contract Performance.

2.5.3. Government Representatives

Contractors will communicate and submit required pest management reports via their government representative. For the pest control and grounds maintenance (FSC/BOS) contracts, the representative is the PMPAR who is responsible for assessing the contract. For Non-Appropriated Fund Instrumentality programs (NAFI) (i.e., NEX, MWR) contracts, the representative is the local NAFI organization manager. In cases where a government representative is not available, the installation IPMC may liaison with a contractor's representative.

2.5.4. Contract Requirements

The application of pesticides on Navy properties by contractors is strictly regulated by Department of Defense and Navy regulations, this IPMP, and host nation regulations. These requirements apply to all pesticide applications including insecticides, herbicides, fungicides, molluscicides, etc. to any area in or outdoors. These requirements apply to any size contract (small purchase or facility support contract generated) and services acquired by any other means including government purchase cards (DoN eBusiness Operations Office Instruction (EBUSOPSOFFINST) 4200.1, chapter 6, paragraph 7). The specific requirements for contracted pest control operators working on Navy properties are:

1. Contractor Work Plan (CWP): If required by the contract, a CWP shall be submitted as part of the contractor's proposal. The CWP specifies how the contractor will meet the contract requirements.
2. Pesticide Applicator Certification: All contractor personnel, who apply pesticides (which include all herbicides), shall be certified/licensed in the appropriate applicator category in accordance with [section 2.4.3](#) of this IPMP. All contractors who will apply pesticides shall, prior to the start of work, supply a copy of the certificate(s)/license(s) in accordance with contract specifications. Pesticide business licenses and pesticide applicator certificates are included in [appendix E](#) of this IPMP.

3. Pesticide Approval: Pesticides used by contractors must be approved and added to the installation pesticide AUL, before use, by the NAVFAC Applied Biology PPMC as described in [section 2.2](#). The list of proposed pesticides shall be included in the CWP or submitted to the designated Government representative using the format designated in the contract specifications. The pesticide AUL is in [appendix D](#) of this IPMP.
4. Pesticide Mixing, Storage, and Disposal: Contractors shall not store, mix, or dispose of pesticides or clean pest control equipment on the installation unless an approved pesticide storage and mixing area is specified in the contract and authorized by the KO. One exception to this is soil treatment for termite prevention during building construction; the contractor must mix the termiticide on-site while the PMPAR or IPMC is there to witness.
5. Pesticide Applications: Only pesticides listed on the pesticide AUL shall be used and applied in a manner consistent with the pesticide label.
6. Pest Management Reporting: Contractors shall submit reports in accordance with the reporting requirements in [section 2.3.3](#).
7. Contractor Vehicles:
 - a. Safety equipment: Vehicles used to transport pesticides shall be equipped with a fire extinguisher and a spill and decontamination kit, and be capable of cleaning up the maximum amount of pesticide transported at any given time. Clean water shall be carried for use in emergency personal decontamination.
 - b. Security: All pesticides carried on the vehicles shall be secured in locked compartments at all times. Vehicles shall not be left unattended at any time unless properly locked and secured.
 - c. Identification: Vehicles will be clearly marked as pest control vehicles.
 - d. Appearance: All vehicles shall be maintained with a clean and orderly appearance, free from observable pesticide spills, residues, or build-up.
 - e. Transporting pesticides: Pesticides shall not be transported in the cab or occupied part of any vehicle. They shall always be carried in a separate compartment from the occupied cab.
8. Compliance Assessment: All contractors are subject to regulatory compliance assessments by the PMPAR, IPMC, environmental compliance staff, and other authorized government personnel. Pest control vehicles, pesticide applications, and administrative requirements are subject to inspection. Authorized government personnel may also require the contractor to stop work if the work is not being performed in a safe manner.

2.5.5. Contract Performance Assessment

Contracts shall be assessed by a trained PMPAR to ensure environmental and contractual compliance. For FSC/BOS contracts, Functional Assessment Plans (FAPs) for pest control and grounds maintenance should be developed and implemented. Functional Assessment Plan templates are available from the KO or NAVFAC Atlantic Applied Biology. Periodic assessments for pests prior to, during, or after pest control operations should be conducted to ensure efficacy

of the services. Pest survey methods for contract performance assessments are found in Appendix K on each of the Integrated Pest Management Sheets. Periodic assessment of the contractors during pesticide application should also be conducted to ensure appropriate safety measures are being taken. The contractors' vehicles and equipment must be made available for inspection when requested. In the absence of a PMPAR, a preventive medicine technician can provide information on the efficacy of pest control in some facilities. The PMT conducts monthly inspections that include pest surveys of food service facilities and child development centers. The PMPAR and the IPMC should liaison and coordinate performance assessment activities with the PMT.

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Chapter 3. Operations

3.1. INTEGRATED PEST MANAGEMENT

United States Code (7 U.S.C. § 136r-1) states, “Federal agencies shall use Integrated Pest Management techniques in carrying out pest management activities and shall promote Integrated Pest Management through procurement and regulatory policies, and other activities.” Department of Defense policy is to, “Use integrated pest management (IPM) to prevent or control pests and disease vectors that may degrade readiness or military operations by affecting the health of personnel or that may damage structures, materiel, or property” (DoDI 4150.07, paragraph 1.2.a).

3.1.1. Integrated Pest Management Defined

Integrated pest management is, “A science-based, sustainable, decision-making process that identifies and reduces risks from pests and pest management-related strategies. IPM coordinates the use of pest biology, environmental information, and available technology to prevent unacceptable levels of pest damage using the most economical means, while minimizing risk to people, property, resources, and the environment. IPM provides an effective strategy for pest management in all arenas from developed agricultural, residential, and public lands to natural and wilderness areas” (DoDI 4150.07, paragraph G.2). There are significant differences between IPM and traditional pest control methods. Table 3-1 lists some of the differences.

In IPM programs, treatments are not made according to a predetermined schedule. Rather, treatments are made only when and where monitoring has indicated that the pest will cause unacceptable economic, medical, or aesthetic damage. Treatments are chosen and timed to be most effective and least hazardous to non-target organisms and the general environment.

Table 3-1. Comparison of traditional pest control and integrated pest management methods.

Pest Management	Traditional Pest Control	IPM
Program Strategy	Reactive	Preventive
Customer Education	Minimal	Extensive
Potential Liability	High	Low
Emphasis	Routine pesticide application	Pesticides used when exclusion, sanitation, and other non-chemical methods are inadequate
Inspection and Monitoring	Minimal	Extensive
Pesticide Application Frequency	By schedule	By need
Pesticide Application Target	Area-wide spraying	Spot treatment of areas where pests are found

Pest Management	Traditional Pest Control	IPM
Customer Involvement in Preventing Pests	Minimal	Extensive

Under an IPM program, execution of individual pest management practices involves the following steps:

1. Identify pests.
2. Establish action thresholds that are sufficient to warrant treatment. In determining threshold levels, the amount of public health, aesthetic, or economic threat that can be tolerated must be correlated with the population size of pests, natural enemies, time in the season, and/or life stage of the pest or host.
3. Develop plans/strategies through an integration of treatment methods that are effective against the pest, least disruptive to natural controls, and least hazardous to human health and the environment.
4. Monitor pest population before and after treatment. Monitoring is an ongoing activity.
5. Implement pest control measures if economic damage or public health threat are above the established action threshold.
6. Document results.
7. Evaluate/redesign plan to determine the outcome of treatment actions.

Controlling pests has traditionally been the responsibility of the pest control operator. Using IPM, preventing and controlling pests is the responsibility of all personnel on the installation.

3.1.2. Integrated Pest Management Compliance

All pesticide applicators are trained in IPM techniques during initial and refresher licensing or certification training. Government representatives shall assess the PMSP's compliance with IPM. This may include:

1. Reviewing the approved pesticide list for use of less toxic pesticides, baits with sustainable control, short-residual and pest-specific products, and products used for spot treatment rather than broadcast application
2. Ensuring contractor work plans and partner pest management plans incorporate IPM
3. Reviewing pest management records to ensure that only approved pesticides are used, spot applications are performed, non-chemical methods are used, and routine surveys are being performed
4. Observing pest control service calls to ensure pest control operators identify conditions conducive to pest infestations, provide information to building occupants on how they can prevent pests, use only approved pesticides, perform spot treatments, properly apply baits, conduct routine surveys, and monitor baits/bait stations/traps

3.1.3. Integrated Pest Management Sheets

The IPM sheets in [appendix K](#) provide general guidelines for the integrated control of pests. They may be used as a reference for surveillance and non-chemical and chemical control alternatives.

3.2. CURRENT PEST MANAGEMENT OPERATIONS

Nuisance and health-related pests and vegetation are managed on the installation. Routine inspections provide early detection of pests. Pest inspections should be conducted routinely at all food service, sales, and storage facilities. Preventive medicine technicians conduct food safety inspections including surveys for pests and pest signs at the galleys and food facilities each month. They also inspect the child development center, exchanges, and barber shops. The preventive medicine department provides monthly and quarterly sanitation inspection reports and, if necessary, immediate recommendations to facility managers when contractor-administered pest control operations are needed to control pests. United States (U.S.) Army Veterinarian food inspectors conduct food quality inspections that include examining food items for pests at the commissary and at food service and sales locations.

3.2.1. General Household and Nuisance Pests

Nuisance pests, such as cockroaches and ants, often account for the most significant pest problems indoors. Non-chemical practices which limit pest access to food, water, and shelter are the primary sustainable means of control. Sanitation, trapping, and pest exclusion are all prime examples of such non-chemical control measures. Low-toxicity insecticidal baits are also used effectively against cockroaches and ants. Most pesticide applications can be effective in immediate reduction of pest populations, but have short residual efficacy and are not sustainable in the long term.

In buildings that are most susceptible to pest infestation, these measures are performed on a scheduled basis through preventive maintenance contracts. Common use areas and food consumption areas of other buildings, such as restrooms, coffee messes, lounge areas, and vending machine rooms, are serviced on either a monthly or quarterly basis, depending on the preventive maintenance contract for that building, with response to call-backs as necessary. This also includes common use areas of the barracks, warehouses, and administration buildings. Buildings where there is no food and where problems occur only occasionally such as shops and storage buildings, are handled on a service call basis. Pesticide treatments in food handling areas shall be confined to crack and crevice placement when using residual aerosol or dust formulations. Insect growth regulators and baits are used to complement other control measures. Self-contained light traps may also be utilized in these areas. Low-odor formulations are used in offices and in other spaces where a pesticide odor would be objectionable.

3.2.2. Grounds Maintenance

Grounds maintenance is performed on improved or landscaped grounds. Pest management during grounds maintenance may involve weed control; control of pests and disease on plants, trees, and turf; and control of vertebrate animals (e.g., squirrels, moles) that may destroy plants and turf. Mechanical removal of weeds and mowing are routinely performed. Grounds maintenance also includes weed control in drainage ditches that may contribute to mosquito control and bird habitat removal. Weed control is performed along roadways, fence lines, and at fuel farms where they pose fire and visibility concerns.

3.2.2.1. Weed Control

A wide variety of herbicides are available for controlling unwanted vegetation. Extreme care should be taken when using herbicides around waterways. Herbicides can be used around mowing obstacles such as signs, fire hydrants, and manholes. Herbicides are used to control weeds along cracks in sidewalks and asphalt parking areas, along fence lines, around buildings, and along ditch banks. Selective herbicides are used to control various weeds that occur in lawns on the installation. Various cultural and chemical controls can be used to deal with these and other weed control problems. When using chemical controls, both selective and non-selective herbicides may be used.

3.2.2.2. Aquatic Weed Control

Aquatic weed control work should be conducted in accordance with the Natural Resource Management Plan. The policy is to control and limit the spread of invasive species of plants, and to protect aquatic plant species. Each pesticide must be approved prior to use to prevent harm to the natural resources that feed in the ponds.

3.2.3. Athletic Fields

Pest management operations include weed, fungus, and insect control on ball fields, and weed control in miscellaneous MWR facilities. Undesirable vegetation in the turf grass are controlled by using pre- and post-emergent selective herbicides.

3.2.4. Structural Pests

Structural pests, which have an impact on activity operations, include termites, powder post beetles, wood borers, and wood destroying (decay) fungi. Of these, subterranean termites and wood destroying fungi cause the most damage. A well-managed structural pest control program includes inspection, prevention, and chemical treatments when needed.

3.2.5. Stored Product Pests

Stored product pests are a potential problem at any installation. Inspection upon receipt of products and rejection of obviously infested materials generally prevents heavily infested material from being placed in the storage area.

3.2.5.1. Dermestid Beetle

If dermestid beetles (Dermestidae: *Trogoderma*, *Desmestes*, etc.) are found in a commodity, the whole lot of food must be condemned. The pointed hairs on the larvae will cause digestive problems if the contaminated food is eaten. An accurate identification of dermestids is required to condemn the lot. For the most part, sanitation (keeping storage areas clean) and stock rotation minimize or prevent pest infestation. If an infestation is found, the most effective way to control dermestids is through deep cleaning, vacuuming, and discarding or segregating the infested product while surveying adjacent areas.

3.2.5.2. Storing Meal, Ready-to-Eat Rations

More stringent controls are required for prevention of stored products pests when storing meal, ready-to-eat (MRE) rations. Guidance on this program can be found in AFPMB Technical Guide No. 38, Protecting Meals, Ready-to-Eat Rations (MREs) and Other Subsistence during Storage.

3.2.6. Health-Related Pests

In accordance with OPNAVINST 6250.4C (paragraph 4c), the Preventive Medicine Department (PREVMED) is responsible for conducting inspections and surveys aboard the installation to determine the species, source, location, and density of medically-important arthropods and provide the results to the public works and facilities departments for use in planning pest control operations. Mosquitoes, ticks, biting flies, and filth flies constitute the most important insect pests for both disease transmission and general annoyance. Controlling these insect pests should be based on a thorough knowledge of the target pest, actionable surveillance data, and compelling evidence of an infestation that poses an emergent public health risk.

3.2.6.1.1. Mosquito Surveillance and Control

Routine mosquito surveys are the responsibility of preventive medicine technicians. However, if a PMT is not available, a pest control provider can conduct mosquito surveillance. If additional assistance is needed, Navy entomologists from NECE or the NEPMUs can provide assistance in establishing mosquito surveillance programs. Survey operations are essential to determine the species present, the population level involved, and the potential risk of disease transmission. Surveys also serve as a valuable tool in evaluating control operations. Mosquito surveillance includes conducting both larval and adult surveys. Larval surveys are important because they determine exactly where mosquitoes are breeding, providing the information necessary to manage or eliminate mosquitoes at the source. Larval surveys involve regular dipping stations that are selected, noted on a map, and inspected periodically throughout the mosquito season. In areas where mosquito control is conducted, random larval sampling should be made to check the effectiveness of the control program.

Adult mosquito surveys may be conducted by either collecting mosquitoes from resting sites or using traps. Traps should be placed near where the mosquitoes are expected to be found, during the periods they are active. Two adult mosquito traps that are available with National Stock Numbers (NSNs) are the CDC light trap and the Biogents (BG) Sentinel trap. Traps should be baited with CO₂ from dry ice, when available or, in the case of the BG traps, with the manufactures' recommended lure. The BG-Sentinel™ trap is specifically designed to collect daytime-feeding mosquitoes, and has been found to collect *Ae. aegypti* and *Ae. albopictus* (Zika vectors) more effectively than the standard CDC light trap. Adult collections are then counted, and identified to genus or species. Samples may be submitted to a local health department for testing, so that the disease transmission risk can be assessed. Adult surveys focus on collecting female mosquitoes because they are the only ones that bite. A high proportion of adult males in a trap collection usually will indicate a nearby larval habitat, and a survey of the area should be done to locate possible breeding sites.

The adult mosquito populations are being monitored by the preventive medicine depart at NASSIG with 4 Center for Disease Control (CDC) light traps. When trap counts are above or at the action threshold, mosquito control is conducted.

Mosquito control methods are either permanent (e.g., eliminating the water source) or temporary (e.g., chemical control) in nature and may be directed against larvae or adults. The most effective way to control mosquitoes is to target the larval stage. Larvicides, pesticides specifically labeled to control mosquito larval stages, should be applied to areas where water stands for longer than 7 days when results of mosquito dip counts exceed 1-2 larvae per dip.

To decrease the amount of standing water, it is important to have a drainage system allowing proper runoff of rain water from roadways. Ditches should be maintained free of weed growth.

This increases water flow in the ditch allowing access of natural mosquito predators. It is also important to educate the public on source reduction when the problem mosquitoes originate from artificial containers (e.g., bird baths, gutters, flower pots) found around homes and other buildings.

Regular testing of ULV aerosol droplet dispersal is required to assure maximum control, minimum insecticide use, and prevention of automobile finish spotting caused by droplets that are too large. This testing must be done at the beginning of each spray season and for every 50–100 hours of operation, or when the pesticide is changed. More information is included in AFPMB TG No. 13, Ultra Low Volume Dispersal of Insecticides by Cold Aerosol and Thermal Fog Ground Equipment. Government personnel can obtain slides for aerosol droplet size testing from the Testing and Evaluation Department at the Navy Entomology Center of Excellence, Jacksonville, Florida.

If mosquito populations are extremely high or infestations occur in hard to reach areas, aerial application of adulticides or larvicides by helicopter or fixed-wing aircraft may be the only effective treatment method available. Refer to [section 3.2.8.10](#) for requirements regarding aerial spraying.

3.2.6.2. Filth Fly Management

Performing routine sanitation is the best method to manage filth flies (houseflies, blow flies, flesh flies, bottle flies, etc.). Removal of refuse and routine cleaning of garbage cans and dumpsters will minimize the problem. Garbage cans and dumpsters should be placed on concrete pads at least 100 feet from facilities to reduce breeding under and around the containers and to minimize access to the facilities. Continuous monitoring of sanitation conditions in and around food service areas helps assure that significant fly breeding will not occur.

Chemical control of filth flies is short-term and unsustainable. The choice of fly control techniques must be based on an on-site evaluation of the problem. Pest control personnel inspect areas where garbage is handled and treat these locations with approved insecticides when flies exceed control limits. Preventive medicine technicians also inspect these areas and report significant findings to facility managers for corrective action. Exclusion devices, such as screens and air curtains, help prevent the entrance of flying insects into buildings when installed and properly maintained. Aerosol insecticide treatments are provided when adult flies become a problem in indoor spaces. Automated pesticide misting devices are not allowed according to DoDM 4150.07 Volume 1 (para 6.3.). Light trap devices are also helpful for filth fly control in food handling areas, but only when they are placed inside of the building. Use only non-contaminating light traps with some way of containing the dead insects. For more information on filth fly management, see AFPMB TG No. 29, Integrated Pest Management In and Around Buildings.

3.2.6.3. Bed Bug Management

Bed bugs belong to a family of blood-feeding, ectoparasitic insects called *Cimicidae*. They have a number of features that make them very effective pests and difficult to control. Their small, flattened body allows them to hide in inconspicuous places such as cracks and crevices. A female can lay several hundred eggs during her lifetime. Bed bugs can survive a long time without feeding; and many insecticides have been rendered ineffective due to resistance development. The most common way bed bugs are introduced is by the movement of infested items (e.g., bedding, clothing, and luggage) from one place to another. The common bed bug is

not known to transmit human disease. For most people, the bite of a bed bug is painless and will usually go unnoticed, though many people can have allergenic skin reactions, ranging in severity from local inflammation and itchiness, to asthmatic symptoms and anemia. Although the common bed bug seems to prefer human hosts, they are also capable of feeding on birds, rodents, or other mammals. While other cimicid species, like bat bugs and swallow bugs, mainly feed on bats or birds, but may incidentally bite people when their usual host abandons the nest or is eliminated from the building.

Bed bugs can be difficult pests to detect without a diligent survey strategy. Bed bugs typically feed at night when the host is asleep, and hide in cracks and crevices during the day. It is very important to thoroughly inspect areas where bed bug infestation is suspected. Typical harborage areas might include mattress seams, box springs, bed frames, night stands, picture frames, loose wallpaper, and curtains. Bed bugs typically travel 5–20 feet from their harborage area to feed. When populations are small, infestations may go unnoticed. Some tell-tale signs of a larger bed bug population include the presence of fecal spotting, shed skins, increased biting frequency, and in serious cases a distinct, obnoxiously sweet, odor produced by the bugs. Persons conducting inspections and surveys should be properly trained on what to look for and where to look for infestations. The NECE and NEPMU personnel are available to provide training on bed bug inspections.

Bed bugs are a public health issue; installation preventive medicine department should be contacted immediately. Bed bug control may be more difficult to achieve today with increased travel and more stringent limitations on available control materials. A successful control program will require a carefully planned and integrated approach. For more information on controlling bed bugs see AFPMB Technical Guide No. 44, *Bed Bugs—Importance, Biology, and Control Strategies*.

3.2.6.4. Rodent Management

Rodent control work is an ongoing program to eliminate the causes of rodent infestations. Major emphasis is placed on sanitation and exclusion to limit the amount of food and harborage available to rodents. Tamper-resistant bait stations should be maintained in high infestation areas. There are specific EPA requirements for first generation anticoagulant products (warfarin, chlorophacinone, and diphacinone), second generation anticoagulant products (brodifacoum, bromadiolone, difenacoum, and difethialone), and non-anticoagulants (bromethalin, cholecalciferol, and zinc phosphide). Bait stations are required for all outdoor, above-ground placements and must be placed within 100 feet of man-made structures. Bait stations are also required indoors if exposure to children, pets, or non-target animals is possible. Mechanical traps (snap traps, glue traps, etc.) are another effective control method. Trapping is an effective way of quickly reducing a large mouse population.

3.2.6.5. Bird Management

Pigeons are the primary bird pests on most installations. Pigeons, English sparrows, and European starlings can be controlled without a permit because they are not covered under the Migratory Bird Treaty Act (MBTA). Other bird species require special permits before any control measures can be taken. Bird control methods vary according to the situation. Bird droppings pose a health hazard, as a possible cause of histoplasmosis and other respiratory problems when airborne. Bird ectoparasites, such as mites, can also fall on installation employees. Ultrasonic devices and plastic owls and snakes, etc. are not effective for bird control. The best alternative for bird control is bird-proofing or the exclusion of birds by closing up all openings. Because of its permanency, bird-proofing (e.g., the placing of hardware cloth and chicken wire

over potential roosts) is considered the most cost effective means of control. Population reduction techniques (e.g., destruction of nests accessible by a ladder or cherry picker) can sometimes be used effectively. According to Italian law, no matter the species nest, removal is not possible if eggs or chicks are present. When the birds have left the nest, then it can be removed. In order to ensure that nest removal does not inadvertently violate local laws or regulations, contact the Environmental Division before removing nests. Repellent chemicals which produce alarm reactions and cause a flock to leave or avoid an area are often used. Control personnel should continually monitor bird population levels and take appropriate control actions when required.

3.2.6.6. Bird/Wildlife Aircraft Strike Hazard

Wildlife populations on the airfield continually present a Bird/Wildlife Aircraft Strike Hazard (BASH) concern and danger to human health and safety. Habitat that may provide food, cover, or water for various bird/wildlife species may need to be addressed. The installation BASH Program is carried out by the Italian Command and installation environmental department (including pest management). Corrective recommendations may include removing unused airfield equipment to eliminate perch sites, placing anti-perching devices on equipment to remain, wiring streams and ponds, brush/tree removal, the use of pyrotechnics, or changing the grass mowing program. The CNIC BASH Program supports the NASSIG BASH Program through communication and coordination that includes a periodic update to the installation Wildlife Hazard Assessment (WHA). The current WHA was completed in 2017.

3.2.6.7. Feral Animal Management

Feral or free-ranging domestic cats and dogs are considered by the professional wildlife management community to be one of the most widespread and serious threats to the integrity of native wildlife populations (e.g., birds, reptiles, bats) and natural ecosystems in North America. Navy commands must prevent feral cat and dog populations, and ensure their humane removal from Navy lands through close coordination and cooperation between natural resources, pest management, security, veterinary, and housing personnel. In accordance with the Chief of Naval Operations Policy Letter Preventing Feral Cat and Dog Populations on Navy Property (10 Jan 2002) and OPNAV M-5090.1, chapter 12, Navy commands shall not allow trap-neuter-release or the release of unwanted house pets on their lands due to the potential of feral or free-ranging cat populations to act as disease reservoirs, threatening human health, native wildlife populations, and natural ecosystems.

Cats may occasionally be found near food handling areas or dwelling in crawl spaces under buildings where they can cause flea problems inside of the buildings. The elimination of available food by keeping garbage cans and dumpsters sealed will decrease the appeal of the area to the cats. Elimination of shelter is also a good means of control. The installation should discourage people from feeding stray cats. Guidance on feral cat management can be found in AFPMB Technical Guide No. 37, Integrated Management of Stray Animals on Military Installations.

3.2.7. Pest Management in Housing

Housing areas on the installation are government owned and under the installation's pest contract. Certificates for the pesticide applicators are located in [appendix E](#). Contractors providing services in the housing areas must follow all local laws.

Pet dogs and cats released or lost by owners on base can become a pest problem. Feral cats and dogs are susceptible to and can carry disease, damage natural habitats, harm protected wild animals, become a vehicle strike hazard, and attack and injure personnel. Pet owners are required to microchip their pets. Microchipping is a permanent pet identification system using a computer chip implant in the skin of the animal. This allows a lost pet to be identified even if the collar tag is missing.

3.2.8. Self-Help Pest Management

Self-help pest management programs on DoD installations are authorized by DoDM 4150.07 Volume 1 (para 5.9.b.) when they are cost-effective and when IPM monitoring indicates the need for control. Self-help pest management allows uncertified personnel to use low-toxicity, ready-to-use (RTU) pesticides for small-scale pest control operations. Examples of self-help programs that may be available are: stinging insect pest control for maintenance personnel, venomous spider control, fire ant control, vegetation control using glyphosate, and barracks/office pest control. Requirements for self-help are:

1. The program shall be reviewed and approved by the IPM coordinator and then by the NAVFAC Applied Biology PPMC
2. A program manager, who will be responsible for the program and be the primary point of contact, shall be designated
3. All personnel that will be applying pesticide must be trained and their training documented
4. Only RTU pesticides approved for use by the NAVFAC Applied Biology PPMC shall be used
5. The area(s) to be treated should be small enough to be practically treated with RTU pesticides
6. All pesticides will be stored in a storage site as described on the pesticide label
7. All pesticide use will be reported

Any personnel or departments conducting unauthorized pesticide applications should be directed to immediately cease applications. To request review of a proposed program and submit a statement of need, the IPMC must contact the NAVFAC Applied Biology PPMC.

3.2.9. Prohibited Operations and Devices

Several operations and devices are prohibited by DoD and DoN regulations.

Prohibited operations and devices include:

1. Occupied spaces—Installations shall not permit liquid spray and dust pesticide formulations in any space occupied by unprotected personnel. However, pesticides contained in gel or paste bait formulation may be applied in occupied spaces (OPNAV M-5090.1, paragraph 24-3.2; MCO 5090.2, paragraph 030608).
2. Preventive or Scheduled Pesticide Treatments—DoD policy prohibits the use of regularly scheduled, periodic pesticide applications except in situations where the installation pest

management plan clearly documents that no other technology or approach is available to protect personnel or property of high value (DoDM 4150.07 Volume 1, para 6.3).

3. Electrically-Operated Devices—"Electromagnetic exclusion or control devices, ultrasonic repellent or control devices, and outdoor devices for electrocuting flying insects are not approved for use on DoD installations" (DoDM 4150.07 Volume1, para 6.1). This does not apply to indoor use of selected devices, carefully placed, for electrocuting flying insects. Pest surveillance traps and monitoring equipment, such as non-electrocuting mosquito light traps, may also be used by trained personnel.
4. Paints and Coatings Containing Pesticides and Other Biocides—DoD policy prohibits the use of paint containing insecticides on DoD property. This guidance applies to interior and exterior pesticide-containing paints intended for application to structural surfaces, such as walls, ceilings, and siding. It also applies to insecticides formulated and labeled for use as paint additives. Paints containing fungicides as mildew inhibitors may be used when application directions specify no special instructions due to the fungicide. Approved marine anti-fouling compounds or coatings may be applied to protect surfaces of watercraft (DoDM 4150.07 Volume1, para 6.2.c.).

3.3. REGULATORY COMPLIANCE

The Department of Defense's policy is to ensure that DoD pest management programs achieve, maintain, and monitor compliance with all applicable executive orders and applicable federal, and local statutory and regulatory requirements. When there is a conflict between federal and local regulations, the installation will comply with the more stringent of the two. This may occur with pesticides limited for use by the host nation, which are not necessarily restricted by the EPA. In this case, the installation must comply with host nation regulations.

3.3.1. Pesticide Regulation and Enforcement

Navy installations must comply with host nation pesticide use regulations. The responsibility for compliance and enforcement lies with the installation's commanding officer. As the installation CO's pest management advisor, the IPMC shall be familiar with federal and host nation pesticide use regulations and ensure that all applicators conduct operations in compliance with these regulations. The environmental division should be familiar with these regulations as well due to the environmental hazards of pesticides. Regulatory enforcement for each of the PMSPs is provided.

1. Commercial contractor applicators: PMPARs shall provide assistance by monitoring contract PMSPs for compliance with all applicable regulations as specified in the contract and will recommend appropriate actions to the contracting officer if the contractor does not comply. Preventive medicine technicians conducting sanitation inspections of food service facility pest management programs can also ensure compliance for safe pesticide use and applicator licensing/certification. Inspection guidelines are found in NAVMED P-5010, chapters 1 and 8. The preventive medicine technicians will notify the IPMC of any potential pesticide application violations observed during the course of routine sanitation inspections.
2. DoD applicators: The pesticide applicator's immediate supervisor, with the assistance of the IPMC, shall also ensure that pesticide use is in compliance. Under the authority of DoDI 4150.07 and DoD Directive 5134.01, and per Department of Defense Manual (DoDM) 4150.07, Volume 2, DoD Pest Management Program Elements and

Implementation: Pesticide Applicator Training and Certification Program, the DoD may deny, suspend, or revoke the certificate of any DoD employee who violates any provision of Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA) or falsifies records under DoDM 4150.07, Volume 2 (para 4.7.c.). In accordance with DoDM 4150.07, Volume 2 (para 4.10.b.(1)), the installation CO may initiate a formal review if FIFRA violations are suspected. Violations shall be reported through appropriate command channels to the NAVFAC certifying authority for review. The certifying authority shall determine if further action is required. That action may include suspension of the applicator's certification.

Naval Facilities Engineering Systems Command, Applied Biology shall provide assistance to the installation IPMC with compliance and enforcement issues and clarification of regulations. The senior pest management consultant is the certifying official for DoD-certified pesticide applicators on the installation.

3.3.2. Pesticide Laws and Regulations

Primary pesticide regulations include:

1. DOD and Navy: Department of Defense Instruction (DoDI) 4150.07, DoD Pest Management Program, Chief of Naval Operations Instruction (OPNAVINST) 6250.4C, Navy Pest Management Programs, Chief of Naval Operations (OPNAVINST) 5090.1E, Environmental Readiness Program.
2. Italy Final Governing Standards (FGS): The Italy FGS is the definitive source of environmental compliance criteria applicable to DOD components at installations in Italy. The Italy FGS includes environmental requirements developed by comparing and adopting the more protective requirements of the DOD's Overseas Environmental Baseline Guidance Document; Italian national, regional, and local environmental laws and regulations; and applicable international agreements (i.e., European Union [EU]). The FGS covers many different environmental areas including pesticides (Italy FGS Chapter 11). The Pesticides chapter provides criteria for regulating the use, storage, and handling of pesticides at installations in Italy.

3.4. PESTICIDE MANAGEMENT

Chemical control of pests using pesticides can be an integral part of an IPM program. Proper management of pesticides will ensure a safe and cost-effective pest management program. Management of pesticides includes the proper selection of pesticides, pesticide approval, procurement, storage, mixing, use of pesticide application equipment, and clean-up. The pesticide label provides most of the information needed to manage pesticide use and must be affixed to the container at all times.

3.4.1. Pesticide Selection

The following criteria should be used when selecting a pesticide:

1. Determine the need for a pesticide. Is a chemical pesticide really needed? In some situations, non-chemical control methods may be more effective or less costly and time-consuming in the long term. Will exclusion or habitat elimination take care of the problem?

2. Choose a pesticide with a low toxicity. Can the pest be sufficiently controlled with a pesticide that has a low toxicity to humans?
3. Choose pesticides and pesticide formulations with minimal environmental impact. Avoid using “Restricted Use” pesticides if possible. The environmental impact of pesticide spills is reduced when using a granular pesticide formulation rather than a liquid. Can attractant bait stations be used instead of broadcast application of a pesticide?
4. Choose pesticides that provide a long-term or sustainable solution. For example:
 - a. Contact insecticides applied to ant trails will only temporarily halt the infestation, and may cause the colony to bud and form new colonies, while baits can kill the entire colony including the queen.
 - b. Over time, pests can develop a significant decrease in sensitivity to a pesticide. This process is known as resistance. Insecticide resistance, for example, is particularly problematic in insect groups with short developmental times and high reproductive rates (e.g., German cockroaches, house flies, mosquitoes, bed bugs, and certain stored product pests), especially when pesticides of one particular class are overused. Combat/delay pest resistance to pesticides by rotating/alternating pesticides from different mode of action groups in conjunction with other IPM methods.

3.4.2. Pesticide Procurement

Pesticides used by contractors are included in the cost of the contract and are procured through commercial sources. Pesticides used by DoD personnel may be purchased through the Federal Stock System. Contractors cannot purchase pesticides through the Federal Stock System. DoD approved pesticides listed in the Federal Stock System can be found on the Armed Forces Pest Management Board’s Pesticides and Equipment [website](#). These are not the only pesticides that may be used on the installation. Pesticides listed on the installation’s pesticide Authorized Use List (AUL) ([appendix D](#)) may be purchased. All pesticide products and pest control services procured via government credit cards must also be pre-approved by the NAVFAC Applied Biology PPMC according to EBUSOPSOFFINST 4200.1A, Department of Navy Policies and Procedures For the Operation and Management of the Government Commercial Purchase Card Program (chapter 6, paragraph 7). For information on requesting new pesticides to the installation pesticide AUL, see [section 2.2](#), Pesticide Approval.

3.4.3. Pesticide Storage and Mixing

Pesticide storage facilities, retail sales, and vehicles each have specific requirements in regards to pesticide storage.

3.4.3.1. Pesticide Storage Facilities

DoDM 4150.07, Volume 1 (para 4.4.a), states that pesticide storage facilities shall comply with all applicable regulatory standards and shall, where feasible, be modified to meet the minimum standards for new pesticide storage facilities. The Department of Defense standards are described in AFPMB Technical Guide 17 Military Handbook, Design of Pest Management Facilities. The NAVFAC Applied Biology PPMC should be consulted during the design phase of new pesticide storage facilities to ensure that the latest requirements are included.

At a minimum, all existing facilities shall meet the following standards:

1. An active ventilation system that provides a minimum of six air changes per hour
2. Backflow prevention on all water sources used for mixing/filling
3. No floor drains and a surrounding berm that provides containment of any pesticide spills
4. Warning signs
5. Surrounded by a climb-proof fence with access only through doors with locks

Niscemi stores pesticides in a ventilated storage facility.

3.4.3.2. Pesticide Mixing

All pesticide mixing conducted by commercial contractors is done off-site. One exception to this is soil treatment for termite prevention during building construction; the contractor must mix the termiticide on-site while the PMPAR or IPMC is there to witness. Pest control operators must mix pesticides in accordance with the pesticide label in appropriate areas that minimize the risk of safety and environmental hazards. Contracted pest control operators must also mix pesticides in accordance with the contract specifications. Persons mixing pesticides with water shall protect the water supply from back-siphoning of the pesticide mixture. They shall also ensure accurate measurement of concentrated pesticide to ensure proper application rate. Precautions must be taken to minimize the risk of a pesticide spill. See section 5.3.4 for pesticide spill prevention measures. Spill kits must be maintained on pest control vehicles and must be available at the mixing site. An eye wash and deluge shower is essential for emergency washing of individuals accidentally contaminated with pesticides (29 CFR 1910.151). The emergency eye wash and deluge shower are to be accessible within 10 seconds from indoor and outdoor mixing areas. If located in the outdoor mixing area, the eye wash and shower must be protected from freezing.

3.4.3.3. Retail Sale Pesticide Storage

Pesticides sold in exchange facilities and commissaries shall be consistent with recommendations of the AFPMB or the responsible PPMC and shall be displayed on shelves separate from food in locations where accidental breakage will not result in contamination. No category 1 pesticide products identified or marked as “Danger” or “Danger-Poison” shall be sold in exchange facilities (OPNAVINST 6250.4C, paragraph 13c; MCO 5090.2, paragraph 040309). Additional guidance can be found in AFPMB TG No. 45.

3.4.3.4. Vehicles

Pest control vehicles must carry pesticide spill kits (OPNAVINST 6250.4C, paragraph 13d; MCO 5090.2, paragraph 030605). Pesticides shall not be transported in the vehicle’s passenger compartment and pesticide containers shall be secured to vehicles to prevent spillage.

3.4.4. Pesticide Application

All pesticides shall be applied in accordance with federal, host nation, and label directions. In accordance with label directions, personal protective equipment should be worn during the application of pesticides. Application of pesticides should be timed to ensure contact with and maximum kill of the pest and to prevent use under adverse weather conditions that can cause drift of the chemical outside the target area. See [section 4.2.2](#) for more information on timing and drift prevention.

3.4.4.1. Service Containers

Containers other than the original pesticide container that are used for transporting pesticides to the job site must have a copy of the label attached. Service containers used for the application of a pesticide must have the following information on a tag attached to the container: name of party responsible for the container, the identity of the chemical in the container, and the signal word of the chemical. Containers commonly used for food, drink, or household products shall not be used to hold pesticides.

3.4.4.2. Equipment

Only pest control equipment that is in good repair and safe to operate shall be used by PMSPs. The equipment should be in good condition, free from corrosion, clean, and free from leaks. The PMPAR shall inspect equipment used by contract applicators. Applicators shall also ensure that they use equipment suitable to ensure proper application of pesticides.

3.4.5. Pesticide Disposal

All pest control equipment shall be properly cleaned. Contract PMSPs are not allowed to dispose of excess pesticide, used containers, or residues on the installation per contract specifications; they must conduct all cleaning off-site. Spray tanks and pesticide containers must be triple-rinsed prior to storage or disposal. Disposal of pesticide spray tank rinse water should be performed by applying to a site listed on the pesticide label, used for future mixing of the same pesticide, or disposed of as hazardous waste. Rinse water shall not be allowed to enter storm drains.

3.4.5.1. Empty Containers

OPNAV-M 5090.1 (paragraph 24-3.12) requires disposal of pesticide wastes be in accordance with 40 CFR § 262, EPA Regulations for Hazardous Waste Generators. The disposal of pesticides, their containers, and related wastes is strictly regulated. Empty liquid pesticide containers will be triple-rinsed with $\frac{1}{4}$ (25%) of the container's capacity divided into 3 doses, shaken and drained according to label instructions. Disposal of empty containers will be coordinated with the installation's environmental division. Empty containers will not be reused. If possible, pesticide containers shall be returned to the manufacturer for recycling.

3.4.5.2. Rinse Water

When cleaned, spray equipment will be triple rinsed in the field using $\frac{1}{4}$ (25%) of the tank capacity divided into 3 doses. The rinse material will be sprayed on the application site in accordance with the pesticide label. Water from rinsing out equipment will be used immediately. If it cannot be sprayed on the application site, rinse water should be stored in marked plastic containers and used as the diluent for the next time the same pesticide is formulated for application. Wastewater formulations that contain pesticides shall not be discharged into any storm or sanitary sewer system.

3.4.5.3. Excess Pesticides

Disposal or redistribution of excess pesticides shall be coordinated through Environmental and the IPMC. Environmental and the Consolidated Hazardous Material Reutilization and Inventory Management Program will determine whether the pesticide can be redistributed or if it needs to be disposed of. Excess pesticides shall never be disposed in any storm or sanitary sewer system.

3.5. MINIMUM RISK PESTICIDES

Minimum risk pesticides may be used by pest management service providers as part of their IPM program. According to the EPA, “Minimum risk pesticides are a special class of pesticides that are not subject to federal registration requirements because their ingredients, both active and inert, are demonstrably safe for the intended use.” These pesticides are exempt from federal registration under 40 CFR 152.25(f) and are not labeled with an EPA registration number. Since there is no federal review of these pesticides or their pesticide label, there is no federal review of the instructions for effective use of these products. Although these pesticides are exempt from federal registration, they still need to be approved prior to use on DoD property, primarily for efficacy and safety reasons.

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Chapter 4. Health and Safety

4.1. PESTICIDE APPLICATOR SAFETY

To ensure the safe use of pesticides, pesticide applicators shall handle and apply pesticides in accordance with the product's label directions.

4.1.1. Potential Occupational Hazards

The following hazards may be encountered by pesticide applicators or Government representatives that may be exposed while inspecting pest management operations. Occupational safety and health guidance is found in the OPNAVINST 5100.23G, The Navy Occupational Safety and Health Program Manual.

4.1.1.1. Direct Contact Chemical Exposure

Many chemicals used as pesticides are also harmful to humans. The three routes of exposure to applicators are dermal, inhalation and ingestion. For applicators, the most common route of exposure is dermal and is frequently due to not wearing the appropriate personal protective equipment. Severity of the harmful effects is determined by duration of exposure and toxicity of the chemical. The effects can be acute (rapid onset due to high-dosage, high-toxicity chemicals) or chronic (slow or delayed onset due to long-term exposure to low-dosage, low-toxicity chemicals). The highest risk for severe acute chemical exposure occurs during pouring and mixing of concentrated pesticide resulting in high-dose, rapid-onset chemical poisoning. Chronic exposure can occur when the applicator fails to use appropriate PPE during frequent pesticide applications and the chemical accumulates in the body of the individual over a period of time leading to delayed or gradual onset of illness or injury. Direct chemical exposure can result not only in pesticide poisoning, but also in skin burns due to corrosive chemicals.

4.1.1.2. Heat

The use of protective equipment such as a respirator, goggles, gloves, and coveralls increases the risk of heat injury especially in warm climates. Heat injury can occur during long periods of work outdoors during warm weather or in enclosed spaces where machinery or equipment may generate heat.

4.1.1.3. Noise

Some pesticide application equipment use gas-powered air compressors or pumps that produce noise hazards. Gas-powered backpack sprayers are particularly hazardous due to the proximity of the noise source to the ears.

4.1.1.4. Eye Hazards

Eye hazards may result from chemical splashed into the eyes causing corrosive, toxic, or impact injury. Some pesticides are labeled "Restricted Use" due to their corrosive nature. The highest risk occurs during pesticide pouring, mixing, and application. During pesticide applications, chemicals may enter the eyes through splash back when applying the chemical under pressure into a crack or crevice or when applying pesticides overhead. Injury may also occur during equipment cleaning.

4.1.1.5. Infectious Zoonotic Disease

Care should be taken when trapping and handling live or dead animals. Hantavirus may be transmitted from rodents to humans through body fluid exposure or when breathing aerosolized rodent excreta. Pest management providers may be exposed when handling rodent carcasses after trapping or handling traps contaminated with rodent urine and feces. Feral dogs, cats, skunks, raccoons, and bats may carry and transmit rabies through a bite.

4.1.1.6. Inhalation Hazards

Many pesticides release hazardous vapors and are particularly hazardous in enclosed spaces. Some pesticides are labeled “Restricted Use” due to the high risk of inhalation injury. Personnel may be exposed during mixing, application, and equipment cleaning.

4.1.1.7. Electrical and Fire Hazards

Spot and crack and crevice applications may require application of a pesticide to areas near motors of refrigerators, compressors, and other machinery where it can become an electrical shock hazard. They may also be applied to areas near pilot lights resulting in an explosion and/or fire hazard.

4.1.1.8. Head Impact and Sharp Hazards

Surveys and pest control procedures may be done in attics, crawl spaces, basements, and other areas with low overheads where head impact hazards exist. Some devices used for bird roosting exclusion and rodent control have sharp edges and can cause cuts, puncture wounds, and abrasions.

4.1.1.9. Trip and Fall Hazards

Trip hazards may occur when applicators are spraying without close attention to where they are stepping. Spraying around buildings where there are various obstacles (e.g., plants, utility boxes, plumbing) in the path of the applicator can be particularly hazardous. Pest control may also need to be performed from ladders, on roofs, in ceilings, and in trees. Wet surfaces on the ground or on elevated surfaces can increase the risk of trips and falls.

4.1.1.10. Exposure to Harmful Animals

Venomous animals such as bees, wasps, rattlesnakes, and spiders are potential hazards when attempting to control them. Some of these are very dangerous due to envenomation and allergic reactions. Feral dogs, cats, coyotes, raccoons, and other large pest animals can inflict serious bites or clawing wounds.

4.1.2. Hazard Abatement

Detecting and reporting unsafe or unhealthful working conditions as early as possible, and then promptly controlling the reported hazards, is essential to a successful safety and occupational health program.

4.1.2.1. Operational Risk Management

Operational risk management (ORM) is a decision-making tool to reduce the risk of mishaps, whether in military contingency or support operations. Pest management operations pose risks to human health and the environment that affect the installation’s mission that can be reduced

and minimized through ORM. Pest management ORM uses the following process to minimize hazards:

1. Identify hazards—the hazards may involve the pesticide or the application equipment (see list of hazards in [section 4.1.1](#)).
2. Assess hazards—determine the degree of risk based on the probability and severity of these hazards. For example, the risk may be high if a highly-toxic pesticide is used daily.
3. Make risk decisions—develop risk control options. Decide whether benefits of control outweigh the risks involved.
4. Implement controls
 - a. Engineering controls—e.g., use a less-toxic pesticide for controlling the pest
 - b. Administrative controls—e.g., place warning placards around pesticide vehicles and pesticide storage areas.
 - c. Personal protective equipment—e.g., wear a respirator when an inhalation hazard exists.
5. Supervise—follow-up to determine effectiveness of controls and monitor changes to hazards

For more information on ORM, go to the Naval Safety Center website at <https://navalsafetycenter.navy.mil/>.

4.1.2.2. Read the Pesticide Label

Pesticide labels are found on all pesticide containers used by installation PMSPs. The pesticide label provides directions for mixing, applying, and disposing of pesticides safely. It also includes a list of hazards to humans and first aid treatment. It may also include a list of personal protective equipment that must be worn and user safety recommendations. The label should always be read completely and thoroughly by the applicator before purchasing and using a pesticide. The label is a legal document mandated by FIFRA.

4.1.2.3. Personal Protective Equipment

Personal protective equipment (PPE) should always be used when applying pesticides. The type and level of protection needed will be determined by the toxicity, formulation, and method of application of the pesticide. The pesticide label provides guidance on what PPE to use.

1. Respirator
2. Chemical-resistant gloves
3. Chemical-resistant coveralls or long-sleeve shirt and long pants
4. Chemical-resistant boots
5. Hard hat
6. Goggles

7. Apron
8. Face shield
9. Self-contained breathing apparatus (for fumigation)

Personal protective equipment must be appropriate for the type and application of the pesticide being used. It is the applicator's responsibility to maintain the PPE. Contractors must provide appropriate PPE to their applicators.

4.1.2.4. Pest Control Vehicle Safety Devices

Pest control vehicles should be equipped with safety devices and information.

1. Labels and SDSs for all pesticides in vehicle
2. Emergency medical information including nearest emergency treatment center
3. Fire extinguisher
4. Spill kit
5. First aid kit
6. Cell phone or radio
7. Drinking water supply
8. Rinse water supply for washing pesticide off skin

4.1.2.5. Pesticides and Equipment

The risk of pesticide exposure can be reduced by selecting the appropriate pesticide and equipment for the job. Applying small amounts of low-toxicity pesticide using appropriate and properly-maintained equipment greatly reduces the risk of harm. Using pesticides that are formulated (e.g., contain emetics) or packaged (e.g., water-soluble packets) to minimize chemical exposure and increase safety should be considered when purchasing pesticides. Pesticide selection is addressed in [section 3.4.1](#). Equipment should be tested with water prior to use to ensure proper application and that it is not leaking. Situational awareness, such as monitoring meteorological conditions and location, may also prevent harmful exposure to pesticides.

4.1.2.6. Protection from Infectious Zoonotic Diseases

Pest control personnel who handle trapped animals or dead animal carcasses should wear gloves to prevent exposure to potentially infectious body fluids. A respirator fitted with a high-efficiency particulate air filter should be worn when entering enclosed spaces with large amounts of rodent feces that might be disturbed and become airborne. Additional protection from hantavirus can be provided by spraying dead rodents and rodent feces with a commercial disinfectant. This will kill hantavirus as well as wet the feces to prevent it from becoming airborne. Detailed guidance on rodent handling is found in AFPMB TG No. 41, Protection from Rodent-borne Diseases with Special Emphasis on Occupational Exposure to Hantavirus. It is also recommended that personnel who handle live mammals receive a rabies vaccination.

4.1.2.7. Hazard Communication

All pesticide applicators must receive Occupational Safety and Health Administration (OSHA) Hazard Communication training (29 CFR § 1910.1200). Contractors must carry safety data sheets in their vehicles or, as appropriate, at their on-base administration office. Applicators must understand all of the hazards associated with the chemicals they will use and be able to communicate those to the customer if necessary.

4.1.2.8. Medical Surveillance Program

Department of Defense pesticide applicators are required to be in a medical surveillance program depending on their hazard exposure. Applicators possibly facing exposure to organophosphate or carbamate pesticides should have their cholinesterase levels tested in accordance with the [Medical Surveillance Procedures Manual and Medical Matrix](#). Medical surveillance is conducted by occupational health personnel in accordance with Navy and Marine Corps Public Health Center Technical Manual [\(NMCPHC-TM\) 6260.9A](#), Occupational and Environmental Medicine Field Operations Manual. Some situations require personnel to wear a respirator to minimize pesticide exposure, in which case it is imperative to ensure that the respirator is fitted properly. Industrial hygiene (IH) provides respirator fit testing for DoD personnel.

4.2. PUBLIC SAFETY

By their nature, many pesticides may pose some risk to humans, animals, or the environment because they are designed to kill or otherwise adversely affect living organisms. Safely using pesticides depends on using the appropriate pesticide and using it correctly.

4.2.1. Potential Hazards to the Public

A potential hazard is the risk of harmful effects from pesticides and the level of risk depends on the toxicity of the pesticide and the exposure a human will receive in any situation.

4.2.1.1. Direct Contact with Pesticides

Pesticide exposure can occur through dermal contact with a pesticide on a surface, inhalation of vapors, or ingestion of pesticide through contaminated food or eating utensils. This type of exposure can occur if a pesticide application is done while unprotected building occupants are present, occupants are allowed entry into buildings before the pesticide has dried, or food and food preparation and serving equipment are not properly protected or cleaned after an application.

4.2.1.2. Pesticide Drift

Pesticide drift occurs when a pesticide leaves the target area and affects unprotected persons outside the area. This commonly occurs outdoors when winds can carry the pesticide off-site. Drift can occur indoors if there is air movement or pesticides are drawn up through ventilation ducts. Pesticide applications that involve small pesticide droplets, such as fogging or ultra-low volume application, or dusts are most susceptible to drift.

4.2.1.3. Contact with Contaminated Water

Some pesticides can move through soil and contaminate groundwater used for drinking. Others, if applied in or close to surface water, can cause contamination of recreational waterways.

4.2.1.4. Injury Due to Animals

The use of an inappropriate pesticide may cause collateral injury due to an insufficient knockdown of the target pest. This can occur with bees and wasps. Some insecticides do not knock down the insects rapidly and may actually excite them causing them to become more aggressively defensive in behavior. Unprotected persons blocks away from the pesticide application may become the target of their aggression. Injury can also occur when persons get too close to or try to release a trapped animal or try to capture feral animals by themselves.

4.2.1.5. Fumigation Exposure

Fumigants are highly toxic and can cause immediate death upon exposure. Fumigations can be performed in the housing area where it poses a potential hazard to neighbors and pets. During fumigation the chemical is injected into a tarped structure and allowed to remain for 24 hours. The highest risk of injury or death occurs if a person or animal were to enter the tarp during this period or after the tarp is removed, but before the building is completely ventilated. The fumigant, when exposed to air, dissipates rapidly and readily.

4.2.2. Hazard Abatement

Pesticide applicators should continually be aware of the hazards associated with pesticide use in order to protect the public from exposure.

4.2.2.1. Proper Timing of Pest Control Operations

Most indoor application of pesticides should be conducted when building occupants are not present. An exception to this is the application of pesticide baits that are enclosed in a tamper-resistant bait station that does not allow exposure to occupants or pets. The building occupants must remain out of the building to allow the liquid pesticide to dry. Some pesticide labels are specific about re-entry times (time after application that occupants are allowed back into the treated building). Some pesticides, such as fumigants, provide specific directions on aeration of spaces to remove pesticide prior to re-entry. Certain operations, such as bee and wasp control or removal, are best conducted after the area has been cleared of unprotected persons. Refer to the product label for specific information.

4.2.2.2. Preventing Pesticide Drift

Pesticide drift from target areas to areas where humans, animals, and plants can be affected can be reduced through the following means (adapted from University of Nebraska publication G1773, Spray Drift of Pesticides).

1. Select low or nonvolatile pesticides.
2. Read and follow the pesticide label. Apply a pesticide only if an application is warranted.
3. Use spray additives that decrease drift within label guidelines. This will increase the droplet sizes and pesticide effectiveness.
4. Use larger spray nozzle orifice sizes. This will give larger droplets and will increase the number of tank refills, but will improve coverage and effectiveness.
5. Avoid high pressure. High pressure creates finer droplets; 45 PSI should be considered maximum for conventional broadcast spraying.

6. Use drift-reduction nozzles. These will produce larger droplets when operated at low pressures.
7. Use wide angle nozzles and low boom heights, and keep the boom stable.
8. Drift is minimal when wind velocity is less than 10 mph. Do not spray when wind is greater or blowing towards sensitive crops, gardens, dwellings, livestock, or water sources.
9. Use shielded spray booms. When banding, use shroud covers to keep chemical from drifting.
10. For indoor applications, turn off ventilation and close doors to prevent air currents.

4.2.2.3. Preventing Water Contamination

In addition to other best management practices highlighted throughout this document, pesticide contamination of surface and groundwater sources can be reduced through the following means:

1. Follow the directions on the pesticide label. Many pesticide labels contain use instructions or precautions designed to avoid surface and groundwater contamination.
2. Evaluate the need, method, and frequency of chemical control. Use pesticides only when necessary and only in amounts that will control pests adequately. Pesticides that are applied in low concentrations and less frequently are less likely to leach into the groundwater.
3. Identify the vulnerability of the soil. Well-drained or sandy soils low in organic matter have a high potential for groundwater contamination. Consider the location of the pesticide application in relation to ground and surface water. Web Soil Survey, operated by the USDA Natural Resources Conservation Service, provides soil data and information that may be useful in evaluating the soil conditions on the installation (see <https://websoilsurvey.sc.egov.usda.gov/App/HomePage.htm>).
4. Become familiar with pesticides that may leach. Pesticides with a high potential for leaching (high solubility, low adsorption, persistent) are more likely to contaminate groundwater. Check the pesticide label for warnings about potential to leach to groundwater.
5. Apply the pesticide at the appropriate time. Pesticides are most susceptible to runoff from heavy rains or irrigation during the first several hours after application. Do not apply to saturated or frozen ground. Also, fewer applications are required if they are carefully timed in relation to appropriate stages in the pest's life cycle.
6. Avoid spills and back-siphoning. Avoid spills, especially near wells or other water sources. Only mix pesticides where adequate containment is available. Prevent back-siphoning of pesticide-contaminated water into the water source by keeping the end of the fill hose above the water level in the spray tank. Install a backflow device (such as an air gap or check valve) on the filling pipe to prevent backflow problems.

4.2.2.4. Prevent Tampering with Animal Traps

Caged animals can be very aggressive. Traps should be placed in areas where they will not be tampered with by humans or pets. Warning signs can be placed on the traps and area occupants can be warned of the risk of injury. Live and dead rodents in traps can also be a hazard for hantavirus. Traps should be placed in areas where humans or domestic animals will not be exposed to the rodents.

4.2.2.5. Protection for Fumigation Sites

Warning signs should be posted at the fumigation site warning of the hazards. Some installation contracts require the contractor to provide a 24-hour roving watchperson to patrol the fumigation site to prevent entry by unauthorized personnel.

4.2.3. Special Safety Considerations

Certain areas require special considerations due to the sensitive nature of the area or the people contained in that area.

4.2.3.1. Child Development Center

Children can be sensitive to pesticides and other chemicals. Parents are also concerned about potential hazards that their children may be exposed to and have a right to know about these hazards. Best practice is to minimize pesticide use in and around child development centers and schools, use only enclosed baits and low-toxicity pesticides, do not apply pesticides when people are present, and inform staff and parents of any pesticides used on the property. Integrated pest management methods should be used to reduce the health risks of pesticides to children.

4.2.3.2. Navy Medicine Readiness and Training Command

Persons undergoing medical treatment may be highly sensitive to pesticides and pesticide odors in the environment. Additionally, medical equipment and supplies may be contaminated during pesticide applications. Alternative IPM methods must be considered prior to using pesticides in medical treatment areas. If pesticides must be used, then only crack and crevice treatments with low toxicity pesticides or enclosed baits can be used. Application of any liquid or dust formulation must only be done when the area is unoccupied. Guidance for pest management operations in medical treatment facilities can be found in AFPMB TG No. 20, Pest Management Operations in Medical Treatment Facilities.

4.2.3.3. Food Service Areas

Food contaminated with pesticides can lead to pesticide poisoning. Sanitation and exclusion should be the primary means of preventing and reducing pest infestations. Pesticide use in food service areas should be limited to low-toxicity pesticides, applied to cracks and crevices, and baits. The area should be properly prepared for treatment by putting away utensils and equipment and covering food preparation services. After treatment, the area should be thoroughly cleaned to prevent contamination.

4.3. PEST CONTROL ACCIDENTS

In the case of a pest control accidents, applicators should be trained in first aid procedures and identify the nearest medical services.

4.3.1. First Aid

First aid for pesticide accidents is included on the pesticide label. The applicator should be familiar with first aid procedures required for the pesticide they are using. A copy of the label must be available at the application site. For some pesticides, immediate first aid and medical treatment may be required.

4.3.2. Medical Emergencies

Pesticide applicators experiencing an acute exposure to hazardous pesticides or significant injuries sustained in control operations should immediately go to the nearest emergency room capable of treating their emergent condition. Pesticide applicators that are government employees enrolled in a medical surveillance program with the occupational health department should schedule a follow-up appointment after their condition has subsided. The name, address, and telephone number of an emergency medical care facility should be posted in the commercial applicator's vehicle. For pesticide poisonings, a copy of the pesticide label should be given to the medical first responders or taken to the emergency medical facility. If cholinesterase-inhibiting pesticides (e.g., malathion) are used, the proper antidotes include atropine and 2-pam chloride.

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Chapter 5. Environmental Considerations

5.1. ENVIRONMENTAL CONSIDERATIONS ON THE PESTICIDE LABEL

If the pesticide is potentially harmful to the environment, information will be provided in the following sections of the label:

1. Directions for Use—If pesticide drift is a potential environmental hazard, the directions may require certain application equipment and/or the addition of an anti-drift agent to the tank mix.
2. Environmental Hazards—This section may indicate the pesticide is particularly hazardous to specific animals (e.g., bees, fish). It will also provide information on how to avoid environmental damage.

5.2. MANAGING ENVIRONMENTAL IMPACT

Air, water, and soil risk contamination from pesticides. Pesticide drift to outside the target application area is the primary reason for contamination. Pesticides that pose the highest risk of contamination are herbicides applied to improved and unimproved grounds. Despite being applied in water, pesticides to control mosquito larvae pose a minimal risk due to the target-specific nature of the pesticide (e.g., the biopesticide, *Bacillus thuringiensis israelensis* (Bti), and insect growth regulators). Many procedures to reduce the impact of pest management practices on vulnerable assets are already in place.

5.2.1. Pesticide Pollution

A pesticide is unique as a potential environmental pollutant. When it is applied properly for the correct target pest and to the target location, it is not considered a pollutant. When a pesticide is applied or it drifts outside of the target area, it becomes a pollutant. Pollution can occur during most pest management practices as the result of accidental spills, air and water emissions, and container disposal. Pollutants can be in the form of pesticide residues from equipment and container cleaning or be waste containers.

5.2.1.1. Synthetic Pyrethroids

Pyrethroids are insecticides that are widely used for household, garden, and agricultural pest control. Most were replacements for more toxic and environmentally-hazardous organophosphate and carbamate insecticides. Surveys have indicated that some pyrethroids are being detected in urban stream sediment and at least one chemical has been shown to be toxic to sediment dwelling organisms. Specific pyrethroids of concern include:

1. Bifenthrin (e.g., Talstar)
2. Cyfluthrin (e.g., Cykick, Tempo)
3. Beta-Cyfluthrin (e.g., Tempo Ultra)
4. Cypermethrin (e.g., Demon, Cynoff)
5. Deltamethrin (e.g., Deltadust)

6. Lambda-Cyhalothrin (e.g., Demand)
7. Permethrin (e.g., Permanone)
8. Tralomethrin

Outdoor operations pose the greatest risk for pyrethroid contamination of surface water and stormwater runoff. Increased risk operations that may use pyrethroids include landscape plant insect control, agricultural insect control, and uniform repellent treatment.

5.2.1.2. Pollinator Protection from Pesticides

The Office of the Under Secretary of Defense distributed a memorandum on the DoD Policy to Use Pollinator-Friendly Management Prescriptions dated 5 Sep 2014 (located in additional source documents). This memorandum is issued in accordance with and references DoDI 4150.07 on DoD Pest Management and DoDI 4715.03 on Natural Resources Conservation Programs. In response to this memorandum, NAVFAC Applied Biology put together the "Pollinator Friendly Pesticide Applicator Best Management Practices" fact sheet (located in additional source documents).

Pollinators, such as bees, bats, birds, and butterflies, are essential to the majority of the flowering plants in the environment and to the production of more than 130 different food crops. Protection of both managed bee colonies that are used in the agricultural outleases and feral bees must be considered in pest management operations. Pollinators are highly sensitive to many pesticides, especially insecticides. Best management practices to protect pollinators include:

1. Read the pesticide label for any precautions for bees and apply the product in a manner consistent with the label directions.
2. Use less hazardous insecticides. Certain classes of insecticides, such as organophosphates, carbamates, and neonicotinoids (i.e., imidacloprid), are highly toxic to bees.
3. Choose the least hazardous insecticide formulation if possible. Granules are the least hazardous. Dusts are the most hazardous because they are similar in size to pollen, stick readily to the hairs on the insect, and can be carried back to the nest.
4. Use insecticides with short residuals. The label will include a residual toxicity (RT) time that is the time after application until there is minimal toxic effect on bees.
5. Avoid applying any bee-toxic pesticides on blooming plants that attract bees.
6. Do not apply insecticides when temperatures are forecast to be unusually low or when the evening forecast is for dew. These conditions extend the period in which the insecticide residue remains toxic.
7. Apply pesticides that are toxic to bees at night when most honeybees have stopped foraging and returned to their hives.
8. Use ground applications instead of aerial applications to reduce pesticide drift out of the target area.

Efforts should be made to conserve bee colonies. If the situation allows, bee swarms and hives should be removed and relocated rather than destroyed. For more on protecting bees and other pollinators from pesticides go to the EPA Pollinator Protection website: <https://www.epa.gov/pollinator-protection>.

5.2.1.3. Pollution Prevention

The following pollution prevention best practices should be used on the installation:

1. Determine the need for pesticide use by conducting surveillance.
2. Apply pesticides and clean equipment away from storm drains to prevent storm water contamination.
3. Do not pour pesticide container rinsate into drains. Apply rinsate to a site listed on the pesticide label, store rinsate to use for future pesticide mixing, or dispose of according to local regulations.
4. Use less-toxic and target-specific pesticides.
5. When applying permethrin repellent to uniforms outdoors, do not mix or apply near storm drains or where water run-off will result in storm water contamination, avoid overspray of pesticide onto the ground, and apply spray tank rinsate to uniforms.
6. Minimize outdoor applications of pyrethroid pesticides.
7. Use targeted spot spraying or crack and crevice applications rather than broadcast or baseboard spraying.
8. Minimize pesticide storage on the installation through proper inventory management and by not allowing contractors to store pesticides on the installation.
9. Use rodent traps rather than rodenticides.

5.2.2. Natural and Cultural Resources Protection

Natural and cultural resources on the installation can potentially be impacted by pest management operations or have an impact on these operations. These pest management operations include, but are not limited to, surveys, trapping, weeding, biological control, and pesticide use. The Sigonella 2019 NRMP provides detailed information on the natural resources found on the installation. The NRMP also lists management objectives and recommendations to protect and enhance the installation's natural resources programs.

Any archeological sites on the installation are at low risk for damage due to pest management operations. Certain historic buildings may require special treatment depending on the pest management activity. The Sigonella 2023 CRMP provides detailed information on the cultural resources found on the installation as well as the areas considered historic and cultural vulnerable assets.

5.2.2.1. Environmentally-Sensitive Areas

Sensitive habitats are declared in the installation NRMP. The IPMC is responsible for knowing the boundaries and restrictions of sensitive habitat(s) on their respective site and communicating this information to any pest control or grounds contractors via the PMPAR.

Although the IPMC should have a general knowledge of these areas, any proposed application of pesticides in any of these areas must first be coordinated and approved by the natural resources manager. Applications of pesticides to wetlands or other environmentally sensitive sites, such as tidal marshes and beaches, or around these areas should be carefully planned. Strict adherence to both the pesticide label and the clearances described in the NRMP are required.

5.2.2.2. Invasive Species Prevention

Invasive species can cause damage to native habitats and introduce diseases to native plants and animals. All military vehicles and materials that have been in contact with foreign soil and returning from foreign locations including Hawaii are required to be cleaned by the deployed unit and inspected by the U.S. Department of Agriculture Plant Pest Quarantine Officer prior to disembarkation onto U.S. soil per OPNAVINST 6210.2A, Quarantine Regulations of the Navy. The purpose of these inspections is to prevent the introduction of disease-causing organisms and plant pests. Although the inspections are generally thorough, the equipment of recently redeployed units should be monitored to ensure that any introduced pests are destroyed properly. Any pests found on this equipment should be reported to the environmental division.

5.2.2.3. Cultural Resources

Pest control operations should be checked for consistency with the Cultural Resources Management Plan. Pest management operations requiring alteration of the historic building or structure, including exclusion modifications and significant changes in landscape, will require consultation with the cultural resources manager (CRM) through the IPMC.

Termites damage wooden structures and incidental wood in steel and concrete buildings, such as trim or molding, paneling, or door and window frames. Annual termite inspections detect termite infestations before significant damage occurs. Any termite inspections of historic buildings or structures should be documented using DD Form 1070 and reported immediately to the CRM.

5.2.3. Spill Prevention and Management

Installation spill prevention guidelines shall be followed. The following spill prevention actions shall be taken:

1. Spill kits shall be readily accessible in all pest management vehicles, mixing sites, and pesticide storage facilities
2. Pesticides shall only be stored in an area with containment to hold a spill and without a floor drain
3. Portable mixing pads shall be used when appropriate
4. All pesticide applicators shall be familiar with the Spill Prevention and Response Plan, if available, and who to notify in case of a spill

All pesticide applicators are trained on spill response procedures as part of their initial pest management certification/licensing training. In accordance with OPNAVINST 3440.17 Navy Installation Emergency Management Program, spills will be managed as described in the installation Spill Prevention and Response Plan. Further information on preventing and

controlling pesticide spills is contained in the AFPMB TG No. 15, Pesticide Spill Prevention and Management.

5.2.4. Hazardous Materials and Hazardous Waste Management

Pesticides, being hazardous materials, shall be managed in accordance with the applicable hazardous materials regulations, instructions and installation specific orders/procedures. Proper inventory management and planning will prevent waste generation. The appropriate use of pesticides produces very little hazardous waste. Rinse water containing pesticide residues usually has very small quantities of chemical and is often applied to the target pest site. Not permitting contractor storage of pesticides and on-site disposal of pesticide waste eliminates the need for hazardous material and waste management. In general, pesticides that are not applied must be disposed of as hazardous waste. Large quantities of hazardous waste may be produced when a pesticide is not used by its expiration date. It may also be produced if a pesticide is not used up before the registration for that pesticide is canceled and the stop-use date has occurred. These pesticides may be disposed of as universal waste only when allowed by the standards for universal waste management found in 40 CFR § 273. Any excess pesticides or absorbent material used for spill clean-up requiring disposal requires evaluation by the hazardous waste coordinator in the environmental division to ensure proper disposition.

5.3. CLIMATE CHANGE CONSIDERATIONS FOR PEST MANAGEMENT

In order to comply with climate change directives Executive Order (EO) 14008, Tackling the Climate Crisis At Home and Abroad, and the DoD Climate Adaptation Plan, all operations and planning activities must include climate change considerations. Installations must monitor changes due to climate change and adapt pest management strategies and this IPMP to reduce the adverse effects of climate change. Within the Department of the Navy (DoN)'s Climate Action 2030 strategy to mitigate harmful effects on the Navy's mission and materiel, pest control can most directly contribute to infrastructure and operations resilience by protecting materiel and real estate assets, and protecting and improving natural resources.

Climate change is most likely to affect pest control on the installation due to increased extreme weather event and changes in the distribution of health-related pests, forest pests, invasive species, and beneficial insects. Extreme weather events make an outbreak of vector borne disease more likely due to potential break downs in infrastructure and sanitation, and increases of mosquito breeding areas when water accumulates following heavy rains. In areas that are experiencing increased drought, trees will become more susceptible to pine beetles and other pests. In some areas changes in the local climate and conditions may favor further expansion of invasive species which have been introduced.

Specific pests which have already increased their distribution in response to the changes in climate and habitat that have accompanied global warming include the red imported fire ant, Asian tiger mosquito, and bark beetles. Other pests which have been able to extend their active season or accelerate development include the deer tick and the hemlock wooly adelgid. In the western U.S., West Nile virus epidemics are associated with periods of drought. Increased rainfall increases the abundance of deer mice and the risk of Hantavirus transmission to humans. Overall, climate change is expected to increase the risk of certain vector borne diseases and damage to native species. Protective measures are dependent on the situation and may require increased pesticide use.

Climate-related impacts on pests can be monitored through the identification of new pests and invasive species, and an increase of reported vector-borne and zoonotic diseases in humans

and host animals. Current pest management operations, particularly pest prevention through cultural methods (e.g., pest habitat modification, natural habitat enhancement, elimination of breeding sites, food source reduction), are effective mitigation measures to counter pest impacts due to climate change.

5.4. PUBLIC PERCEPTION

The misuse of pesticides that lead to animal or human injury can lead to negative publicity for the installation. This is also the case with accidental pesticide spills, especially if they occur off-base or cause contamination of a local natural or cultural resource.

Chapter 6. Emergency Pest Management

6.1. PUBLIC HEALTH EMERGENCIES

Pests create a public health emergency when the pests increase in number and/or are found to carry human disease pathogens. A public health emergency, or potential emergency, requiring pest management action may be indicated in several ways. See the Emergency Vector-Borne Disease Response Plan (EVDPR) for more information ([appendix H](#)).

6.1.1. Natural or Manmade Disaster

Usually pest problems do not develop immediately after a disaster, such as earthquakes, wildfires, floods, vehicle accidents and terrorist attacks. Public health pest problems may be the result of increased amounts of refuse, collapse of local infrastructure (e.g., lack of garbage pick-up), decay of human and animal bodies, and accumulation of standing water. The potential pest-related consequences are vector-borne or zoonotic disease outbreaks and increased contact with rodents and feral animals that may cause injury.

6.1.2. Vector-Borne or Zoonotic Disease

The report of human cases of vector-borne or zoonotic disease or the detection of infected mosquitoes or sentinel animals is an indicator of a public health emergency or potential emergency and often warrants an increase in pest management activities.

Reports of human cases—Many human cases of vector-borne and zoonotic disease identified in local medical facilities are reportable to the local and/or state health agencies. A report of a human case of West Nile virus or other vector-borne disease may initiate an investigation and result in alerts going out to other hospitals and clinics if it appears that the case was locally acquired. Immediate vector control may be necessary to prevent further transmission.

6.1.3. Animal Attack

Attacks on humans by vertebrate animals almost always require an emergency medical response. If a person is bitten or scratched by a mammal such as a dog, cat, skunk, coyote, fox, raccoon, opossum, or bat, they are at risk of contracting rabies and should begin a treatment program. If the animal that was involved can be positively identified and safely captured, it should be held for testing to determine if it is infected with rabies or other zoonotic diseases.

Bites by venomous snakes are always emergencies, and the victim should be immediately transported to the nearest medical treatment facility. If the snake can be identified or killed/captured, it may help in the selection of the proper antivenin for treatment.

Certain ants, bees, and wasps can cause painful stings and, in some cases, severe allergic reactions. The local fire department is usually the primary responder to bee sting incidents. Fire department personnel have been trained to protect and manage bee sting victims. A stinging incident is not considered a pest control response issue, but rather, an emergency response and any and all appropriate bee control measures can be used. If fire department response is delayed, installation first responders should be trained how to protect themselves and victims from bee stings.

6.2. AGRICULTURAL EMERGENCIES

Agricultural emergencies are the result of the introduction of insects or other animals that can cause extensive damage to agriculture or forestry in the country. Examples of introduced agricultural pests include the Mexican fruit fly and gypsy moth. Military installations can be a conduit for the introduction of these pests due to the movement of military equipment and personnel in and out of the country. The military's role in preventing introduction of these pests is described in OPNAVINST 6210.2A, Quarantine Regulations of the Navy. Inspections to prevent importation of pests are normally conducted at the port of debarkation in the foreign country.

6.3. EMERGENCY PEST MANAGEMENT RESOURCES

Contract PMSPs can be used for emergencies if it is written in the contract specifications. The U.S. Naval Hospital, Sigonella should develop an EVDRP to manage public health emergencies when applicable. It includes additional Navy and local government contingency vector surveillance and control resources.

Chapter 7. Program Resources

The installation has access to the following support agencies and organizations for pest management assistance. Contact information specific to the installation is included in [appendix A](#).

7.1. NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND, APPLIED BIOLOGY

Naval Facilities Engineering Systems Command, Applied Biology is currently staffed by full-time, civilian DoD professional pest management consultants certified in DoD pesticide applicator categories 3, 5, 6, 7, 8, and 11. NAVFAC provides the following products and services:

1. Review and technically approve installation IPMPs in accordance with DoD and Navy policies
2. Provide technical assistance to the installation IPMCs, natural resources and other environmental managers, safety officers, medical officers, and other regional and installation personnel regarding IPM, pesticide regulatory compliance, and pesticide safety
3. Review and approve or reject pesticides and equipment use on installations
4. Conduct on-site program reviews and environmental compliance program external assessments to ensure compliance with the regulations and IPMPs in accordance with Navy policies and procedures
5. Compile and report actual pesticide use and pest management operations to appropriate DoD agencies
6. Provide IPM recommendations and pest identification
7. Assist installations with writing or re-writing IPMPs
8. Review and approve contract specifications or lease agreements that include the use of pesticides
9. Provide recertification training for DoD-certified applicators, as well as initial and recertification training for PMPARs/IPMCs

7.2. NAVY ENVIRONMENTAL AND PREVENTIVE MEDICINE UNIT SEVEN

The Navy Environmental and Preventive Medicine Unit Seven (NEPMU-7) is staffed by two full-time, active-duty Navy entomologists. The entomologists are certified in DoD pesticide applicator categories 3, 5, 6, 7, and 8 and are assigned the following responsibilities:

1. Act as BUMED's professional pest management consultants to provide BUMED review of installation IPMPs and Emergency Vector-Borne Disease Response Plans
2. Provide technical assistance on the surveillance and control of vectors on permanent and contingency installations

3. Review and approve contract specifications that include the use of pesticides
4. Conduct vector-borne disease risk assessments and disease prevention recommendations when requested
5. Provide disease vector management consultation and identification services
6. Provide pest management during contingency operations and for installations in the event of a disaster or disease outbreak

The NEPMU-7 (Rota, Spain) website is at: <https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/Field-Activities/Navy-Environmental-Preventive-Medicine-Unit-7/>.

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Appendices

Appendix A. Points of Contact

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10 **A.1. Installation Pest Management**

Name	Title	Phone Number	E-mail
Alessandra Contarino	IPMC, Natural/Cultural Resources Manager	DSN 624-2719	alessandra.contarino.ln@us.navy.mil
CDR Seth Cochran	Public Works Officer	624-2906	seth.d.cochran.mil@us.navy.mil
Alexandra Ebejer	FSC Contracting Branch Head	DSN 624-2948	alexandra.ebejer.ln@us.navy.mil
Justin Lancaster	Installation Environmental Program Director	DSN 624-2725	justin.j.lancaster2.civ@us.navy.mil
Helen Haase	Hazardous Waste/Materials	DSN 624-2722	helen.k.haase.civ@us.navy.mil
LCDR R. Christoff	Preventive Medicine Department Head		robert.g.miller210.mil@health.mil
Sebastiano Bongiovanni	Pest Management PAR	DSN 624-2920	sebastiano.bongiovanni.ln@us.navy.mil
Joe Fiorenza	Realty Specialist/Lease Administrator	DSN 624-6889	Giuseppe.fiorenza.ln@us.navy.mil
Peter Faulk	Navy Gateway Inns and Suites (NGIS) Manager	DSN 624-4070	Peter.faulk@us.navy.mil

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13 **A.2. Naval Facilities Engineering Systems Command Atlantic Applied Biology**

NAVFAC Atlantic Applied Biology			
Code EV51		6506 Hampton Blvd	Norfolk, VA 23508-1278
COMMERCIAL		DSN	
Commercial: (757) 322-XXXX		DSN: 262-XXXX	
NAME	TITLE	EXT	EMAIL
Sabra Scheffel	Applied Biology Program Manager	4320	sabra.l.scheffel.civ@us.navy.mil
Steven Holmes	Pest Management Consultant	8295	steven.p.holmes.civ@us.navy.mil
Chris Martin	Pest Management Consultant	4611	christopher.d.martin108.civ@us.navy.mil
Steve Robertson	Pest Management Consultant	4796	steven.b.robertson2.civ@us.navy.mil
Kirk Williams	Pest Management Consultant	4254	kirk.p.williams4.civ@us.navy.mil

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15 **A.3. Navy Environmental and Preventive Medicine Unit Seven**

Navy Environmental and Preventive Medicine Unit SEVEN (NEPMU-7)		
NEPMU-7	PSC 819 Box 67, FPO AE 09645-0085	Rota, Spain
COMMERCIAL	DSN	FAX
Local: 727-2230 International: 011-34-956-2230	International: 94-314-727-2230	727-xxxx
E-MAIL	WEBSITE	
NEPMU7@eu.navy.mil nepmu7ombudsman@gmail.com	https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/Field-Activities/Navy-Environmental-Preventive-Medicine-Unit-7/	

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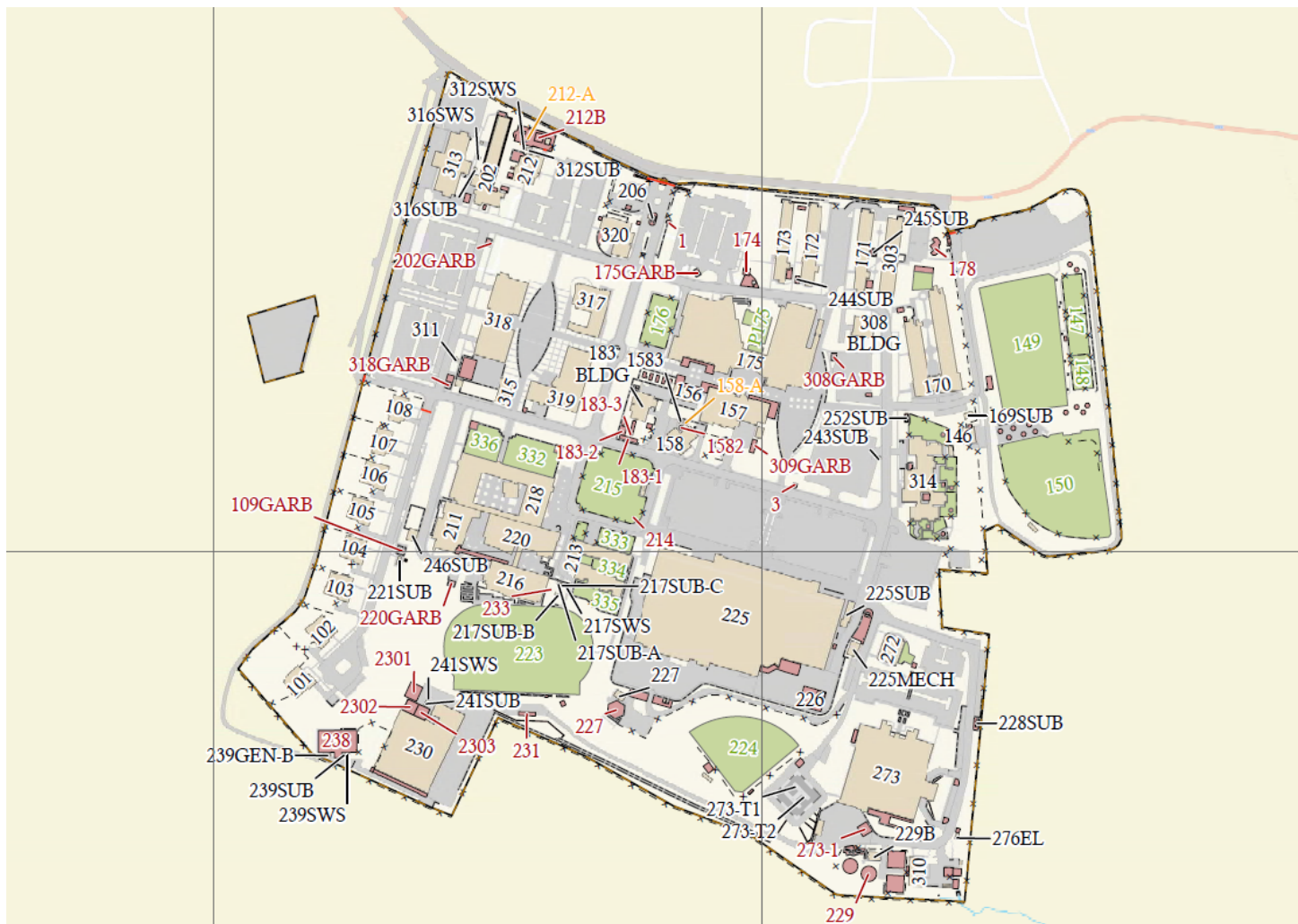
Appendix B. Maps

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B.1. General Location Maps



B.2. Map of Installation NASSIG I



B.3. Map of Installation NASSIG II



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Appendix C. Program Review

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C.1. Pest Management Program Self-Assessment Checklist

Installation Name: _____ Date: _____

Name of Person Completing Checklist: _____

Review Item	Reference	Verification and Documentation	Y	N	N/A
Pest Management Coordinator					
Is IPMC designated and appointed by CO/CG by letter?	6250.4C: Encl. (1), Para.4.h.(6) 5090.1: 24-5.3 5090.2: 040313	Copy of appointment letter.			
Is IPMC properly qualified and trained? If an IPMC selects or applies pesticides, he or she must be certified as a DoD pesticide applicator.	6250.4C: Encl. (1), Para.6 M4150.07-V2 Para.3.4	Copy of course completion certificate or DoD pesticide applicator certificate.			
Does the IPMC oversee the installation pest management program and pest management plan and remain aware of and familiar with all pest management operations on the installation?	M4150.07-V1: Para.3.1 and 4.1.c	Operations documented in the installation integrated pest management plan; IPMC has copies of pesticide approvals and pest management reports; IPMC is actively involved in pest management decision making.			
Pest Management Plan					
Does installation have a current comprehensive IPMP? IPMPs remain current for 5 years.	M4150.07-V1: Para. 3.2 Table 1 6250.4C: Encl. (1), Para.9.a 5090.1: 24-3.9 5090.2: 0303	Copy of IPMP.			
If installation does not have an IPMP, has command planned and budgeted for development and maintenance of an IPMP?	6250.4C: Encl. (1), Para.4.h.(1) 5090.2: 040313	Environmental division should have IPMP listed as a deficiency and submit an EPR.			
Is IPMP signed by CO/CG?	M4150.07-V1: 3.4 Table 2 6250.4C: Encl. (1), Para.9.a 5090.2: 040313. C and E	IPMP signature page.			
Is IPMP reviewed and signed by IPMC, medical department, and BUMED and NAVFAC pest management consultants?	M4150.07-V1: Para. 3.2 Table 1 6250.4C: Encl. (1), Para.6.a 5090.2: 040313. C and E	IPMP signature page.			
Is IPMP updated annually by qualified personnel (trained or certified IPMC or PMPAR and current (contains current POCs, contracts, applicator licenses, list of approved pesticides, etc.)?)	M4150.07-V1: 3.2 Table 1 5090.2: 040313.G	View applicator licenses, dates of pesticide approvals, and other items that indicate the information is not outdated. Review NOPRS IPMP information.			
IPMP includes the following sections: - List of program objectives - Description of all pest management requirements and programs and staffing	M4150.07-V1: 3.2 and 3.3 5090.2: Appendix B	IPMP contains information and sections as outlined in M4150.07-V1, Section 3.			

Review Item	Reference	Verification and Documentation	Y	N	N/A
requirements (including in-house, contract, agricultural outlease, golf course, NAFI, GOCO, experimental, and natural resources) • Description of IPM procedures for all pest and disease vectors • Identification of program resources (facilities, equipment, etc.) to support program • List of pesticides approved by NAVFAC pest management consultant • Procedures for managing spills • Identification of planned measures to comply with DOD MOA and with state pesticide regulatory office regarding use or application of pesticides • Description of contracted pest management operations • Description of operations with special environmental considerations • Identification of animal control efforts • Identification of potential vector-borne diseases and collaboration with local health agencies • Applicable laws and regulations • Agricultural outlease operations					
Location and information on pesticide applications (e.g., mosquito control, forestry, right-of-way, aquatic, aerial) likely to affect threatened and endangered species and associated critical habitat are coordinated with installation natural resources managers and other appropriate stakeholders and included in the IPMP.	M4150.07-V1: 5.10.a.(2)	Information is in the IPMP.			
All stakeholders (including IPMC, PMPARs, environmental division, medical department, pest control shops, NAFIs, agricultural outlease program manager, and golf course superintendent) have copy of or ready access to current IPMP.	BMP	IPMP readily available to stakeholders as hard or electronic copy.			
Program Maintenance					
Have BUMED and/or NAVFAC conducted program reviews in order for the installation to maintain program and IPMP?	6250.4C: Encl. (1), Para.10.a 5090.1: 24-3.10 5090.2: 030401	This is a NAVFAC requirement that is coordinated with the installation. IPMC should maintain copies of program review reports. EMS audits may fulfill requirement.			
Have deficiencies and recommendations from past reviews	6250.4C: Encl. (1), Para.10.a	Documentation of corrections on file and/or corrections made were noted in follow-up inspection or review.			

Review Item	Reference	Verification and Documentation	Y	N	N/A
been resolved or addressed in order to maintain and improve program?	5090.1: 24-4.4 5090.2: 030401				
Do DoD pest management personnel remain current in IPM technologies?	BMP	Personnel attend training workshops, are provided in-service training and/or have access to pest control trade journals.			
Training and Certification					
Do all installation pest management personnel who apply or supervise the application of pesticides have current DoD certification or EPA-approved certification or license?	4150.07: Section 4 5090.1: 24-3.19 5090.2: 030501	Copies of all licenses and certificates on file, preferably in IPMP, and applicators have cards while applying.			
If DoD applicator certification expired, has applicator received a six-month extension from a NAVFAC pest management consultant?	M4150.07-V2: 5.3	Correspondence from NAVFAC approving extension.			
If DoD applicators are not certified (i.e., apprentices), are they under the direct supervision of a certified applicator while performing pesticide applications?	M4150.07-V2: 3.2.b	Observe operations to ensure proper supervision, if necessary.			
Was evidence of contractor pesticide applicator licensing or certification provided to contracting officer prior to award?	4150.07: Section 4 6250.4C: Encl. (1), Para. 15.b 5090.2: 030702	Copies of all licenses and certificates on file, preferably in IPMP, and applicators have cards while applying. Certifications are listed in NOPRS IPMP section.			
Are PMPARs trained in performance assessment evaluation and pest management technology?	M4150.07-V1: 4.6.d 6250.4C: Encl. (1), Para. 15.c 5090.2: 030502.D	Copies of training course certificates on file, preferably in IPMP.			
Do pest management personnel seek and attend continuing education courses?	BMP	Copies of course completion certificates on file.			
Staffing					
Is staffing sufficient to effectively control pests and manage program?	BMP	Interview applicators, supervisors, and managers.			
If personnel indicate that staffing is insufficient, then what indicators or data are being collected to show that staffing levels are insufficient?	BMP	View indicators or data.			
Pesticide Procurement					
Does installation use only pesticides approved by NAVFAC pest management consultant?	M4150.07-V1: 3.2 Logistics 5090.1: 24-3.16 5090.2: 030601	IPMC maintains approved pesticide list. Inspect pesticides in pest control storage and on vehicles to ensure they are listed on the current pesticide approval list. Review NOPRS records.			
Pest Management Records and Reporting					
Are records kept for <u>all</u> pest management operations conducted on the installation, including those by NAFIs and for agricultural operations and environmental protection?	7 U.S.C. § 136i-1(a)(1) 6250.4C: Encl. (1), Para. 23 5090.1: 24-3.16 M4150.07-V1: 4.5 5090.2: 031002	Review NOPRS records.			
Are records retained indefinitely?	6250.4C: Encl. (1), Para.23.a 5090.1: 24-3.4	Review NOPRS records. Does IPMP download files annually and maintain onsite?			

Review Item	Reference	Verification and Documentation	Y	N	N/A
	M4150.07-V1: 3.2 Pesticides				
Do personnel and regulatory agencies have ready access to records? (e.g., able to access records by location, pesticide, applicator, etc.)	7 U.S.C. § 136i-1(b)	Review NOPRS records.			
Is the installation using the NAVFAC Online Pesticide Reporting System?	BMP				
Are reports of pest management operations being sent to NAVFAC?	6250.4C: Encl. (1), Para. 23.b 5090.2: 031002	NOPRS records			
Contracting					
Do properly trained PMPARs inspect the performance of contractors?	M4150.07-V1: 4.6.d 6250.4C: Encl. (1), Para. 15.c 5090.2: 030502.D	Training certificates and contract monitoring documents are on file.			
Are all pest management contracts on the installation monitored by PMPARs?	M4150.07-V1: 4.6.d 6250.4C: Encl. (1), Para. 15.c 5090.2: 030502.D	Check that MCCS, MCX, NEX, and MWR contracted services are monitored by trained personnel.			
Do PMPARs measure efficacy and ensure safety and environmental compliance of contract pest control?	M4150.07-V1: 4.6.d 6250.4C: Encl. (1), Para. 15.c 5090.2: 030502.D	Interview PMPARs to identify method and frequency of inspections. List methods of measurement.			
Are pest management contracts sent to the NAVFAC PMC for review?	4150.07: 2.5.b M4150.07-V1: 4.6.b 5090.1: 24-3.16 M4150.07-V1: 4.6.b	Correspondence with NAVFAC PMC.			
Are all contract pesticide applicators currently licensed in the state in which they operate?	M4150.07-V1: 4.6.a.1 5090.2: 030701	Copies of current certificates or licenses are on file, preferably in IPMP.			
Is the pest control contractor currently registered with the Structural Pest Control Board or the equivalent state pest control business registration agency?	4150.07: 1.2.b	Copy of current registration certificate on file, preferably in IPMP.			
Pest Control Shop					
Does pesticide storage area pose a hazard to personnel in adjacent areas or buildings?	M4150.07-V1: 4.4.a TG17:2.4	Inspect building to see that exhaust vapors will not move into adjacent occupied areas.			
Does storage area have sufficient security to prevent unauthorized entry?	M4150.07-V1: 4.4.a TG17:2.8	Conduct inspection to ensure doors can be locked, equipment storage areas can be secured, and that applicators lock doors when leaving premises.			
Does building have clean area for office?	M4150.07-V1: 4.4.a TG17: 3.1.3.1				
Are separate laundry facilities (designated only for cleaning of clothing potentially contaminated with pesticide) available for work clothing?	BMP				
Are shower facilities available for employees?	BMP				
Is separate space or cabinets provided for storage of PPE?	M4150.07-V1:4.4.a TG17: 3.1.3.1.3				

Review Item	Reference	Verification and Documentation	Y	N	N/A
Are pesticides stored off the floor and with sufficient access so that all labels are visible?	M4150.07-V1: 4.4.a TG17: 3.1.4.1.2				
In areas where pesticide concentrates are stored or mixed, are floor drains sealed or not present and is containment provided (bermed or sloped floors)?	M4150.07-V1: 4.4.a TG17: 3.1.4.1.2 5090.1: Para.24-3.11				
Are all surfaces on which pesticides are stored and mixed and on which pesticide application equipment is serviced made of non-absorbent materials?	M4150.07-V1: 4.4.a TG17: 3.1.4.1.2				
Are pesticides stored in a dry room or building with a temperature between 50 °F and 100 °F?	M4150.07-V1: 4.4.a TG17: 3.1.4.1.2				
Are fire extinguishers provided and easily accessible to occupants?	M4150.07-V1: 4.4.a TG17: 3.1.4.1.2	Inspect inspection record and see that fire extinguishers are fully charged.			
For large pesticide containers with spigots, is a drip pan containing absorbent material placed below spigot?	M4150.07-V1: 4.4.a TG17: 3.1.4.1.2				
Are backflow prevention devices installed on faucets used to fill pesticide tanks?	M4150.07-V1: 4.4.a TG17: 3.5.2.10				
Are emergency decontamination facilities (i.e., eye wash, deluge shower) provided onsite and readily accessible?	M4150.07-V1: 4.4.a TG17: 3.5.2.12	Check to see that it is functional and that inspection records are up-to-date. Ensure that, in an emergency, personnel can easily access and operate the devices.			
Are ventilation fans available in storage and mixing areas and do they function and provide adequate ventilation (six changes of air per hour)?	M4150.07-V1: 4.4.a TG17: 3.5.4.2 5090.1: 24-3.11	Operate fans. Check IH survey ventilation results (Copy may be available in shop or contact installation IH).			
Are identification signs clearly visible on building and fences to advise personnel of the contents and warn of their hazardous nature?	M4150.07-V1: 4.4.a TG17: 3.8				
Are only pesticides listed on approved pesticide list stored?	M4150.07-V1: 3.2 Logistics 5090.1: 24-3.16 5090.2: 030601	Compare approved pesticides list with items stored on shelves.			
Do all pesticide containers have EPA-approved labels attached?	5090.1: 24-3.5. 5090.2: 030602				
Are spill kits provided and readily accessible?	6250.4C: Encl. (1), Para. 13.d 5090.1: 24-3.12(b) 5090.2: 030605	Inspect to ensure contents are suitable for pesticide spills.			
Are SDSs and labels for each pesticide stored and used maintained and readily accessible in the pest control shop?	5090.1: 24-3.5 5090.2: 030602	Review SDS/label book and compare with pesticides stored in shop and on vehicles.			
Pest control equipment					
Is equipment properly maintained and clean (no evidence of leakage and spillage)?	BMP	Inspect			

Review Item	Reference	Verification and Documentation	Y	N	N/A
Are different sprayers used for herbicides and insecticides?	BMP	Inspect that sprayers are properly marked.			
Is equipment routinely calibrated to ensure proper delivery of pesticide?	BMP	Calibrations, if needed, are recorded in a logbook or other recordkeeping system.			
Is application equipment stored in a secure area?	M4150.07-V1: 4.4.a TG17: 3.4.6	Inspect			
Pest Control Vehicles (DoD and Contract)					
Are pesticides stored in a lockable compartment on the vehicle?	BMP	Inspect			
Does applicator ensure that pesticides are not stored in passenger compartment of vehicle?	BMP	Inspect			
Is the vehicle clean and maintained (no evidence of leakage and spillage)?	BMP	Inspect			
Does the vehicle have a properly stocked spill kit?	6250.4C: Encl. (1), Para. 13.d 5090.2: 030605	Inspect to ensure contents are suitable for pesticide spills.			
Is the vehicle properly identified to warn of pesticides on vehicle?	BMP	Inspect			
Are all containers on vehicle, including service containers, properly labeled?	5090.1: 24-3.5 5090.2: 030602	Inspect			
Is PPE properly stored on vehicle?	BMP	Inspect			
Are SDSs for pesticides carried on vehicle?	BMP	Inspect			
Are appropriate wash racks provided for cleaning vehicles (i.e., does not drain into stormwater system)?	BMP	Inspect			
Integrated Pest Management					
Is integrated pest management practiced in order to minimize pesticide use when non-chemical alternatives are available and cost effective?	7 U.S.C. § 136r-1 6250.4C: Encl. (1), Para. 3.c. (5) 5090.1: 24-3.8 5090.2: 030103 4150.07: 1.2.a	Pest management service providers have survey devices (i.e., sticky traps) and less toxic and sustainable pesticides (i.e., baits) in their inventory. Records include surveys and the application of less toxic pesticides and use of non-chemical methods.			
Does the installation pest management plan emphasize and describe the use of IPM to provide sustainable pest management?	7 U.S.C. § 136r-1 6250.4C: Encl. (1), Para. 3.c.(5) 4150.07: Section 4 M4150.07-V1: 3.1 5090.2: 030103	Review IPMP sections that list pest control methods. Review installation instructions, orders, or policies, especially for housing, that encourage IPM practices.			
Does the installation use practices that demonstrate IPM?	7 U.S.C. § 136r-1 6250.4C: Encl. (1), Para. 3.c. (5) 4150.07: 1.2.a 5090.2: 030103	Identify and list practices.			
Does the installation promote IPM?	7 U.S.C. § 136r-1 6250.4C: Encl. (1), Para. 3.c. (5) 4150.07: 1.2.a 5090.2: 030103	Identify and list promotion practices.			
Pesticide Application					

Review Item	Reference	Verification and Documentation	Y	N	N/A
Are pesticides applied in accordance with the label directions?	7 U.S.C. § 136(j) M4150.07-V1: 4.4.b 5090.2: 030606	Interview applicators. Observe application if possible. Wear appropriate PPE.			
Are special precautions taken for operations at child development centers, housing, medical treatment facilities, and food preparation areas?	M4150.07-V1: 3.4 5090.2: 030806, 030807, 030808	Interview applicators and review records to see if steps are taken to minimize pesticide use or use less toxic pesticides in these areas.			
Are liquid and dust formulations of pesticides applied only when unprotected personnel are not occupying the work space to be treated?	5090.1: 24-3.2 5090.2: 030608	Interview applicators. Observe application if possible. Wear appropriate PPE.			
Are preventive pesticide treatments prohibited unless approved by a pest management consultant?	M4150.07-V1: 6.3 5090.2: 030608	Interview applicators regarding practices. Review pest management records to see if there is any indication of routine pesticide applications.			
Are all applicators familiar with the installation's spill response procedures?	6250.4C: Encl. (1), Para. 13.d 5090.2: 030605	Get copy of installation instruction on spill response procedures. Interview applicators.			
Are all feasible efforts and management controls used to avoid production of hazardous wastes and to ensure use of pesticides before shelf-life expiration?	6250.4C: Encl. (1), Para. 13.f 5090.2: 030604	Ask applicators how they clean equipment and dispose of rinsate. Interview shop supervisor to determine container disposal methods.			
Is the installation aware of and do they enforce pesticide "stop sale, use, or removal" orders issued by the EPA?	FIFRA 5090.2: 030604	Check EPA Web site regarding the provisions of pesticide orders. Check records to see if pesticides that have a "stop sale, use, or removal" order are being used contrary to the provisions of the order.			
Aerial Pesticide Applications					
If conducted or proposed, has a plan for the aerial application of pesticides been reviewed and approved by a BUMED (medically-important pests) or NAVFAC (economic pests) aerial spray-certified pest management consultant?	6250.4C: Encl. (1), Para. 13.e 5090.2: 030801	Ask for and review signed validation statement.			
Safety					
Are applicators provided with the appropriate PPE?	5090.1: 24-3.7	Ask applicators to show you PPE in shop and on vehicles.			
Do applicators maintain and wear appropriate PPE when applying pesticides?	6250.4C: Encl. (1), Para. 12.b M4150.07-V1: 4.4.b 5090.2: 030606	Ask applicators to show you PPE in shop and on vehicles. Observe application, if possible.			
Do all applicators receive training on use of PPE? Are applicators physically qualified to wear respirators?	M4150.07-V1: 4.4.b M4150.07-V2: APP 5A.g 5090.2: 040312	Review training record or rosters. Ask to see respirator fit test cards.			
Medical					
Are all preventive medicine technicians currently DoD certified to apply pesticides?	M4150.07-V1: Table 1	IPMC should maintain and update their certification information in the IPMP.			
Does the preventive medicine department (PMD) provide guidance, recommendations and on-site assistance on technical matters	M4150.07-V1: Table 1	Reports			

Review Item	Reference	Verification and Documentation	Y	N	N/A
related to disease vectors and other medically-important pests?					
Has PMD developed and maintained an Emergency Vector-Borne Disease Response Plan?	M4150.07-V1: Table 1	Copy of plan should be included in the IPMP			
Is PMD prepared to provide area-wide operational pest management services in the event of a vector-borne disease outbreak or elevated risk of disease?	M4150.07-V1: Table 1	Details should be provided in the Emergency Vector-Borne Disease Response Plan and PMD should be familiar with it.			
Does PMD provide guidance on the use of PPE to pesticide applicators and medical surveillance program managers?	M4150.07-V1: Table 1	Only applies when the installation has a DoD pest control shop and applicators.			
Does PMD conduct inspection and surveys to determine species, source, and location of medically-important pests.	M4150.07-V1: Table 1	Reports and survey data.			
Does PMD evaluate the effectiveness of control measures, communicate surveillance data to pesticide applicators, and coordinate with civilian and government agencies on disease vector issues?	M4150.07-V1: Table 1	Surveys should be conducted to evaluate efficacy of control and survey data maintained. PMD should have a list of agencies in the Emergency Vector-Borne Disease Response Plan.			
NAFI Operations (NEX/MCX/MCCS/MWR)					
Are all NAFI pest management operations described in the IPMP? This includes operations conducted at: - NEX/MCX retail stores - NEX/MCX and MWR/MCCS food service facilities MWR/MCCS athletic fields and golf courses	M4150.07-V1: 3.3.c, d, and h	Review IPMP.			
Are pesticides used by NAFI pest control providers included on the installation approved pesticide list?	M4150.07-V1: 3.3.e 5090.2: 030601	Pesticides used by NAFIs are included on pesticide approval list.			
Are pesticide use records maintained at each facility?	7 U.S.C. § 136i-1(a)(1) 6250.4C: Encl. (1), Para. 23.a 5090.2: 031001	NAFI maintains records.			
Are all pest management operations reported to the installation IPMC so that it can be reported to NAVFAC?	6250.4C: Encl. (1), Para. 23.b 5090.2: 031002	Records and reports contain operations conducted by NAFIs			
If NAFIs purchase pest control services with the DoN purchase card, are the services in compliance with DoD and DoN pest management requirements?	4200.1A: Ch.6, Para.7 5090.2: V14, 030704	Identify pest management services that may have been purchased with a purchase card, i.e. one time pest control service in a club.			
Pesticide Retail Sales in the Navy Exchange, Commissary, and Veterinary Clinics					
Are only pesticides that are <u>not</u> Category I pesticides labeled "Danger, Poison" displayed for retail sale?	6250.4C: Encl. (1), Para. 13.c 5090.2: 040309	Inspect pesticide display.			

Review Item	Reference	Verification and Documentation	Y	N	N/A
Are pesticides properly displayed to prevent contamination of food, equipment, utensils, linens, and single-service and single-use articles? (i.e., separated by partition or located in an area not above items)	FOOD: 7-301.11	Inspect pesticide display.			
Are spill containment items available?	6250.4C: Encl. (1), Para. 13.d 5090.2: 030605	Inspect spill containment kits.			
Are employees familiar with spill procedures?	6250.4C: Encl. (1), Para. 13.d 5090.2: 030605	Ask employees to describe procedures.			
Does the preventive medicine department assist NEX and commissary in selecting, storing, and displaying pesticides in retail outlets?	6250.4C: Encl. (1), Para. 4.c.(8)	Interview preventive medicine personnel.			
Does the installation monitor pesticides available for purchase in Marine Corps commissaries and exchanges to ensure that the pesticides available for sale are least hazardous pesticides that are compatible with DoD IPM programs and are pesticides that comply with applicable Federal, state, and local laws.	5090.2: 14302.12	Find out who does that?			
Is the retail store aware of and do they enforce pesticide "stop sale, use, or removal" orders issued by the EPA?	FIFRA	Check EPA Web site regarding the provisions of pesticide orders. Check retail shelves to see if pesticides that have a "stop sale, use, or removal" order are being displayed for sale contrary to the provisions of the order.			
Environmental Programs					
Does the installation have a noxious weed management program developed in accordance with Section 2814 of Title 7, U.S.C. to comply with the Federal Noxious Weed Act and is conducted in cooperation with State and local agencies?	M4150.07-V1: 5.5	Review any plans (INRMP, invasive weed management plan) or contracts for noxious weed control. Specific control/prevention measures are listed in IPMP			
If the installation has an active airfield, does the IPMP reference the BASH plan?	M4150.07-V1: 3.1	Plans, directives, or contracts for BASH or references should be included in the IPMP. A list of specific control/prevention measures should be in IPMP.			
Does the IPMP reference the INRMP? Are appropriate portions of the IPMP implemented in accordance with the INRMP?	5090.1: 24-3.9 M4150.07-V1: 5.10 5090.2: 040313.D	Review IPMP. Copy or link to INRMP should be included in IPMP.			
Are pesticides used in invasive weed control, BASH, depredation and other environmental programs included in the installation approved pesticide list?	M4150.07-V1: 3.3.e 5090.1: 24-3.16 5090.2: 030601	Review NOPRS records or plans.			
Are pesticides used in these operations recorded and reported to the IPMC so that they can be reported to NAVFAC?	7 U.S.C. § 136i-1(a)(1) 6250.4C: Encl. (1), Para. 23.b 5090.2: 031002	Review NOPRS records.			

Review Item	Reference	Verification and Documentation	Y	N	N/A
Have pest management operations that may have an impact on natural resources been coordinated with and reviewed by the natural resources program manager?	M4150.07-V1: 5.10 (2) and (4) 6250.4C: Encl. (1), Paragraphs 20, 21, and 22 5090.2: 010302.B	Review IPMP for environmental impacts of pest management operations and for environmental manager signature. Interview natural resources manager to ensure if he/she is aware of pest management impacts on natural resources.			
If feral cats and dogs are present or potentially present, does the installation have a program to capture and remove them from the installation?	6401.1B: Para. 4-2c(4) CNO policy letter 5090 N456M/ 1U595820 of 10 Jan 2002 5090.2: 030814	Visual observations. Review installation policies or directives. Identify and review procedures.			
Are installation pest management and environmental personnel and the installation commander aware of the CNO policy on feral animals?	CNO policy letter 5090 N456M/ 1U595820 of 10 Jan 2002	Applies only to Navy installations. Does the installation have an instruction, order, or policy to prevent feral animals?			
Is the CNO feral animal policy communicated to installation personnel and enforced?	CNO policy letter 5090 N456M/ 1U595820 of 10 Jan 2002	Applies only to Navy installations. Identify practices that support the presence of feral animals.			
Agricultural Outlease Program					
Does the IPMP describe the agricultural outlease pest management program?	M4150.07-V1: 3.3.h 6250.4: Encl. (1), Para. 9.a.	Review IPMP.			
Has agricultural outlease agreement (including soil and water conservation plan) been reviewed by the NAVFAC PMC	6250.4: Encl. (1), Para. 6.k. 5090.2: V14, 030702	Record of review.			
Are agricultural pesticides included on the installation list of approved pesticides?	M4150.07-V1: 3.2 Logistics 5090.2: V14, 030601	Review NOPRS.			
Are the pest management operations reported to the installation IPMC reported to the NAVFAC PMC?	7 U.S.C. § 136i-1(a)(1) M4150.07-V1: 4.5 6250.4C: Encl. (1), Para. 23.b. 5090.2: V14, 0310	Review NOPRS			
Is on-base agricultural pesticide storage in compliance with local and State regulations?	M4150.07-V1: 4.4.a	Obtain State regulations and inspect pesticide storage or review agricultural commissioner inspection records.			
If lessee(s) use aerial pesticide application, has the aerial spray project been reviewed and approved by NAVFAC?	M4150.07-V1: 5.1 6250.4C: Encl. (1), Para. 13.e. 5090.2: V14, 030801	Review aerial spray statement of need and validation letter.			
Pest Management Operations					
What are the installation's top five pests and what are their impacts on the installation?	BMP	Interview pest management service providers and complete pest management project sheets for each pest.			
What survey methods are used to detect, assess, and prescribe treatment for the top five pests?	BMP	Interview pest management service providers and complete pest management project sheets for each pest.			

Review Item	Reference	Verification and Documentation	Y	N	N/A
What non-chemical control methods are used to prevent and control the top five pests?	BMP	Interview pest management service providers and complete pest management project sheets for each pest.			
Key to references: • 7 U.S.C. § 136 FIFRA • 4150.07: DoDI 4150.07, DoD Pest Management Program • M4150.07- V1: DoD Manual 4150.07, Volume 1, DoD Pest Management Program Elements and Implementation: Structure and Operation • M4150.07- V2: DoD Manual 4150.07, Volume 2, DoD Pest Management Program Elements and Implementation: Pesticide Applicator Training and Certification Program • TG17: Armed Forces Pest Management Board, Technical Guide No. 17 Military Handbook, Design of Pest Management Facilities • 6401.1B: SECNAVINST 6401.1B, Veterinary Health Services • 6250.4C: OPNAVINST 6250.4C, Navy Pest Management Programs • 5090.1: OPNAV M-5090.1, Environmental Readiness Program Manual • 5090.2: MCO 5090.2-V14, Environmental Compliance and Protection Program, Volume 14, Integrated Pest Management • 4200.1A: EBUSOPSOFFINST 4200.1A (DON EBusiness Operations Office Instruction), Department of the Navy Policies and Procedures for the Operation and Management of the Government Commercial Purchase Card Program • FOOD: U.S. Food Code 2017 Notes regarding directive authority. • DoDI 4150.07 applies to both Navy and Marine Corps installations • SECNAVINSTs apply to both Navy and Marine Corps installations • OPNAVINST 6250.4C applies only to Navy installations • MCO 5090.2 applies only to Marine Corps installations					

C.2. Installation Technical Review (August 2023)



DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ATLANTIC
6506 HAMPTON BOULEVARD
NORFOLK VA 23508-1278

6250
Ser EV51KW/00230
3 OCT 23

From: Commander, Naval Facilities Engineering Systems Command, Atlantic
To: Commanding Officer, Naval Air Station Sigonella

Subj: PEST MANAGEMENT PROGRAM REVIEW

Ref: (a) DoDM 4150.07, Volume 1, DoD Pest Management Program Elements and Implementation: Structure and Operation
(b) OPNAVINST 6250.4C, Navy Pest Management Programs
(c) OPNAV M-5090.1, Environmental Readiness Program Manual, Chapter 24, Pesticide Compliance Ashore
(d) Environmental Final Governing Standards Italy
(e) DoDM 4150.07, Volume 2, DoD Pest Management Program Elements and Implementation: Pesticide Applicator Training and Certification Program
(f) AFPMB Technical Guide 17, Military Handbook—Design of Pest Management Facilities
(g) CNO Letter 5090 Ser N456M/1U598920
(h) NAVMED P-5010-1, Tri-Service Food Code

Encl: (1) 2023 Pest Management Program Review at Naval Air Station Sigonella

1. As required by references (a) through (c), your program was reviewed 21–24 August 2023 in conjunction with rewriting the integrated pest management plan. The purpose of the review was to evaluate compliance with references (a) through (h) and to recommend needed program changes due to new technology or varying field conditions.
2. Your installation pest management program does not meet all of the basic requirements of references (a) through (h). The main areas of concern are that not all pest management operations are being reported, and the Performance Assessment Representative's pest management certification is expired. Findings and recommendations were discussed with on-site personnel prior to departure and detailed recommendations for improvements are included in enclosure (1). In accordance with reference (b), please respond within sixty days advising us of action planned or taken on recommendations contained in the enclosed report.
3. We are grateful for the assistance provided by your staff. Questions may be referred to Mr. Kirk Williams, Entomologist at DSN 262-4254, (757) 322-4254, or by e-mail: kirk.p.williams4.civ@us.navy.mil.


J. R. CIRVELLO, P.E.
By direction

Copy to:
NEPMU-7
NAVFAC EURAFCENT (EVBLL)
NAS SIGONELLA (PWO, IEPD, IPMC, PAR)

Pest Management Program Review

Naval Air Station Sigonella
Sigonella, Italy

Date of Review: 21–24 August 2023

Reviewer: Kirk Williams, Applied Biology Branch,
Naval Facilities Engineering Systems Command, Atlantic

References: (a) DoDM 4150.07, Volume 1, DoD Pest Management Program Elements and Implementation: Structure and Operation
(b) OPNAVINST 6250.4C, Navy Pest Management Programs
(c) OPNAV M-5090.1, Environmental Readiness Program Manual, Chapter 24, Pesticide Compliance Ashore
(d) Environmental Final Governing Standards Italy
(e) DoDM 4150.07, Volume 2, DoD Pest Management Program Elements and Implementation: Pesticide Applicator Training and Certification Program
(f) AFPMB Technical Guide 17, Military Handbook—Design of Pest Management Facilities
(g) CNO Letter 5090 Ser N456M/1U598920
(h) NAVMED P-5010-1, Tri-Service Food Code

Personnel Contacted:

CDR S. Cochran	Public Works Officer
Mr. J. Lancaster	Installation Environmental Program Director
LCDR M. Leone	Facilities Engineering Acquisition Divisions Director
Ms. A. Ebejer	Facility Support Contracts Branch Head
Mr. S. Bongiovanni	Pest Management Performance Assessment Representative
LCDR R. Miller	Environmental Health Officer
HM2 D. Christoff	Preventive Medicine Technician
HM2 T. Tackett	Preventive Medicine Technician
Ms. A. Grillo	Environmental Protection Specialist
Mr. T. Rauth	Environmental Water Program Manager
Mr. D. Latino	Environmental Tanks/Spills Manager
Mr. L. Arendola	Navy Exchange (NEX) Service Supervisor
Mr. W. Reed	Youth Sport Coordinator
Mr. L. Sangiorgio	Galley Kitchen Manager
Mr. P. Faulk	Navy Gateway Inns and Suites (NGIS) General Manager
Ms. P. Locanda	NGIS Assistance Manager
Mr. L. Ionta	Valiant/Alca JV, LLC
Mr. P. Lucifora	Midtown Food Court Manager
Ms. K. Peraze	NEX Food Court
Ms. T. Berry	Child Development Center (CDC) Director
Ms. A. Kunkle	Department of Defense Dependents School Vice Principle

Personnel Contacted:

CPL K. Lewis	Army Food Inspector
Mr. J. Friedberg	Housing Facility Director
Mr. C. Giovanni	Niscemi Pesticide Applicator

SUMMARY OF COMPLIANCE RECOMMENDATIONS

Compliance Recommendation 1: Send Mr. Bongiovanni to an Initial Pest Management Performance Assessment Representative (PMPAR) certification course as required per references (a) through (e).

Compliance Recommendation 2: Have Marinai Housing submit all pest management records to the NAVFAC pest management consultant as required per references (a) through (d).

Compliance Recommendation 3: Provide water for the decontamination shower at the Naval Radio Transmitter Facility Niscemi pesticide mixing facility at as required per reference (f).

Compliance Recommendation 4: Stop feeding the feral cats on the installation as required per reference (g).

Compliance Recommendation 5: Repair the air curtain over the rear exit doors of the NEX Food Court (Building 225) as required per reference (h).

Compliance Recommendation 6: Provide an air curtain over the rear exit door of the CDC kitchen (Building 314) as required per reference (h).

SUMMARY OF BEST MANAGEMENT PRACTICE RECOMMENDATIONS

Best Management Practice Recommendation 1: Place a bottom sweep on the front exterior doors of the NAS II Galley (Building 533).

Best Management Practice Recommendation 2: Increase the frequency of trash collection at unaccompanied housing and NGIS on NAS II to prevent nuisance pests.

Best Management Practice Recommendation 3: Place a bottom sweep on the exterior kitchen door of the Child Development Center (Building 314).

Best Management Practice Recommendation 4: Provide pest management services to control ants in the kitchen of the Niscemi administration facility (Building 2100).

Questions or comments regarding this report should be addressed to the assigned Naval Facilities Engineering Systems Command, Atlantic professional pest management consultant.

Name: Kirk Williams
Phone: 757-322-4254
DSN 262-4254
E-mail: Kirk.p.williams4.civ@us.navy.mil

INTRODUCTION

Periodic on-site reviews are required by references (a) through (c) to maintain the program and to ensure regulatory compliance with these references and federal, state, and local regulations. This review focused on pest control operations, pesticide safety, and pesticide program compliance at Naval Air Station (NAS) Sigonella including NAS I, NAS II, and NRTF Niscemi.

PROGRAM STATUS

General

General pest control and grounds maintenance pest services are provided by the Base Operations Support (BOS) contract with Valiant/Alca JV, LLC. Department of Defense Dependents School (DoDDS) pest control services are provided by SKE Pest Control. Navy Gateway Inns and Suites (NGIS) pest control services are provided by TriStar. Pest control services for Marinai Housing are provided by Pizzarotti. Grounds maintenance pest control service is provided by in-house applicators at NRTF Niscemi.

Program Administration

The integrated pest management coordinator (IPMC) is the installation commanding officer's advisor on pest management issues and has the authority to oversee all pest management operations, including non-appropriated activities. The designated IPMC for NAS Sigonella is Alessandra Contarino. IPMC certification for Ms. Contarino is valid through February 2024. The IPMC maintains and annually updates the NAS Sigonella integrated pest management plan (IPMP), which is currently being rewritten by NAVFAC Atlantic.

Sebastiano Bongiovanni serves as the Pest Management Performance Assessment Representative (PMPAR) to oversee the BOS contract with Valiant/Alca JV, LLC. Mr. Bongiovanni's certification expired in June 2022. Send Mr. Bongiovanni to the next Initial PMPAR certification course.

Compliance Recommendation 1: Send Mr. Bongiovanni to an Initial PMPAR certification course as required per references (a) through (c).

Pesticide Records and Reporting

All pest management service providers are required to report pest management records to NAVFAC Atlantic Applied Biology through the NAVFAC Online Pesticide Reporting System (NOPRS). NAVFAC Atlantic has not received pest management records from Marinai Housing.

Compliance Recommendation 2: Have Marinai Housing submit all pest management records to the NAVFAC pest management consultant as required per references (a) through (d).

Pesticide Storage and Mixing Facility

There is a pesticide storage and mixing facility located at NRTF Niscemi. The water tank attached to the decontamination tank is empty. No water is available for the decontamination shower at the mixing facility. The rest of the facility is in compliance.

Compliance Recommendation 3: Provide water for the decontamination shower at the NRTF Niscemi pesticide mixing facility as required per reference (f).

Mosquito Control

Not only are mosquitos an annoyance, but they also pose a serious health threat to everyone on base. Surveillance and monitoring of adult mosquitoes are critical to any successful mosquito management program. The adult mosquito populations are being monitored by the preventive medicine depart at NAS Sigonella with 4 Center for Disease Control (CDC) light traps. When trap counts are above or at the action threshold, mosquito control is conducted.

Feral Cats

Feeding of feral cats (Fig. 1) was observed along the NEX Food Court (Building 225) during this visit. Feral cats can harbor and transmit fatal and non-fatal diseases to humans. Health issues caused by feral cats include rabies, plague, ringworm, internal and external parasites, toxoplasmosis, cat scratch fever, allergies to cat hair, and secondary bacterial infections from cat scratches and bites. Feral cats living on the installation can damage government property, contaminate food supplies, and kill birds and other wildlife. Parasites such as fleas are often a problem in areas inhabited by feral cats. Feeding feral cats is prohibited.



Fig 1. Feeding of feral cats observed at NAS I Sigonella.

Compliance Recommendation 4: Stop feeding the feral cats on the installation as required per reference (g).

Nuisance Pest Control

The reviewer inspected a number of facilities for nuisance pest problems. The most common nuisance pests seen during this review were ants. Nuisance pests need only a small opening to enter a building. The first defense against pests is keeping doorways properly sealed. An efficient seal can also reduce energy costs over time. When weather stripping or seals on doors wear out, pests can gain entry into the facility. Rubber door seals warp more easily than bristled sweeps, so rubber door seals should be replaced with sweeps when possible. Additionally, gaps around pipes or electrical conduits, as small as 1/4 inch, can provide enough space for rodents to enter.

Food establishments, a focal point of this review, are particularly sensitive to pest incursion and must be protected against pest entry using tight-fitting doors, windows, screens, air curtains, or other effective means [See reference (h)]. Once pests are in the building, food droppings and grease provide food sources; any water leaks in equipment provide water sources; and any clutter, cracks, or holes provide harborage areas. Proper sanitation as well as prompt equipment and facility repair can decrease pest populations. Creating a barrier zone (18 inches recommended) between vegetation outside and buildings can also help reduce the migration of pests indoors. German cockroaches (*Blattella germanica*) are known for breeding prolifically in cardboard boxes. Avoid storing items in corrugated cardboard boxes long-term and discard unnecessary cardboard as quickly as possible. A robust pest management program is multifaceted, relying on concerted efforts by installation personnel and pest management professionals alike.

The NEX Food Court (Building 225)

The NEX Food Court was good in regard to sanitation. No pests were observed at the food court during this review. The air curtain above the rear exit doors was not working. Filth flies can transmit disease organisms, contaminate food, and be an annoyance. Properly working air curtains over the doors will limit filth fly access.

Compliance Recommendation 5: Repair the air curtain over the rear exit doors of the NEX Food Court (Building 225) as required per reference (h).

Big Al's (Building 2569)

Dead American cockroaches were observed underneath the kitchen equipment. Big Al's was in poor condition with regard to sanitation. Food debris was observed underneath the kitchen equipment. Employees of Big Al's began cleaning the kitchen during this inspection.

NAS II Galley (Building 533)

The overall sanitation of the NAS II Galley was great. This is extremely important to maintain good pest control. The front exterior doors to the dining hall need a bottom sweep.

Best Management Practice Recommendation 1: Place a bottom sweep on the front exterior doors of the NAS II Galley (Building 533).

Garbage bins at NAS II Sigonella

Garbage bins located near the NGHS (Fig. 2) and unaccompanied housing (Fig. 3) on NAS II were overflowing with garbage bags. The installation has scattered trash debris from the overfilled garbage bins. The trash debris is not only unsightly but can also cause an increase of nuisance pests in the area.



Fig 2. Trash debris from the overfilled garbage bins at NGIS.



Fig 3. Trash debris from the overfilled garbage bins at unaccompanied housing.

Best Management Practice Recommendation 2: Increase the frequency of trash collection at unaccompanied housing and NGIS on NAS II to prevent nuisance pests.

Child Development Center (Building 314)

The overall sanitation of the CDC was great. This is extremely important to maintain good pest control. The exterior kitchen door needs a bottom sweep. There is no air curtain above the rear kitchen door. An air curtain over the kitchen door of this facility will limit filth fly access.

Best Management Practice Recommendation 3: Place a bottom sweep on the exterior kitchen door of the Child Development Center (Building 314).

Compliance Recommendation 6: Provide an air curtain over the rear exit door of the CDC kitchen (Building 314) as required per reference (h).

Niscemi Administration Facility (Building 2100)

Ants were observed in the kitchen area of the administration facility during this inspection.

Best Management Practice Recommendation 4: Provide pest management services to control ants in the kitchen of the Niscemi administration facility (Building 2100).

Kirk Williams
KIRK WILLIAMS
Applied Biology Branch
Naval Facilities Engineering Systems Command,
Atlantic

C.3. DD Form 1070: Termite and Wood Decay Inspection Form

See appendix C files for fillable pdf file

TERMITE AND WOOD DECAY INSPECTION						DATE INSPECTED		BUILDING NUMBER	
INSTALLATION				TYPE BUILDING		INSPECTOR			
				☐ PERM ☐ TEMP					
I. FAVORABLE TERMITE AND FUNGI INFESTATION CONDITIONS									
WOOD IN CONTACT WITH SOIL				POOR VENTILATION UNDER BUILDING					
FORM BOARDS LEFT IN CONCRETE				WATER COLLECTIONS UNDER BUILDING					
WOOD MEMBERS SET IN CONCRETE FLOOR				VINES AND SHRUBS AGAINST BUILDING					
WOOD STEPS IN CONTACT WITH SOIL				LEAKY PLUMBING IN BUILDING					
WOOD STEPS WITHOUT SHIELDS				WOOD SCRAP PILED UNDER BUILDING					
WOOD SIDING IN CONTACT WITH SOIL				LOOSE WIRE IN CONTACT WITH SOIL					
PIPES IN CONTACT WITH SOIL AND WOOD				OTHER (Specify)					
NO SHIELDS ON FOUNDATION									
FAULTY TERMITE SHIELD									
II. LOCATION OF INFESTATIONS									
FOUNDATION TIMBERS				BASE BOARDS					
WOOD PILLARS				DOOR FRAMES					
SILLS				WINDOW FRAMES					
CROSS BEAMS				STEPS					
FURNITURE				ROOF					
FLOOR JOIST				OTHER (Specify)					
FLOOR									
STUDS									
III. TYPE OF TERMITE			IV. TYPE OF FUNGI			V. DAMAGE			
☐ SUBTER-RANEAN		☐ NONSUBTER-RANEAN	☐ WOOD DECAY		☐ WOOD STAINING	☐ STRUCTURAL WEAKENING		☐ SUPERFICIAL	ESTIMATED COST
VI. REPAIR AND TREATMENT									
TYPE			RECM	ACCOMP	TYPE			RECM	ACCOMP
REMOVAL OF WOOD FROM SOIL CONTACT					REPAIR OF TERMITE SHIELDS				
SEALING CRACKS IN CONCRETE					REMOVAL OF CONCRETE FORMS				
POINTING UP POOR MORTAR					REMOVAL OF VINES AND SHRUBS				
LOWERING GRADE LEVEL					REMOVAL OF WOOD TRASH				
CAPPING CONCRETE FOUNDATION					OTHER (Specify)				
IMPROVING DRAINAGE UNDER BUILDING									
IMPROVING VENTILATION UNDER BUILDING									
VII. CHEMICAL CONTROL									
APPLICATION OF POISON DUST TO SHELTER TUBES ☐ YES ☐ NO					CHEMICAL USED				
SOIL POISON ☐ YES ☐ NO					TRENCH DEPTH	LINEAR FEET	CHEMICAL USED		
REPLACEMENT OF DAMAGED WOOD ☐ YES ☐ NO	NO. OF M BD FEET REPLACED		UNTREATED		DIPPED	SOAKED	SPRAYED	PRESSURE TREATED	
DRILLING AND FLOODING TREATMENTS ☐ YES ☐ NO					CHEMICAL USED				
WOOD INJECTION FOR DRY WOOD TERMITE ☐ YES ☐ NO					CHEMICAL USED				
VIII. COST									
LABOR		MATERIAL			OTHER		TOTAL		
							0.00		
IX. TREATMENT EFFECTIVENESS									
DATE	REMARKS						INSPECTOR		
DATE	REMARKS						INSPECTOR		
DATE	REMARKS						INSPECTOR		
DATE	TITLE OF INDIVIDUAL AFFECTING REPAIR AND TREATMENT				SIGNATURE				

DD Form 1070, FEB 58

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Appendix D. Pesticide Authorized Use List

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D.1. Pesticide Authorized Use List

The following pesticides are authorized for use at NAS Sigonella as of December 2023

The most current list of approved pesticides for each installation can be found on the NAVFAC Online Pesticide Reporting System at <https://noprs.pestlogics.com/>.

Pesticide Type	Formulation	Pesticide Name	Reg. Number	Active Ingredient
Herbicide	Concentrate	Apache	Italian Registration	Quizalotop-ppethyl
Herbicide	Concentrate	Glifene HP	Reg. # 8656	glyphosate
Herbicide	Concentrate	Hyvar X-L	5481-634	bromacil
Herbicide	Solution	MISTRAL	Italian Ministry of Health n.10306	Acetic acid, Dimethylamine salt
Herbicide	Solution	Turfene L	7368-Italy	Mecoprop-P sale potassico / Dicamba
Insecticide	Suspension Concentrate	Bleu Delta	IT/2017/00404/AUT	Deltamethrin
Insecticide	Emulsifiable Concentrate	Exit 100	MoH n.19350-Italy	Cypermethrin
Insecticide	Gel	Fourmidor	Reg#19817 (Italy)	fipronil
Insecticide	Gel	Goliath Gel	Italian Ministry of Health n.18274	fipronil
Insecticide	Granules/Pellets	NEXA ANTS	Italian Ministry of Health N.18587	FIPRONIL
Insecticide	Concentrate	ROMAL 65	Italian Ministry of Health n.10466	Cypermethrin / Tetramethrin / Piperonyl butoxide
Insecticide	Gel	Solfac Gel Scarafaggi	18504-Italy	imidacloprid
Insecticide	Solution	Toxolin Super	Ministry of Health N. 11068	Pyrethrum Extract / Tetramethrin Pure / Piperonyl Butoxide
Insecticide	Aerosol	VESPA MAYER	Italian Ministry of Health n.18143	Tetramethrin / Permethrin / Piperonyl butoxide
Insecticide	Aerosol	Vespa Spray	18143-Italy	Permethrin / tetramethrin / piperonyl butoxide
Repellant	Granules/Pellets	Snake Out	25 (b) exempt	Camphor
Rodenticide	Bait - Solid	Brocum blocks	Italian Ministry of Health N. IT/2014/00	brodifacoum / denatonium benzoate

Rodenticide	Paste/Glue	BRODIBLEU	IT/2014/00228/AUT-Italy	Brodifacoum / Denatonium benzoate
Rodenticide	Bait - Solid	Notrac Blox	ES/RM-2018-14-00332 (Spain)	Bromadiolone
Rodenticide	Granules/Pellets	Rogar X Plus Extra	IT/2014/00217/AUT-Italy	Bromadiolone / Denatonium benzoate
Rodenticide	Paste/Glue	ROGAR X PLUS PASTA	IT/2013/00077-Italy	Bromadiolone / Denatonium benzoate
Rodenticide	Bait - Solid	SOLO	IT/2015/0-Italy	Brodifacoum

Appendix E. Certifications

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E.1. Table of Applicator Certifications

DoD or Host	Organization or Contractor Name	Name	Applicator Cert Number	Expiration Date
Italy	Valiant/Alca JV, LLC	Carmelo Marchese	56621/SFR	05/13/2027
Italy	Valiant/Alca JV, LLC	Damiano Carlo Rosso	56617/SFR	05/13/2027
Italy	Valiant/Alca JV, LLC	Dario Castello	54587/SFR	03/24/2026
Italy	Valiant/Alca JV, LLC	Davide Tomaselli	56624/SFR	05/13/2027
Italy	Valiant/Alca JV, LLC	Felice Cavallaro	56622/SFR	05/13/2027
Italy	Valiant/Alca JV, LLC	Michele Musumeci	56623/SFR	05/03/2027
Italy	Valiant/Alca JV, LLC	Salvatore La Mastra	56615/SFR	05/13/2027
Italy	Valiant/Alca JV, LLC	Salvatore Nasello	56754/SFR	05/13/2027
Italy	Valiant/Alca JV, LLC	Salvatore Roberto	29661/SFR	03/11/2027
Italy	Valiant/Alca JV, LLC	Sebastiano Roberto	32894/SFR	03/11/2027
Italy	Valiant/Alca JV, LLC	Simone Raineri	12792/SFR	05/13/2027
Italy	Valiant/Alca JV, LLC	Vincenzo Varrica	56677/SFR	05/19/2027
Italy	NRTF Niscemi	Carmelo Di Giovanni	51898/SFR	04/30/2024
Italy	NRTF Niscemi	Mario Amato	51895/SFR	04/30/2024
Italy	TriStar (NGIS)	Drago Antonino	56492/SFR	07/10/2028
Italy	TriStar (NGIS)	Giunta Flavio	56493/SFR	07/10/2028
Italy	Pizzarotti (Marinai Housing)	Livio Germano	48316/SFR	06/29/2028
Italy	SKE Pest Control (DoDDs School)	Giuseppe Monaco	48454/SFR	06/01/2028
Italy	SKE Pest Control (DoDDs School)	Giuseppe Musumeci	13908/SFR	05/12/2027

E.2. Table of IPMC/PAR Certifications

Name	Function	Expiration Date
Alessandra Contarino	IPMC	02/28/2027
Sebastiano Bongiovanni	PAR	01/31/2027

E.2.1. IPMC Appointment Letter



DEPARTMENT OF THE NAVY
U.S. NAVAL AIR STATION SIGONELLA, ITALY
PSC 812 BOX 1000
FPO AE 09627-0001

5300
N00
23 Mar 20

From: Commanding Officer, U.S. Naval Air Station Sigonella, Italy
To: Ms. Alessandra Contarino, CIV

Subj: DESIGNATION AS INTEGRATED PEST CONTROL MANAGEMENT
COORDINATOR

Ref: (a) Executive Order 13423
(b) OPNAV M 5090.1
(c) ISO 14001:2004
(d) Executive Order 13514

1. Per references (a) through (d), you are hereby designated as the integrated pest control management coordinator for U.S. Naval Air Station Sigonella, Italy. As such, you will familiarize yourself with reference (a) through (d) and all other instructions pertaining to your designation.

2. This designation will remain in effect until you transfer or upon written notification.


K. PICKARD, JR

Appendix F. Laws, Regulations, Policies, and Guidance Related to Pesticides and Pest Management

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F.1. Federal

Laws		
Title/Reference	Date	Relevant Requirements / Guidance
Federal Insecticide, Fungicide, and Rodenticide Act (7 U.S.C 136 et seq.) (codified at 40 CFR Parts 152-180) Labeling Requirements for Pesticides and Devices (40 CFR 156) Pesticide Management and Disposal , (40 CFR 165) Exemption of Federal and State Agencies for Use of Pesticides under Emergency Conditions (40 CFR 166) Certification of Pesticide Applicators , (40 CFR 171)	1947, and amendments	Requires pesticide and applicator registration with the U.S. EPA, properly labeled containers, pesticide application records, adequate worker safety, and the proper disposal of unused products. Pesticides are also classified under this act as general use or restricted use. Integrated Pest Management for Federal agencies (7 USC § 136r-1) requires Federal agencies to use IPM techniques in carrying out pest management activities and promote IPM
Animal Damage Control Act (7 USC 426-426c) http://www.animallaw.info/statutes/stusfd7usc426.htm	1931, amended in 1987 and 1991	Gives the Secretary of Agriculture broad authority to investigate and control certain predatory or wild animals and nuisance mammal and bird species.
Clean Air Act (CAA) (42 USC 7401 et seq.) http://www.epa.gov/air/caa/	1955, amended in 1970, 1977, and 1990	Mandates the prevention and control of air pollution from toxic emissions including pesticides.
Clean Water Act (Amended the Federal Water Pollution Control Act of 1972) (33 USC 1251-1387) http://www.law.cornell.edu/uscode/text/33/chapter-26	1977, reauthorized in 1987	Calls for the restoration and maintenance of the chemical, physical, and biological integrity of our nation's waters, including sensitive environments like wetlands. This Act prohibits non-storm water discharges from entering surface waters.

Endangered Species Act (16 USC 1531-1544), (50 CFR Part 402) Federal list of endangered/ threatened plants and wildlife is at 50 CFR §§ 17.11 & 17.12 Interagency Cooperation (16 USC 1536) Interagency Cooperation—Endangered Species Act of 1973, as Amended (50 CFR 402)	1973, amended in 1978	Dictates that all federal agencies must protect listed plants and animals and their habitats from harm. Indicates that pesticide formulations and application methods be reviewed by the U.S. Fish and Wildlife Service to determine whether there could be adverse effects.
Endangered Species Protection Bulletins (ESPB)		Bulletins set forth geographically-specific <i>pesticide use limitations</i> for the protection of endangered or threatened species and their designated critical habitat.
Federal Food, Drug, and Cosmetic Act (FFDCA), (21 USC 301-399d)	25 Jun 1938, amended in 1951, 1962, 1990, 1994, 1997, 2007	Mandates tolerances and exemptions for pesticide chemical residues in food.
Federal Noxious Weed Act (7 USC 2814) incorporated into the Plant Protection Act	FNWA 1974 PPA 2000	Requires federal agencies to develop and implement noxious weed management programs on federal land. Federal Noxious Weed List
Food Quality Protection Act (FPQA), Section 303 – Integrated Pest Management (Public Law 104-170)	1996, amendment to FIFRA and FDCA	Mandates that federal agencies use IPM techniques in pest management activities and promote IPM through procurement and regulatory policies. Primarily established safety standards for pesticides applied to foods.
Migratory Bird Treaty Act (16 USC 703)	1918, with numerous amendments	Requires permits to take migratory birds.
National Environmental Policy Act (NEPA) (42 USC 4321-4347)	1969	Requires a detailed environmental impact statement for any major federal action that can significantly affect the environment. This may include pest management operations that involve large areas of land, application of chemicals to waterways and aerial application of pesticides.
National Oil and Hazardous Substances Pollutant Contingency Program (40 CFR 300)	15 Sep 1994	Provides the organizational structure and procedures for preparing for and responding to discharges of oil and releases of hazardous substances, pollutants, and contaminants.

National Pollution Discharge Elimination System (NPDES)—EPA General Permit Covers the following states: MA, ID, NH, NM, DC, and federal facilities in WA, CO, DE, and VT.	31 Oct 2021	Operators that apply pesticides that result in discharges from the following use patterns may have to submit a notice of intent (NOI) and create a Pesticide Management Discharge Plan (PMDP) if they meet certain criteria: (1) mosquito and other flying insect pest control; (2) weed and algae control; (3) animal pest control; and (4) forest canopy pest control
Non-indigenous Aquatic Nuisance Prevention and Control Act (16 USC 4700 et seq.)	1990	Espouses taking preventive management measures nationwide to prevent and control unintentionally introduced non-indigenous aquatic species and prevent further distribution of these species.
OSHA Hazard Communication Standards (29 CFR 1910)	1970	Stipulates the requirements for applicable and adequate training of all employees regarding hazardous substances (including pesticides) and providing access to SDSs for all chemicals.
Recordkeeping on Restricted Use Pesticides by Certified Applicators; Surveys and Reports (7 CFR 110)	9 Apr 1993	Requirements for recordkeeping on restricted use pesticides by all certified applicators, both private applicators and commercial applicators
Resource Conservation and Recovery Act (RCRA) (42 USC 6901 et seq.) (40 CFR § 260-265)	1976, amended in 1986	Requires proper disposal of waste pesticides and pesticide containers.
Sikes Act Improvement Act (SAIA) (16 USC 670)	31 Dec 2003	Authorizes the Secretary of Defense to develop cooperative plans for conservation and rehabilitation programs on military reservations and modify or improve habitat for endangered species and migratory birds. This includes authorizing the elimination of noxious weeds in efforts to rehabilitate native species.
Standards for Universal Waste Management (40 CFR 273)	11 May 1995	Establishes requirements for managing pesticides
Toxic Chemical Release Reporting: Community Right-to-Know (40 CFR 372)	30 Nov 1994	Requirements for the submission of information relating to the release of toxic chemicals under section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986
Toxic Substances Control Act (TSCA) (15 USC 2601 et seq.)	1976	Requires that new chemicals, including pesticides, be registered and that testing for human health and environmental hazards be performed.

F.2. Department of Defense

Pest Management		
Title/Reference	Date	Relevant Requirements/Guidance
DoD Instruction 4150.07 , DoD Pest Management Program	26 Dec 2019	Establishes policy, assigns responsibilities, and prescribes procedures for the DoD Pest Management Program
DoD 4150.07-M, Volume 1 , DoD Pest Management Program Elements and Implementation: Structure and Operation	22 Jan 2020	Implements policy, assigns responsibilities, and provides procedures for the design and operation of the DoD Pest Management Program.
DoD 4150.07-M, Volume 2 , DoD Pest Management Program Elements and Implementation: Pesticide Applicator Training and Certification Program	22 Jan 2020	Implements policy, assigns responsibilities, and provides procedures for: • The DoD Pesticide Applicator Training and Certification Program • DoD training and certification of pesticide applicators pursuant to Section 136 of Title 7, United States Code, also known and referred to in this volume as the “Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA).”
DoD Directive 4715.1E , Environment, Safety, and Occupational Health (ESOH)	30 Dec 2019	Establishes the AFPMB which provides information, guidance, and publications related to pest and pesticide management. Also advocates implementing IPM into DoD acquisition, procurement, maintenance, and repair processes for systems, equipment, facilities, and land.
DoD Directive 5134.01 w/CH-1 , Under Secretary of Defense for Acquisition, Technology, and Logistics (USD(AT&L))	9 Dec 2005	Establish policies and procedures for the management of DoD installations and environment to support military readiness with regard to
TG-18: Installation IPM Program Guide	Sep 2022	The installation pest management program supports the installation’s mission to ensure military readiness. Appendix A-6 contains the format for the Emergency Vector-borne Disease Response Plan

F.3. Navy, USMC, and UFGS

DEPARTMENT OF THE NAVY		
Title/Reference	Date	Relevant Requirements/Guidance
CNICINST 3750.1E, Navy Bird and Animal Aircraft Strike Hazard (BASH) Program Implementing Guidance	9 Aug 2017	Establishes policy and procedures for implementing the Commander, Navy Installations Command (CNIC) Bird/Animal Aircraft Strike Hazard (BASH) Program
CNIC-M-BASH, Bird/Animal Aircraft Strike Hazard (BASH) Manual Appendix 1: BASH Self-Assessment Checklist (Annual)	1 Apr 2018	Supports Navy Bird/Animal Aircraft Strike Hazard (BASH) policy, identifies key BASH statutory and regulatory requirements, and provides advisory information for management of a BASH program at Navy airfields. The intent is to support the Navy mission by safeguarding air operations assets and flight crews by decreasing the probability of bird or wildlife strikes with aircraft.
eBusiness Operations Office Instruction (EBUSOPSOFFINST) 4200.2 , Department of Navy Nonappropriated Fund (NAF) Policies and Procedures for the Operation and Management of the Government Commercial Purchase Card Program	18 June 2004	Government Purchase Card policy regarding the purchase of pesticides, pesticide application equipment, and pesticide application services
NAVFAC Online Pesticide Reporting System Tutorial		Guide on how to use the online pesticide reporting system
NAVMED P-5010-1, Manual of Naval Preventive Medicine — Tri-Service Food Code	1 Mar 2019	Establishes standardized military food safety standards, criteria, procedures, and roles for the sanitary control and surveillance of food to mitigate risk factors known to cause foodborne illness
NAVMED P-5010, Manual of Naval Preventive Medicine Chapter 8 —Navy Entomology and Pest Control Technology	9 Nov 2004	Guidelines and procedures on the prevention, surveillance, and control of medically important pests.
NMCPHC-TM 6260.9A , Occupational and Environmental Medicine Field Operations Manual	Apr 2017	Standardizes medical surveillance and job certification procedures of employees, the management of occupational injuries and illnesses, their reporting and recordkeeping requirements, and training and certification requirements for OEM service providers

NMCPHC-TM OM 6260 , Medical Surveillance Procedures Manual and Medical Matrix (Edition 12)	Aug 2015 Revised 16 Nov 2016	Procedures for the systematic assessment of employees exposed or potentially exposed to occupational hazards
OPNAVINST 5090.1E , Environmental Readiness Program	03 Sep 2019	Overarching document implementing OPNAV M-5090.1. Provides requirements, delineates responsibilities, and issues implementing policy guidance for the management of the environmental resources for all Navy ships and shore activities.
OPNAV M-5090.1 , Environmental Readiness Program Manual (Chapter 24—Pesticide Compliance Ashore)	25 Jun 2021	Implements the policy set forth in Chief of Naval Operations Instruction (OPNAVINST) 5090.1E, Environmental Readiness Program. Generally requires all pesticide applicators to be DoD- or state-certified. In addition, pest management records must be kept and a pest management plan developed, implemented, and maintained that stresses the importance of IPM.
OPNAVINST 5100.23G , Navy Occupational Safety and Health (NAVOSH) Program	11 July 2011	Requires that pest control operations be thoroughly evaluated to identify and quantify potential health hazards.
OPNAVINST 6210.2A, Quarantine Regulations of the Navy	22 Sep 2015	Delineates the Navy and Marine Corps responsibility to prevent the introduction of medically and economically important pests into the United States
OPNAVINST 6250.4C , Pest Management Programs	11 April 2012	The Navy policy that implements DoD's Pest Management Program. Provides policy specific to Navy operations. This includes record keeping, reporting, safety, management of contracted operations, pest management plans, and environmental protection. Provides responsibilities for preventive medicine.
Policy Letter Preventing Feral Cat and Dog Populations on Navy Property	10 Jan 2002	Clarifies the application of policy regarding the prevention of free roaming (also called wild, feral or stray) cat and dog populations on Navy installations
Marine Corps		
MCO 5090.2, Environmental Compliance and Protection Manual Volume 14 : Integrated Pest Management	11 Jun 2018	Volume 14 establishes Marine Corps policy and responsibilities for complying with the legal use of pesticides at Marine Corps installations in accordance with the Department of Defense (DoD) pest management specifications outlined in DoD Instruction 4150.07

Unified Facilities Guide Specifications (UFGSs)		
UFGS-31 31 16.13 (USACE / NAVFAC / AFCEC / NASA), Chemical Termite Control	1 Aug 2016	This guide specification covers the requirements for termiticide treatment measures for subterranean termite control.
UFGS-31 31 16.19 (USACE / NAVFAC / AFCEC / NASA), Termite Control Barriers	1 Feb 2016	This guide specification covers the requirements for termite control barrier systems, including meshes and basaltic sand.

F.4. Installation

NAS Sigonella		
Title/Reference	Date	Relevant Requirements/Guidance
Natural Resources Management Plan	August 2019	Provides information and procedures on the protection of natural resources including the removal of noxious weeds and animal damage prevention. Includes a species list of plants, insects, and animals.
Cultural Resources Management Plan	July 2021	Provides information and procedures on the protection of cultural resources.
Bird/Animal Aircraft Strike Hazard Program	01 September 2020	Establishes requirements and procedures to prevent strike hazards between aircraft and birds and animals. It includes risk assessment, responsibilities, education, reporting procedures, and bird and wildlife management to eliminate or decrease hazard.

F.5. Special Interest Items

Bird/Wildlife Aircraft Strike Hazard (BASH)		
Title/Reference	Date	Relevant Requirements/Guidance
Airport Wildlife Population Management , Airport Cooperative Research Program (ACRP) Synthesis 39 (sponsored by FAA)	March 2013	Provides airport managers and biologists with a working reference document that reviews the tools, methods, techniques, procedures, and considerations for reducing aircraft collisions associated with wildlife population control management on airports and in the immediate surrounding areas
Airport Wildlife Handbook	July 2012	Assess an airport's wildlife hazards and to make recommendations to resolve any wildlife issues
Wildlife at Airports , USDA	Feb 2017	Focused on management efforts to reduce wildlife hazards at airports
Vector-Borne Diseases		
Title/Reference	Date	Relevant Requirements/Guidance
West Nile Virus Surveillance and Control Guide for U.S. Navy and Marine Corps Installations	2014	Explains the importance of WNV surveillance, describes the steps involved in establishing a surveillance and control program, and provides a threshold when control measures can be carried out, based on observed WNV infection in birds, mosquitoes and humans
Chikungunya Vector Surveillance and Control Plan for U.S. Navy and Marine Corps Installations	Jun 2014	Informs individuals about the disease and provides the knowledge and tools for the surveillance and control of the mosquito species that transmit CHIKV
Aedes Surveillance and Control Plan for U.S. Navy and Marine Corps Installations	Feb 2016	Provide basic knowledge for the surveillance and control of <i>Aedes</i> mosquitos on military installations
Controlling <i>Aedes aegypti</i> and <i>Aedes Albopictus</i>: Information for Vector Control Programs	31 Mar 2016	CDC factsheet providing basic information about <i>Aedes</i> vectors
Surveillance and Control of <i>Aedes aegypti</i> and <i>Aedes albopictus</i> in the United States	31 Mar 2016	CDC factsheet providing basic information about the surveillance and control of <i>Aedes</i> vectors
Invasive Species		

Executive Order 13751 , Safeguarding the Nation From the Impacts of Invasive Species	5 Dec 2016	Prevent the introduction, establishment, and spread of invasive species, as well as to eradicate and control populations of invasive species that are established. Amends EO 13112, Invasive Species.
Inspection and Cleaning Manual for Equipment and Vehicles to Prevent the Spread of Invasive Species , US Dept of Interior Technical Memorandum No. 86-68220-07-05	Sep 2009	This manual provides guidance for inspecting and cleaning vehicles and equipment to help prevent the spread of noxious invasive species.
National Invasive Species Council Management Plan 2016-2018	11 Jul 2016	Identifies the interdepartmental actions that the Federal government and its partners can take to prevent, eradicate, and control invasive species, as well as recover species and restore habitats and other assets adversely impacted by invasive species.
Pesticide Application		
NebGuide G1773 , Spray Drift of Pesticides	Nov 2013	Discusses conditions that cause particle drift and methods to reduce drift potential.
Pollinators		
Title/Reference	Date	Relevant Requirements/Guidance
Presidential Memorandum – Creating a Federal Strategy to Promote the Health of Honey Bees and Other Pollinators	20 Jun 2014	Develop a National Pollinator Health Strategy, which shall include explicit goals, milestones, and metrics to measure progress
DoD Policy to Use Pollinator-Friendly Management Prescriptions - Memo	5 Sep 2014	Expands DoD policy to use current best management practices, as appropriate, specifically to protect pollinators and their habitats, and establishes policy to coordinate with partners on pollinator issues
Pollinator Friendly Pesticide Applicator Best Management Practices	Oct 2014	NAVFAC guidance on proper pesticide use to avoid harming pollinators and their food sources, water, and habitats
National Strategy to Promote the Health of Honey Bees and Other Pollinators	19 May 2015	Identify and recommend, as appropriate, priority conservation needs for native plants and their habitats, and to coordinate implementation of programs for addressing those needs

F.6. International

INTERNATIONAL		
Title/Reference	Date	Relevant Requirements / Guidance
DoDI 4715.05 , Environmental Compliance at Installations Outside the United States	Nov 2013	Establishes policy and assigns responsibilities for managing environmental compliance to protect human health and safety outside the United States on installations under DoD control.
DoDM 4715.05, Overseas Environmental Baseline Guidance Document (OEBGD) Volume 1 : Conservation Volume 2 : Air and Toxics Volume 3 : Water Volume 4 : Hazardous Material, Storage Tanks, Spills, and Pesticides Volume 5 : Waste	29 Jun 2020	For OCONUS installations that do NOT have an FGS, Volume 4, Section 8 contains standards on the prevention of threats to human health and the environment from the storage and use of pesticides.
DoDD 3000.10 , Contingency Basing Outside the United States	10 Jan 2013	Policy for integrating comprehensive risk management into pest management planning, design, and operations
Italy: Final Governing Standards	7 Sept 2012	Chapter 11 – Pesticides. Contains standards regulating the use, storage and handling of pesticides, herbicides, and defoliants at DoD installations

Appendix G. Environmental

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G.1. Sample Pesticide Management Program Environmental Impact Log

Practice	Practice Owner	Aspects	Impacts	Vulnerable Assets
Pesticide storage	Pest control shop supervisor	Potential spill	Degradation of water quality Human exposure to chemicals	Environmental resources Human health and safety
	Pest control shop supervisor	Fire	Risk of human injury due to fire and chemicals Degradation of air quality	Human health and safety Environmental resources
Pesticide transportation	Pesticide applicator / transporter	Potential spill due to container damage	Degradation of water and soil quality Property contamination	Real property Mission Human health and safety Public perception
	Vehicle owner	Air emissions from vehicle	Degradation of air quality	Environmental resources
Pesticide mixing	Pesticide mixer / applicator	Potential spill	Degradation of water and soil quality	Environmental resources
	Pesticide mixer / applicator	Hazardous waste generation	Cost of disposal	Mission
	Pesticide mixer / applicator	Chemical mixing	Exposure to toxic chemicals	Human health and safety
Pesticide application	Pesticide applicator	Potential spill	Degradation of water and soil Exposure to toxic chemicals	Natural resources Human health and safety
	Pesticide applicator	Pesticide drift	Killing of non-target plants and animals	Natural resources
	Pesticide applicator	Stormwater discharge	Degradation of water	Natural resources
	Pesticide applicator	Air emissions from gas powered application equipment	Degradation of air quality	Environmental resources
	Pesticide applicator	Vehicle use for pesticide application	Damage to plants and animal habitats if off road	Natural resources
	Pesticide applicator	Hazardous waste generation	Cost of disposal Exposure to toxic chemicals	Mission Human health and safety

Practice	Practice Owner	Aspects	Impacts	Vulnerable Assets
	Pesticide applicator	Chemical usage	Exposure to toxic chemicals	Human health and safety Environmental resources
Cleaning pesticide application equipment at application site and in pest control shop	Pesticide applicator	Waste water generation	Degradation of water quality Cost of treatment/disposal	Mission
Pesticide container disposal	Pest control shop supervisor	Solid hazardous waste generation	Cost of disposal	Mission
	Pest control shop supervisor	Waste water generation	Degradation of water quality Cost of treatment/disposal	Environmental Resources Mission
Non-chemical control: Mechanical weed removal	Grounds Maintenance workers	Root damage to native plants	Destruction of natural resources	Natural resources
Non-Chemical control: Animal trapping	Pest control operator	Animal relocation	Potential destruction of natural resources	Natural resources
	Pest control operator	Trapped animal	Human or domestic animal injury caused by trapped animal	Human and domestic animal health and safety

Appendix H. Medical

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H.1. Emergency Vector-Borne Disease Response Plan

PREVMED shall develop and maintain an Emergency Vector-Borne Disease Control Plan (EVDCP). NEPMU-7 can provide assistance with developing this plan.

Installation Emergency Vector-Borne Disease Response Plan



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Appendix I. Glossary

Acaricide. An agent used to kill mites and ticks.

Applied Biology Program. A network of NAVFAC Pest Management Consultants (PMCs) in the Environmental Business Line that assist Navy and Marine Corps installations with FIFRA and Final Governing Standards-based compliance and provide Integrated Pest Management solutions that protect operations, war-fighters, quality of life, property, materiel and the environment from the adverse effects of living organisms.

Arachnid. An arthropod that has eight legs and two body segments in the adult stage.

Arthropod. Invertebrate animals (insects, arachnids, and crustaceans) that have jointed appendages and a segmented body.

Avicide. An agent used to kill or repel birds.

Broad spectrum. A classification of pesticide that will kill a wide range of pests.

Broadcast application. The application of a pesticide to a wide area.

Crack and crevice treatment. Application of a pesticide to cracks and crevices where pests are known to live, feed, and/or breed.

DoD-certified pesticide applicator. Military or civilian personnel certified per the “DoD Plan for Certification of Pesticide Applicators” in the pest management categories that are appropriate for their type of work.

Drift. The movement of a pesticide through air, ground, or water out of the control target area.

Exclusion. A pest control method that prevents the entry of a pest into an area to be protected from the pest.

Functional area. Installation personnel, agencies, departments, contractors and facilities that use or store pesticides, conduct pest management operations, provide for safety or security of pest control operations, or have the responsibility of preventing pests.

Fungicide. An agent used to destroy or inhibit growth of fungi.

Herbicide. An agent used to destroy or inhibit plant growth.

Insecticide. An agent used to destroy insects.

Integrated pest management (IPM). A planned program incorporating education, continuous monitoring, record keeping, and communication to prevent pests and disease vectors from causing unacceptable damage to operations, people, property, materiel, or the environment. IPM uses targeted, sustainable (effective, economical, environmentally sound) methods including habitat modification; biological, genetic, cultural, mechanical, physical, and regulatory controls; and, when necessary, the judicious use of least-hazardous pesticides.

Integrated pest management coordinator. The individual officially designated by the installation commander to coordinate and oversee the installation pest management program and installation IPM plan. IPM coordinators must be certified as pesticide applicators if their job responsibilities require them to apply or supervise the use of pesticides.

Integrated Pest Management Plan. A detailed document for the design, implementation, and maintenance of all pest management and pesticide storage and use on an installation or group of installations.

Invasive species. A species of animal, plant or organism that is not native to a geographic area and can potentially cause harm to native organisms and their habitats.

Leach. The movement of a pesticide through soil.

Molluscicide. An agent used to kill snails.

Noxious or invasive weed. A weed that, if introduced, into a habitat can cause damage or injury to other organisms in that habitat. They may cause deprivation of water to other plants, physical injury to animals, or increased risk for wild fire.

Personal relief. Pest management control efforts made by DoD personnel or their family members at their own expense for control of pests consistent with DoD and Navy policy.

Pest. Any organism (except for microorganisms that cause human or animal diseases) that adversely affects operations, preparedness, the well-being of humans or animals, real property, materiel, equipment or vegetation, or is otherwise undesirable.

Pest management performance assessment representatives (PMPARs). Installation personnel trained in contract performance assessment and pest management, whose duties include surveillance of commercial pest management services to ensure that the performance complies with contract specifications and legal requirements. [Formerly known as Pest Control Quality Assurance Evaluators (PCQAE).]

Pest management. The prevention and control of disease vectors and pest that may adversely affect the DoD mission or military operations; the health and well-being of people; or structures, materiel, or property.

Pesticide. Any substance or mixture of substances registered by EPA under FIFRA, intended to destroy, repel, or mitigate pests. Includes, insecticides, rodenticides, herbicides, fungicides, plant regulators, defoliants, desiccants, disinfectants, antifouling paints and biocides (such as water treatment chemicals). NAVFAC PPMCs do not approve disinfectants or biocides.

Pesticide applicator. Any individual who applies pesticides.

Pesticide cancelation. An action by EPA that may limit the use of a pesticide. EPA often issues instructions with the pesticide cancelations providing information on the disposition of canceled pesticides.

Pesticide facility. The building and areas designated for handling and storing pesticides.

Pre-treatment. A termiticide applied to the soil during the construction of a new building or addition.

Professional pest management consultant. Degreed technical specialists, such as NAVFAC civilian entomologists (Applied Biologist) and BUMED commissioned medical entomologists, who have command program oversight responsibilities and provides guidance and information on the management of pest management programs for commands and installations.

Registered pesticide. A pesticide registered by the EPA for sale and use within the United States.

Residual pesticide. The application of a pesticide that will remain effective on to the surface to which it is applied for a long period of time.

Rodenticide. An agent used to destroy rodents.

Safety Data Sheet. A document (OSHA form 174, or equivalent) that accompanies a pesticide product, providing the handler with chemical information on ingredients, handling instructions, potential hazards, and manufacturer address and emergency contact information.

Space spray. The application of a pesticide as a fine airborne mist to kill flying insects. This includes ultra-low volume application and fogging.

Stakeholder. A person, agency, organization, or department that has an interest in the installation's pest management program.

State-certified commercial pesticide applicators. Personnel certified in accordance with FIFRA by a State (in which the work will be performed) with an EPA-approved certification plan and certified in the category in which a pesticide will be applied.

Subsistence. Stored food items.

Surveillance. The use of surveys over a period of time to monitor the increase and decrease of pest populations over time. Often used as a means of "early warning" of increase in pests or risk of disease and as a means of determining efficacy of pest management operations.

Survey. Observing, collecting, quantifying, identifying and analyzing a pest population.

Ultra-low volume (ULV). A method of applying a pesticide as a space spray. This method involves applying fine droplets of concentrated pesticide.

Uncertified pesticide applicators. Individuals who have not successfully completed certification training. Uncertified military and DoD civilian personnel who are in training to become certified pesticide applicators may apply pesticides when under the direct line-of-sight supervision of a DoD-certified pesticide applicator. Uncertified personnel may apply self-help or personal relief pesticides when the operation has been approved by a command pest management consultant.

Vector/Disease Vector. Any animal capable of transmitting the causative agent of human disease; serving as an intermediate or reservoir host of a pathogenic organism; or producing human discomfort or injury, including (but not limited to) mosquitoes, flies, other insects, ticks, mites, snails, and rodents. It is recognized that certain disease vectors are predominantly economic pests that as conditions change may require management or control as a disease vector.

Vector-borne disease. A disease transmitted by a vector.

Zoonosis. A disease that normally occurs in animals that can be transmitted to humans.

Appendix J. List of Acronyms & Abbreviations

AFPMB	Armed Forces Pest Management Board
APHIS	Animal and Plant Health Inspection Service
AUL	authorized use list
BASH	Bird/Wildlife Aircraft Strike Hazard
Bti	<i>Bacillus thuringiensis israelensis</i>
BUMED	Navy Bureau of Medicine and Surgery
CFR	Code of Federal Regulations
CO	commanding officer
COR	contracting officer representative
CNIC	Commander, Navy Installations Command
CRM	cultural resources manager
CWP	Contractor Work Plan
DoD	Department of Defense
DoDI	Department of Defense instruction
DoDM	Department of Defense Manual
DoN	Department of the Navy
E	endangered
EA	environmental assessment
EEE	Eastern Equine Encephalitis
EHS	extremely hazardous substance
EMS	Environmental Management System
EO	executive order
EPA	Environmental Protection Agency
ESA	Endangered Species Act
EVDRP	Emergency Vector-borne Disease Response Plan
FAP	Functional Assessment Plan
FIFRA	Federal Insecticide, Fungicide, and Rodenticide Act
FSC/BOS	Facilities Support Contract/Base Operation Support
GRX	GeoReadiness Explorer
IAP	Internal Assessment Plan

ICRMP	Integrated Cultural Resources Management Plan
IH	industrial hygiene
INRMP	Integrated Natural Resources Management Plan
IPM	integrated pest management
IPMC	integrated pest management coordinator
IPMP	Integrated Pest Management Plan
KO	contracting officer
LEPC	Local Emergency Planning Committee
MCCS	Marine Corps Community Services
MCX	Marine Corps Exchange
MoM	measure of merit
MRE	meal, ready to eat
MWR	morale, welfare, & recreation
NAFI	Non-Appropriated Fund Instrumentality
NAVFAC	Naval Facilities Engineering Systems Command
NAVMED	Navy Medical (Command)
NAVSUBASE	naval submarine base
NECE	Navy Entomology Center of Excellence
NEPMU	Navy Environmental and Preventive Medicine Unit
NEX	Navy Exchange
NISH	National Institute of Severely Handicapped
NMCI	Navy and Marine Corps Intranet
NMCPHC	Navy and Marine Corps Public Health Center
NPDES	National Pollutant Discharge Elimination System
NOPRS	NAVFAC Online Pesticide Reporting System
NOSC	naval operational support center
OPNAVINST	Chief of Naval Operations instruction
OPNAV M	Chief of Naval Operations manual
ORM	operational risk management
OSHA	Occupational Safety and Health Administration
PAR	performance assessment representative
PMPAR	Pest Management Performance Assessment Representative
PMRS	Pest Management Record Spreadsheet
PMSP	Pest Management Service Provider

PMT	preventive medicine technician
POC	point of contact
PPE	personal protective equipment
PPMC	professional pest management consultant
PPV	public-private venture
RTU	ready-to-use
SDS	safety data sheet
SPCCP	Spill Prevention, Control, and Countermeasures Plan
T	threatened
TG	technical guide
T&ES	threatened and endangered species
UFGS	Unified Facilities Guide Specifications
ULV	ultra-low volume
U.S.	United States
U.S.C.	United States Code
USDA	United States Department of Agriculture
USDA-APHIS	United States Department of Agriculture-Animal and Plant Health Inspection Service
USFWS	U.S. Fish and Wildlife Service
USN	United States Navy
WNV	West Nile Virus

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Appendix K. Integrated Pest Management Sheets

The integrated pest management sheets are located in the directory Appendix_K_IPM_Sheets in the file Appendix_K_IPM_Sheets.docx.

The sheets provide general guidance for control of common pests and are not installation specific. They should be used as a basis for pest management action, but should not be considered “regulations” for the job. Management sheets should be used as guidelines to help implement reasonable, cost effective, safe, environmentally responsible control of pests. The integrated pest management coordinator (IPMC) or other pest control personnel may choose to establish different thresholds and use IPM methods that are more appropriate to their local circumstances. Write in any new ideas or programs to maintain a document that will remain applicable over time. Any suggested pesticides from these sheets are required to be approved before use.

NUISANCE PESTS

- American Cockroaches
- Cockroaches in Food Preparation Areas
- Drain Flies
- Fruit Flies
- Nuisance Ants
- Stored Product Pests in Food Storage Areas

HEALTH-RELATED PESTS

- Bed Bugs
- Filth Flies
- Fire Ants
- Fleas In and Around Buildings
- Mites
- Mosquitoes, Adult Control
- Mosquitoes, Larval
- Spiders
- Stinging Insects
- Ticks

STRUCTURAL PESTS

- Drywood Termites
- Subterranean Termites

TURF AND ORNAMENTAL PESTS

- Ornamental Plant Pests
- Snails and Slugs

VERTEBRATE PESTS

- Bats
- Nuisance Birds
- Feral Animals
- Rodents

VEGETATION MANAGEMENT

- Invasive Plants in Natural Areas
- Terrestrial Weeds
- Weeds in Rights-of-Way

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